

काय चिकित्सा 1 (B)

ज्वर	Swine flu	Gout
Typhoid	पाण्डु	प्लीहादोष
Pneumonia	Anemia	यकृतदोष
Pleurisy	आमवात	Hepatitis
Influenza	R.A.	Liver Cirrhosis
Mumps	मदात्यय	Leukemia
Meningitis	हृद्रोग	कुष्ठ
Encephalitis	C.V. diseases	श्वित्र
Tetanus	हृदशूल	शीतपित्त उदर कोठ
Yellow fever	Angina pectoris	विसर्प
Plague	Hypertension	क्षुद्र रोग
Dengue fever	Hypotension	National Health Programme
Chukun guniya	रक्तपित्त	
Leptospirosis	Hemorrhagic dis.	Alternative med.
Viral fever	कामला	Diseases of Endocrine gland
Anthrax	Jaundice	
Small pox	दाह	Immunodeficiency Disorders
Chicken pox	मद मूर्च्छा सन्यास	
Measles	वातरक्त	Emergency med.

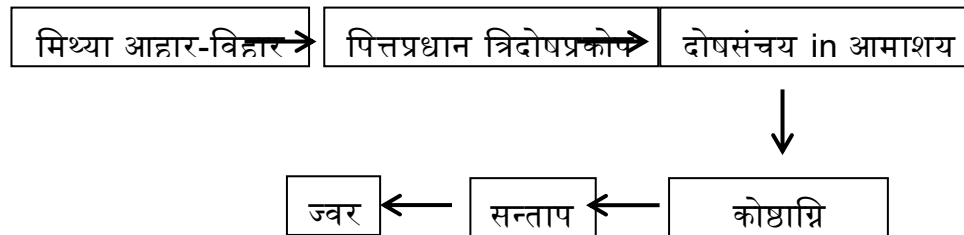
Diseases of रसवह स्रोतस

ज्वर (Fever)

देहेन्द्रियमनस्तापी सर्वरोगाग्रजो बली । ज्वरः प्रधानो रोगाणाम्..... ॥

सामान्य निदान (cause)	पूर्वरूप (prodromal Symptoms)	सामान्य लक्षण (clinical features)	साध्यासाध्यता (prognosis)
मिथ्या आहार-विहार निज & आगतुक कारण	आलस्य नयनस्त्राव जृम्भा गौरव क्लम आस्यवैरस्य अविपाक	देहसन्ताप मनःसन्ताप (increased body temperature & irritation to mind)	Incurable if it has multiple signs & symptoms, arises due to many causes, pt is weak, developed edema & other complications.

सम्प्राप्ति (Pathogenesis):



ज्वर भेद (Types of Jwara)	
द्विविध	A) 1.शारीरिक 2.मानसिक. B) 1.सौम्य 2.आग्नेय. C) 1.बहिर्वेग 2.अन्तर्वेग. D) 1.प्राकृत 2.वैकृत. E) 1.साध्य 2.असाध्य
पञ्चविध (विषमज्वर)	1.सन्तत 2.सतत 3.अन्येद्युष्क 4.तृतीयक 5.चतुर्थक ज्वर
सप्तविध (धातुगत ज्वर)	1.रसगत 2.रक्तगत 3.मांसगत 4. मेदोगत 5.अस्थिगत 6.मज्जागत 7.शुक्रगत ज्वर
अष्टविध	1.वातज 2.पित्तज 3.कफज 4. वात-पित्तज 5.पित्त-कफज 6.वात-कफज 7.सन्निपातज 8.आगन्तुक ज्वर
सन्निपातज्वर	1.वातपित्तोल्बण हीनकफ 2.वातकफोल्बण हीनपित्त 3.पित्तकफोल्बण हीनवात 4.वातोल्बण हीनकफपित्त 5. पित्तोल्बण हीनवातकफ 6.कफोल्बण हीनवातपित्त 7.कफोल्बण मध्यपित्त हीनवात 8.पित्तोल्बण मध्यकफ हीनवात 9.वातोल्बण मध्यकफ हीनपित्त 10.कफोल्बण मध्यवात हीनपित्त 11.वातोल्बण मध्यपित्त हीनकफ 12.पित्तोल्बण मध्यवात हीनकफ 13.समवृद्ध वात पित्त कफ
आगन्तुक ज्वर	1.अभिघातज 2.अभिषंगज 3.अभिचारज 4.अभिशापज

Synonyms of ज्वरः

ज्वर – विकार – रोग – व्याधि – आतंक

Nature (स्वभाविक प्रकृति) of ज्वरः

Jwara is क्षयकारक that is why it is called as क्षय. Jwara is also called as तम, as it causes मोह which is function of तमोगुण. Jwara is the result of पापकर्म, so it is called as पाप्मा. Jwara is the cause of मृत्यु, so it is called as मृत्यु. Jwara kills the person like यम, so it called as यमात्मक.

Cardinal features (प्रत्यात्म लक्षण) of ज्वरः

ज्वरप्रत्यात्मिकं लिंगं सन्तापो देहमानसः ।

The cardinal feature of Jwara is the increased body temperature and irritated mind.

ज्वर भेद	लक्षण
शारीरिक ज्वर	If Jwara manifests in the body first, due to affliction of शारीरिक दोष. ताप to the body produces अरुचि (anorexia), तृष्णा (thirst), अंगमर्द (bodyache), स्वेदावरोध (absence of perspiration) etc.
मानसिक ज्वर	Manifests in the mind first, due to affliction of मानसिक दोष. ताप to the mind produces वैचित्र्य (mental distraction), अरति (restlessness), ग्लानि (feeling of tiredness or weakness) etc.

ज्वर भेद	लक्षण
सौम्य ज्वर	Jwara produced by वात-कफ is Sheeta or Saumya jwara. Patient desires for hot things.
आग्नेय ज्वर	Jwara produced by वात-पित्त is Ushna or Agneya Jwara. Patient desires for cold things.
Due to योगवाही गुण (Synergetic action) of वात, it produces both types of effects viz. in combination with कफ it produces शीतता (cold) & in combination with पित्त it produces उष्णता (heat).	

ज्वर भेद	लक्षण
अन्तर्वेग ज्वर	Characterized by अन्तर्दाह (burning sensation inside the body), अति-तृष्णा (excessive thirst), प्रलाप (delirium), श्वास (dyspnea), भ्रम (giddiness), सन्धिशूल (pain in joints), अस्वेद (absence of perspiration), and दोषवर्चोविनिग्रह (non-elimination of dosha and mala)
बहिर्वेग ज्वर	Characterized by बाह्यशरीर सन्तापोऽधिक (excessive temperature in the exterior parts of the body), तृष्णा (mild thirst) etc. It is curable.

ज्वर भेद	लक्षण
प्राकृत ज्वर	The Jwara manifested in particular season that is responsible for vitiation of involved dosha is called as Prakrita Jwara, e.g. Pittaja Jwara in Sharad ritu & Kaphaja Jwara in Vasanta ritu etc.
वैकृत ज्वर	Jwara manifest in seasons other than that responsible for vitiation of the concerned Dosha is known as Vaikrita jwara, e.g. Kaphaja Jwara in Sharad ritu etc.

ज्वर भेद	लक्षण
साध्य ज्वर	In case of बलवान रोगी, अल्पदोष अवस्था, and अनुपद्रव (no complications) → Jwara is curable.
असाध्य ज्वर	Following factors indicates incurability of Jwara → बहुहेतु → Jwara due to many etiological factors

	बहुलक्षण → Appearance of many signs and symptoms शीघ्रं इन्द्रिय नाशन → Quick destruction of sense organs सप्रलाप-भ्रम-श्वास → If Jwara associated with delirium, giddiness and breathlessness (dyspnea), pt. dies either on 7th day or 10th day or 12th day क्षीण or शून रोगी → Weak or edematous patients गम्भीर ज्वर → Deep seated Jwara दैर्घरात्रिकज्वर → Persistence of Jwara for several nights केश सीमन्तकृत ज्वर → Hairs of the head fall apart to produce a straight line
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ज्वर भेद	लक्षण
वातज ज्वर	→ विषमारम्भ विसर्गित्वं (irregular onset, rise & fall of fever) → ऊष्मणो वैषम्यं (irregular temperature) → परूष अरुणवर्णत्वं नख-नयन-वदन-मूत्र-पुरीष-त्वचा (excessive roughness and redness of nails, eyes, face, urine, stool and skin) → कषायास्यता आस्यवैरस्यं वा (astringent taste or altered taste in mouth) → मुखतालुकण्ठशोष (dryness of mouth, palate and throat) → Various types of localized or shifting pain in different body parts & Aggravation of Jwara at the end of digestion, day, night and in summer season.

पित्तज ज्वर	→ अत्यर्थ उष्ण (hyperpyrexia) → अतिमात्रं दाह (intense burning sensation) → कटुकास्यता (pungent taste in mouth) → घ्राणमुखकण्ठौष्ठतालुपाकः (inflammation of nose, mouth, throat, lips and palate) → हरित हारिद्रत्वं नख-नयन-वदन-मूत्र-पुरीष-त्वचा (greenish or yellowish discolouration of nails, eyes, face, urine, stool and skin) → तृष्णा मद भ्रम मूर्च्छा पित्तछर्दन अतिसार & Aggravation of Jwara at the mid of digestion, mid-day, midnight and autumn season.
कफज ज्वर	→ गुरुगात्रता अनन्नाभिलाषः हृल्लास छर्दि हृदयोपलेप → कास श्वास प्रतिश्याय मुखमाधुर्य (sweetness in mouth) → श्वेत्यं नख-नयन-वदन-मूत्र-पुरीष-त्वचा (whitish discolouration of nails, eyes, face, urine, stool and skin) → Aggravation of Jwara especially immediately after meals, fore-noon, early night and spring season.

निदान of द्वन्द्वज्वर & सन्निपातज्वर
विषमाशन (irregular diet), अनशन (excessive fasting), ऋतुव्यापत्त (seasonal influence), असात्म्यगन्ध (unpleasant smell) विषोपहत (poisonous substance or water), स्नेहस्वेदवमनादिकर्म-मिथ्यायोग (improper administration of panchakarma), मिथ्यासंसर्जन (improper dietetic regimen after shodhana), विषमप्रजनन (abnormal labour), प्रजातानां मिथ्योपचार (improper or unwholesome regimen after delivery) etc.

द्वन्द्वज्वर	लक्षण
वात-पित्तज्वर	शिरोरुक् (headache), पर्वणां भेद (pain in joints), रोमहर्ष (horripilation), कण्ठास्यशोष (dryness of throat and mouth), स्वप्ननाश (sleeplessness), दाह, छर्दि, तृष्णा, मूर्च्छा, भ्रम.
वात-कफज्वर	मध्यवेग सन्ताप (mild increase in temperature), शीतक (feeling of coldness), स्तैमित्य (feeling of covered with wet cloth), गौरव, तन्द्रा, प्रतिश्याय, कास, पर्वणां-रुक्, शिरोग्रह.
पित्त-कफज्वर	मुहुर्दाह मुहुःशीत (frequent alternate feeling of burning sensation and coldness), स्वेद स्तम्भः मुहुर्मुहुः (sometimes sweating and sometimes arrest of sweating), लिप्तित्कास्यता (coating and bitter taste in mouth), तन्द्रा, कास, अरुचि, तृष्णा.

क्र.	सन्निपातज्वर	लक्षण
1	वातपित्तोल्बण हीनकफ ज्वर	भ्रम पिपासा दाह गौरव शिरसोऽतिरुक्
2	वातकफोल्बण हीनपित्त ज्वर	शैत्य कास अरुचि तन्द्रा पिपासा दाह रुक् व्यथा
3	पित्तकफोल्बण हीनवात ज्वर	छर्दि मुहुर्दाह-मुहुःशैत्य तृष्णा मोह अस्थिवेदना
4	वातोल्बण हीनपित्तकफ ज्वर	सन्धि-अस्थि-शिरःशूल प्रलाप गौरव भ्रम

5	पित्तोल्बण हीनवातकफ ज्वर	रक्त-विण्मूत्रता स्वेद तृष्णा बलक्षय मूर्च्छा
6	कफोल्बण हीनवातपित्त ज्वर	आलस्य अरुचि हृल्लास दाह वमन अरति भ्रम तन्द्रा कास
7	कफोल्बण मध्यपित्त हीनवात ज्वर	प्रतिश्याय छर्दि आलस्य तन्द्रा अरुचि मन्दाग्नि
8	पित्तोल्बण मध्यकफ हीनवात ज्वर	हारिद्र-मूत्रनेत्रत्वं दाह तृष्णा भ्रम अरुचि
9	वातोल्बण मध्यकफ हीनपित्त ज्वर	शिरोरुक् वेपथु श्वास प्रलाप छर्दि अरुचि
10	कफोल्बण मध्यवात हीनपित्त ज्वर	शीतक गौरव तन्द्रा प्रलाप अस्थिशिरोऽतिरुक्
11	वातोल्बण मध्यपित्त हीनकफ ज्वर	श्वास कास प्रतिश्याय मुखशोष अतिपार्श्वरुक्
12	पित्तोल्बण मध्यवात हीनकफ ज्वर	वर्चभेद अग्निमान्द्य तृष्णा दाह अरुचि भ्रम
13	सम त्रिदोषज सन्निपातज्वर	क्षणे दाहः क्षणे शीतं अस्थि-सन्धि-शिरोरुजा Mixed lakshanas of V-P-K

आगन्तुक ज्वर	निदान	लक्षण
1. अभिघातज्वर	Due to trauma of शस्त्र (weapons), लोष्ट (stone), कशा (hunter), काष्ठ (wood-stick), मुष्टि (fist) etc. Vata gets aggravated & vitiates rakta resulting in the manifestation of Abhigataja Jwara.	व्यथा (distress), शोथ (swelling), वैवर्ण्य (discolouration) and रुज (pain).

2. अभिषंगज ज्वर	Due to काम क्रोध भय शोक Vata dosha gets aggravated which manifests Jwara. Due to भूत-प्रेत-पिशाच all the Tridosha get aggravated which manifest Jwara. Contact with विषवृक्ष (poisonous tree) & विषानिल (poisonous air) can also manifest Abhishangaja Jwara.	<u>कामज्वर</u> → ध्यान (concentrating or thinking of the desires) निःश्वासबहुलं (tachypnea) <u>क्रोधज्वर</u> → बहुसंरम्भ (red eyes, red face etc.) <u>भयज्वर</u> → त्रास (afraid) <u>शोकज्वर</u> → वाष्पबहुलं (lacrimation) <u>भूतावेशजन्य ज्वर</u> → अमानुषं (super natural activities like more strength, inhuman voice etc.) <u>विषज ज्वर</u> → मूर्च्छा मोह मद ग्लानि
3. अभिचारज ज्वर	Jwara caused by evil tantric rituals or chanting mantras is called Abhicharaja Jwara.	Sannipatajwara lakshana along with affliction of mind, sense organs and body.
4. अभिशापज ज्वर	Jwara caused by cursing of the saints is called Abhishapaja Jwara.	Sannipatajwara lakshana along with affliction of mind, sense organs and body.

सप्तधातुगत ज्वर	लक्षण
1. रसधातुगत ज्वर	गौरवता दैन्यता सदन अंगमर्द छर्दी अरोचक
2. रक्तधातुगत ज्वर	पिडका रक्तष्ठीवन तृष्णा दाह राग भ्रम मद
3. मांसधातुगत ज्वर	अन्तर्दाह गात्रविक्षेप दौर्गन्ध्य सृष्टविट तृष्णा मोह ग्लानि
4. मेदोधातुगत ज्वर	अतिस्वेद स्वगंध-असहत्वं तृष्णा प्रलाप छर्दी
5. अस्थिधातुगत ज्वर	अस्थिभेद (वेदना) गात्रविक्षेप कण्ठकूजन अतिसार छर्दी
6. मज्जाधातुगत ज्वर	तमःप्रवेश मर्मच्छेद बहिर्शैत्य-अन्तर्दाह श्वास कास हिक्का
7. शुक्रधातुगत ज्वर	शुक्रमोक्ष शुक्रनाश मृत्यु

रस-रक्त-मांस-मेदोधातुगत→ साध्य,

अस्थि-मज्जागत→ कृच्छ्रसाध्य,

शुक्रगत→ असाध्य

विषमज्वर :

Vishamajwara is characterized by irregular onset, inconsistent symptoms and appearance in irregular time; and it persists for longer duration.

विषमज्वर	धात्वाश्रय	ज्वर रूप
1. सन्तत ज्वर	रस	Continuous or remittent for 7, 10 or 12 days
2. सतत ज्वर	रक्त	Once in 12 hours (twice in 24 hours)
3. अन्येद्युष्क	मांस	Once in 24 hours
4. तृतीयक	मेद	Every third day (Tertian fever)
5. चतुर्थक	अस्थि-मज्जा	Every fourth day (Quartan fever)

ज्वर-अवस्था (Stages of fever):

- तरुण ज्वर → Jwara upto 7 days is called तरुण- or नव-ज्वर.
- आम ज्वर → Jwara associated with Aama
- पच्यमान ज्वर → Aama is being digested
- निराम ज्वर → Aama has been digested i.e. Jwara free from Aama, usually from 8th day Jwara is Nirama.
- जीर्ण ज्वर → Chronic fever
- पुनरावर्तक ज्वर → Relapsing fever: Jwara which is recurred within few days.

आमज्वर लक्षण :

अरुचिश्च अविपाकश्च गुरुत्वमुदरस्य च ।
हृदयस्याविशुद्धिश्च तन्द्रा च आलस्यमेव च ॥
ज्वरोऽविसर्गी बलवान् दोषाणां अप्रवर्तनम् ।
लालप्रसेको हृलासः क्षुब्धशो विरसं मुखम् ॥
स्तब्धसुप्तगुरुत्वं च गात्राणां बहुमूत्रता ।
न विट् जीर्णा न च ग्लानिः ज्वरस्यामस्य लक्षणम् ॥

Aama Jwara is characterized by anorexia, indigestion, heaviness in abdomen, impurities in heart or epigastrium, drowsiness, laziness, continuous and high fever, non-elimination of dosha, salivation, nausea, loss of appetite, distaste in mouth, stiffness, numbness, heaviness, polyuria, improper formation of stool, and absence of tiredness.

पच्यमानज्वर लक्षण :

ज्वरवेगोऽधिकः तृष्णा प्रलापः श्वसनं भ्रमः ।
मलप्रवृत्तिः ऊत्क्लेशः पच्यमानस्य लक्षणम् ॥

It is characterized by high fever, thirst, delirium, breathlessness, giddiness, elimination of stool and kapha, and nausea.

निरामज्वर लक्षण :

क्षुत्क्षामता लघुत्वं च गात्राणां ज्वरमार्दवम् ॥
दोषप्रवृत्तिः अष्टाहो निरामज्वरलक्षणम् ।

It is characterized by appearance of appetite, lightness of the body, reduction in body temperature (mild fever), normal elimination of waste products through stool-urine-sweat, and fever of 8 days.

पुनरावर्तक ज्वर :

Relapsing fever: Jwara which is recurred within few days

Punaravartaka Jwara manifests if person consumes avoidable food and activities (मिथ्या आहार-विहार) during recovery stage. It may also relapse if Jwara comes down but Doshas are not eliminated properly.

नवज्वर-चिकित्सा सूत्र →

(For the first 7 days, it is called navajwara.)

लंघनं स्वेदनं कालो यवाग्वस्तिक्तको रसः ।

पाचनान्यविपक्वानां दोषाणां तरुणे ज्वरे ॥

General line of treatment of नवज्वर is → उपवास (fasting) – स्वेदनिर्गमन उपाय (measures to induce sweating) – काल (passing of time of 8 days) – तिक्तसरससिद्ध यवागु पान.

लंघन or उपवास (Fasting) is the first line of treatment in Jwara.

But it is contraindicated in case of वातज, क्षयज, कामज, क्रोधज, भयज, शोकज, and श्रमजन्य ज्वर.

त्याज्य in नवज्वर (Prohibitions in nava-jwara) →

नवज्वरे दिवास्वप्न स्नानाभ्यंगान्नमैथुनम् ।

क्रोधप्रवात व्यायामान् कषायांश्च विवर्जयेत् ॥

During the initial stage of Jwara, one should avoid day-sleep, bath, massage, heavy food, sexual intercourse, anger, exposure to wind, exercise and decoctions or medicines having astringent taste.

षडंग पानीय →

मुस्ता पर्पटकोशीर चन्दनोदीच्यनागरैः ॥

शृतशीतजलं दद्यात् पिपासाज्वरशान्तये ।

मुस्ता, पर्पट, उशीर, चन्दन, उदीच्य, and शुण्ठी are taken in equal quantity (i.e. 1 part each) and 64 parts of water is added. Boil it and reduce it to ½, then cool it.

This is given to patient to pacify thirst in Jwara.

वमन in ज्वर →

कफप्रधानानुत्क्लिष्टान् दोषान् अमाशयस्थितान् ।

बुद्ध्वा ज्वरकरान् काले वम्यानां वमनैः हरेत् ॥

If vitiated doshas are Kapha predominant or situated in Amashaya, Vamana is indicated in Jwara.

यवागू-प्रयोग in ज्वर →

After वमन & लंघन, the patient should take मण्ड cooked in decoction of Jwarahara dravyas, and then he should take यवागू.

यवागू सेवन should be continued until fever comes down or becomes low grade. Or it should be continued for 6 days.

Yavagu enhances the Agni of the patient, eliminates stool, urine, dosha etc. and does Anulomana. Being ushna Yavagu produces sweating, and being drava (liquid) it relieves thirst.

यवागू पान is contraindicated in case of मदात्यय, उर्ध्व रक्तपित्त, पित्त-कफ predominance, and ग्रीष्म ऋतु.

कषाय-प्रयोग in ज्वर →

पाचनं शमनीयं वा कषायं पाययेद् भिषक् ।

ज्वरितं षडहेऽतीते लघ्वन्नं प्रतिभोजितम् ॥

After 6 days of Jwara, light food like Yavagu is given to patient on 7th day, and then from 8th day कषाय should be administered which is processed with पाचनीय and शमनीय द्रव्य.

कषाय is contraindicated in नवज्वर.

कषायपान is avoided in navajwara because it has property of स्तम्भन & ग्राही that can cause विबद्धता & स्तब्धता of दोष i.e. Aggravated dosha will not be eliminated properly. Stagnant or adherent dosha (लीन दोष) can cause विषमज्वर.

घृतपान in ज्वर →

अत ऊर्ध्वं कफे मन्दे वातपित्तोत्तरे ज्वरे ।

परिपक्वेषु दोषेषु सर्पिष्पानं यथाऽमृतम् ॥

When Kapha dosha is less and Vata-Pitta are more, and when doshas are pakwa, administration of ghrita is beneficial like amrita.

If a patient having ruksha sharira, is not relieving from fever even after vamana, upavasa, laghu ahara, and kashaya-pana, he should be treated with Ghrita.

दुग्धपान in ज्वर →

दाहतृष्णापरीतस्य वातपित्तोत्तरं ज्वरम् ।

बद्धप्रच्युतदोषं वा निरामं पयसा जयेत् ॥

In which jwara, the patient is suffering from दाह & तृष्णा, and if वात & पित्त are more, and in case of बद्ध मल or निराम दोष, administration of दुग्धपान is beneficial.

विरेचन in ज्वर →

If Jwara is not relieved by above methods, विरेचन should be administered to strong patients.

निरुह बस्ति in ज्वर →

If patient is weak, Vamana & Virechana are contraindicated, then Niruha basti should be administered after dugdhapana.

पटोलादि निरुहबस्ति – आरग्वधादि निरुहबस्ति – गुडूच्यादि निरुहबस्ति

अनुवासन बस्ति in जीर्णज्वर →

In case of strong patients, if Jwara is जीर्ण (chronic), कफ पित्त are less, अग्नि is strong, and मल is रुक्ष or विबन्ध (constipation) → अनुवासन बस्ति (स्नेहबस्ति) should be administered.

जीवन्यादि अनुवासनबस्ति – पटोलाद्यनुवासनबस्ति – चन्दनाद्यनुवासनबस्ति

नस्य in जीर्णज्वर : Shirovirechana (nasya) should be administered in case of Jeerna-jwara, if patient suffers from shirashula, shirogaurava, and vibaddha-indriya.

बाह्य उपचार in जीर्णज्वर →

In दाहज्वर → शीत अभ्यंग, प्रदेह, परिषेक, and अवगाह.

e.g., अभ्यंग with चन्दनाद्य तैल – परिषेचन or अवगाहन with मधु, काज्जी, दुग्ध, दधि, घृत, and जल

In शीतज्वर → उष्ण अभ्यंग, प्रदेह, परिषेक, and अवगाह.

e.g., अभ्यंग with अगुर्वादि तैल – परिषेचन or अवगाहन with उष्ण वीर्य द्रव्य

In आगन्तुकज्वर due to अभिचार, अभिशाप etc. → धूपन and अञ्जन should be administered.

सन्निपातज्वर चिकित्सा →

वर्धनेन एकदोषस्य क्षपणेनोच्छ्रितस्य वा ।

कफस्थानानुपूर्व्या वा सन्निपातज्वरं जयेत् ॥

Increase the diminished dosha & decrease the aggravated dosha, and treat the kapha dosha sthana first (i.e. amashaya).

कर्णमूलशोथ : a complication (उपद्रव) of sannipataja jwara.

Treat it with रक्तमोक्षण – घृतपान – प्रदेह – नस्य – कवल

धातुगत ज्वर चिकित्सा →

→ रसगत → वमन – उपवास

→ रक्तगत → रक्तमोक्षण – प्रलेप – परिषेक

→ मांस & मेदोगत → विरेचन – उपवास

→ अस्थि & मज्जागत → निरुह – अनुवासन वस्ति

→ शुक्रगत → असाध्य (incurable)

विषमज्वर चिकित्सा →

Vishamajwara is usually associated with आगन्तुक कारण like जीवाणु, विष, वायु, भूत, प्रेत etc. it should be treated with युक्तिव्यपाश्रय चिकित्सा as well as दैवव्यपाश्रय चिकित्सा.

In वातप्रधान विषमज्वर → स्नेहपान (घृतपान) – निरुह – अनुवासन वस्ति – स्निग्ध & उष्ण अन्नपान

In पित्तप्रधान विषमज्वर → विरेचन – पित्तहर द्रव्य सिद्ध घृत & दुग्ध – तिक्त & शीत द्रव्य

In कफप्रधान विषमज्वर → वमन – पाचन द्रव्य – कषायपान – उपवास – रुक्ष & उष्ण द्रव्य

दैवव्यपाश्रय चिकित्सा → मणिधारण पूजन यज्ञ स्तुति बलि मंगल होम उपहार ब्रह्मचर्य तप सत्य नियम साधुदर्शन etc.

विषमज्वरघ्न पञ्चकषाय →

In सन्तत → कलिंगादि कषाय → कलिंग(कुटज) पटोल कटुकरोहिणी
 In सतत → पटोलादि कषाय → पटोल सारिवा मुस्ता पाठा कटुकरोहिणी
 In अन्येद्युष्क → निम्बादि कषाय → निम्ब पटोल त्रिफला मृद्विका मुस्ता कुटज
 In तृतीयक → किराततित्कादि कषाय → किराततित्त गुडूची चन्दन शुण्ठी
 In चतुर्थक → गुडूच्यामलकादि कषाय → गुडूची आमलकी मुस्ता

आगन्तुकज्वर चिकित्सा → दैवव्यपाश्रय चिकित्सा

पुनरावर्तक ज्वर चिकित्सा → मृदुशोधन – यापन बस्ति – लघु यूष – जांगल मांसरस –
 तित्त द्रव्य सिद्ध घृत – किराततित्कादि क्वाथ – अभ्यंगादि बाह्य उपचार

ज्वरहर पञ्चकषाय →

1. Either क्वाथ or शीतकषाय (फाण्ट) of: मुस्ता – पर्पट
2. Either क्वाथ or शीतकषाय (फाण्ट) of: शुण्ठी – पर्पट
3. Either क्वाथ or शीतकषाय (फाण्ट) of: दुरालभ – पर्पट
4. Either क्वाथ or शीतकषाय (फाण्ट) of: मुस्ता – शुण्ठी – गुडूची – किराततित्त
5. Either क्वाथ or शीतकषाय (फाण्ट) of: पाठा – उशीर – उदीच्य

शमनौषधि →

1. **रसौषधि →** मात्रा = 125 - 250 mg. अनुपान = ऊष्णोदक / मधु
 त्रिभुवनकीर्ति रस – हिंगुलेश्वर रस – कस्तूरीभैरव रस – सूतशेखर रस – गोदन्ती भस्म
 2. **वटी →** मात्रा = 250 - 500 mg. अनुपान = ऊष्णोदक
 संजीवनी वटी – सौभाग्य वटी – अमृता सत्व
 3. **चूर्ण →** मात्रा = 3 - 6 g. अनुपान = ऊष्णोदक / मधु
 सितोपलादि चूर्ण – तालीसादि चूर्ण – सुदर्शन चूर्ण – लवंगादि चूर्ण – पंचकोल चूर्ण
 4. **क्वाथ →** मात्रा = 10 - 20 ml. अनुपान = जल
 ज्वरहर पंचकषाय – पटोल पंचक क्वाथ – गुडूच्यादि क्वाथ
 5. **घृत/तैल →** मात्रा = 10 - 20 ml. अनुपान = ऊष्णोदक
 पंचतित्त घृत – अमृतादि घृत – चन्दनादि तैल & अगुर्वादि तैल (बाह्य अभ्यंगार्थ)
 6. **अवलेह →** मात्रा = 20 - 30 g. अनुपान = दुग्ध
 द्राक्षावलेह – च्यवनप्राश
- पथ्य आहार-विहार →** पुराण शालि – मुद्ग – पटोल – कारवेल्लक – शिगु – गुडूची –
 जीवन्ती – मृद्विका – दाडिम – मण्ड – यवागु – लंघन – विश्राम
- अपथ्य आहार-विहार →** गुरु विदाही विष्टम्भी आहार – दुषित जल सेवन – तिलकुट –
 दिवास्वप्न – स्नान – अभ्यंग – मैथुन – क्रोध – प्रवात – व्यायाम

TYPHOID		
Typhoid fever, also known as enteric fever is caused by bacteria Salmonella typhi.		
Cause → Salmonella typhi / Salmonella paratyphi I.P. → 10-14 days; Transmitted by faeco-oral route	Types → Typhoid Paratyphoid	
Clinical Features → Fever in step ladder pattern (gradual onset in 1st week, remittent in 2nd week and falling in 3rd week), Anorexia, Headache, Vomiting, Diarrhea, Rose spots	Investigations → WBC ↓ (Leucopenia) Widal test positive	Treatment → Antipyretics Antibiotics (Ampicillin 650 mg 6 hourly; Ciprofloxacin 500 mg 12 hourly; Azithromycin 500 mg OD; Ceftriaxone or Cefotaxime I.V.) Fluid therapy

PNEUMONIA		
Pneumonia is an infection that inflames the alveoli in one or both lungs.		
Cause → Streptococcal pneumonia bacteria or other virus I.P. → Few hours to several days; Spread by droplet infection	Types → Community or Hospital acquired or those occurring in immuno-compromised hosts	
Clinical Features → Pulmonary → Breathlessness and cough (initially dry, later expectoration of bloodstained or mucopurulent sputum), Pleuritic chest pain may be referred to the shoulder, Upper abdominal tenderness General → Fever with chills, rigors, vomiting, headache	Diagnosis → Bronchial breathing and whispering pectoriloquy, Crackles are often detected Investigations → Chest x-ray, Sputum, PCR, Blood (Neutrophil Leucocytosis)	Treatment → Oxygenation Fluid balance Antibiotic therapy (Amoxicillin 500 mg 8 th hourly; if allergic to penicillin – Clarithromycin / Erythromycin 500 mg 12 hourly; Ceftriaxone or Cefuroxime or Co-amoxiclav 1 gm daily I.V.)

PLEURISY		
Pleurisy or Pleuritis is characterized by pleuritic chest pain that worsens with breathing.		
Cause → Viral infections, Bacterial infections, Pulmonary infarction, Chest trauma, Serositis due to connective tissue diseases (e.g. Rheumatoid arthritis, SLE), Pleural malignancy, Pleural embolism, Pleural effusion etc.	Types → Dry pleurisy Pleurisy with effusion	
Clinical Features → Pleuritic sharp pain, worsening with breathing, Dyspnea, If the effusion causes inflammation of diaphragm the pain may be referred to the shoulder or the upper abdomen. Other associated symptoms are due to the underlying disease.	Diagnosis → On the affected side, movements of the thorax are reduced, the chest wall muscles are tender and a pleural friction rub may be heard. Chest x-ray, CT scan, MRI, Thoracocentesis, Blood test	Treatment → Analgesics, Antipyretics and Bronchodilators for symptomatic relief Antiinflammatory drugs to reduce inflammation Thoracocentesis to remove the fluid, air, or blood from the pleural space Treat the underlying cause like viral and bacterial infections, Pneumonia, Tuberculosis etc.

INFLUENZA		
Influenza, commonly known as 'the flu' is an acute systemic infection caused by influenza virus that primarily affects the respiratory tract.		
Cause → Influenza A virus / Influenza B virus I.P. → 1-3 days ; Spread by droplet infection	Types → Seasonal flu H1N1, H3N2, Avian flu, Ebola etc.	
Clinical Features → Fever with chills, Aches, Cough, Running nose, Sore throat, Generalized weakness	Investigations → Diagnosed by viral culture or by antigen or RNA detection in a nasopharyngeal sample	Treatment → Antiviral drugs – Neuraminidase inhibitors oral Oseltamivir 75 mg 12 hourly or inhaled Zanamivir 10 mg 12 hourly for 5 days. Seasonal or annual vaccination

MUMPS

Mumps is a systemic viral infection characterized by swelling of the parotid glands (parotitis).

Cause → RNA Myxovirus I.P. → 14-21 days; Spread by droplet infection		Types →
Clinical Features → Bilateral painful swelling of parotid glands, Fever, Coryza, Sore throat	Diagnosis → Diagnosis is usually clinical Serology – Mumps specific IgM or IgG seroconversion confirms the diagnosis	Treatment → Analgesics Mumps vaccine is one of the component of the combined MMR vaccine

MENINGITIS

Acute inflammation of meninges present with a characteristic combination of pyrexia, headache, and meningism. (Meningism can also occur in subarachnoid haemorrhage)

Cause → Bacteria (Brucella, E.coli, Proteus, Streptococci, Haemophilus influenza, Neisseria meningitidis, Streptococcus pneumonia) Virus (Enterovirus- Coxsackie, Polio etc., Mumps, Influenza, Herpes simplex, Varicella zoster, HIV etc.) Protozoa, Fungi etc. Non infective causes like malignancy (CA), SLE etc.		Types → Viral meningitis Bacterial meningitis Meningococcal meningitis Tubercular meningitis Parasitic meningitis Fungal meningitis Non-infectious meningitis etc.
Clinical Features → Meningism – Headache, Neck stiffness, Photophobia Fever, Drowsiness, Coma, Focal neurological signs	Investigations → Lumbar puncture CT Scan of head Blood culture, PCR	Treatment → With meningococcal rash: Benzylpenicillin 2 gm I.V. 6 hourly Without meningococcal rash: Cefotaxime 2 gm I.V. 6 hourly or Ceftriaxone 2 gm I.V. 12 hourly

ENCEPHALITIS

Encephalitis is an acute inflammation of the brain usually resulting from either a viral infection or due to the body's own immune system mistakenly attacking brain tissue.

Cause → Virus (Herpes simplex, Enterovirus, HIV etc.)		Types →
Clinical Features → Acute onset of headache, Fever, Seizures, Focal neurological signs	Diagnosis → Lumbar puncture CT Scan, MRI, PCR	Treatment → Anticonvulsant treatment Raised intracranial pressure is treated with Dexamethasone 8 mg 12 hourly Antiviral drug (Aciclovir 10 mg/kg I.V. 8 hourly)

TETANUS

Tetanus, also known as lockjaw, is an infection characterized by muscle spasm.

Cause → Bacteria – Clostridium tetani Spread by soil and dust & infection enters body through wounds		Types → Generalized tetanus Localized tetanus
Clinical Features → Lock-jaw, Fever, Headache, Dysphagia, Muscle spasm, Stiffness in neck & back, etc. (Symptoms appear from 2 days to several weeks after injury, the shorter I.P. the more severe attack.)	Diagnosis → Clinical diagnosis	Treatment → Anti-tetanus serum 3000 U I.V. Wound care Antibiotic (Benzylpenicillin 600 mg I.V. 6 hourly or Metronidazole) Inj. TT 0.5 ml on regular interval (repeated in 1 month and 6 month later)

YELLOW FEVER		
An acute viral haemorrhagic disease, caused by a flavivirus, spread by infected mosquitoes.		
Cause → Flavivirus (Transmitted by mosquitoes- Aedes aegypti) I.P → 3 to 7 days	Types →	
Clinical Features → Mild to moderate fever, chills, back pain, abdominal pain, muscle pain (myalgia), Jaundice (Icterus), Hepatic failure, Renal failure	Diagnosis → Clinical features in endemic area Viral isolation from blood in first 24 hours Leucopenia	Treatment → Isolation Symptomatic treatment Fluid therapy Blood transfusion Preventive vaccination

PLAGUE	
An infectious disease caused by Yersinia pestis, a gram -ve bacterium that is spread between rodents (rats) by their fleas.	
Cause → Bacteria – Yersinia pestis Vector – Rat fleas	Types → Bubonic plague Septicemic plague Pneumonic plague
Clinical Features → Fever, Weakness, Headache Bubonic plague → High fever, Rigors, Dry skin, Severe headache, Aching and swelling at the site of the affected lymph nodes. Septicemic plague → Vomiting, Diarrhea, Abdominal pain, Hypotension, Shock, Renal failure, ARDS, Death. Pneumonic plague → Dyspnea, Cough, Expectoration of copious blood stained sputum, Chest pain, Cyanosis, ARDS, Death.	Treatment → Antibiotic Streptomycin 1 g 12 hourly or Gentamycin 1 mg/kg 8 hourly Fluoroquinolone (Ciprofloxacin and Levofloxacin) Symptomatic treatment

DENGUE FEVER		
Dengue fever is mosquito-borne tropical disease caused by the dengue virus. It is a fast emerging endemic viral disease in India.		
Cause → Flavivirus (four serotypes- DEN-1, DEN-2, DEN-3, DEN-4) I.P. → 2-7 days Vector – Aedes aegypti (day-biting)	Types → Dengue haemorrhagic fever Dengue shock syndrome	
Clinical Features → Fever (Break bone fever), Bodyache, Arthralgia, Rash, Severe headache, Nausea, Vomiting etc.	Investigations → Leucopenia Thrombocytopenia Isolation of dengue virus from blood PCR	Treatment → Symptomatic treatment Antipyretics, Antibiotics, Fluid therapy (Aspirin should be avoided due to bleeding risk)

CHIKUNGUNYA		
An infection caused by chikungunya virus, characterized by fever, rash and joints-pain.		
Cause → Chikungunya virus I.P. → 3-12 days Vector – mosquito- Aedes aegypti	Types →	
Clinical Features → A period of fever may be followed by an afebrile phase and then recrudescence of fever. Arthralgia (Painful & swollen joints), Chills, Severe headache	Investigations → Diagnosis by serology but cross reactivity between alpha viruses occurs.	Treatment → Symptomatic treatment

LEPTOSPIROSIS		
A bacterial disease spread through the urine of infected animals.		
Cause → Leptospira interrogans (I.P → 1-2 weeks) Spread by Contaminated water, soil etc.	Types → 4 syndromes → Bacteraemic / Aseptic meningitis / Icteric / Pulmonary leptospirosis	
Clinical Features → Fever of sudden onset, Chills, Headache, Myalgia, Hemorrhagic manifestation	Investigations → CBC (Thrombocytopenia), Urine analysis, PCR, ELISA	Treatment → Symptomatic treatment Blood transfusion Liver and kidney care Antibiotic- Doxycycline 100 mg 12 hourly 1 week or Ceftriaxone 1 g I.V.

VIRAL FEVER		
Fever caused by viral infection is called as viral fever.		
Cause → Virus MoT → Inhalation, Direct contact, Vector, Contaminated water and food.		Types →
Clinical Features → Fever varying from low grade to high grade, Nausea, Vomiting, Running nose, Sore throat, Bodyache, Myalgia, Arthralgia, Headache, Photophobia, Rash	Diagnosis → Viral isolation and culture in blood, Rule out malaria and typhoid by M.P. and Widal test	Treatment → Symptomatic treatment (Atipyretics/Analgesics) Advice to take rest, cover the face with mask or handkerchief, avoid crowded area. Maintain hygiene and take plenty of fluids

ANTHRAX		
A life-threatening infectious disease caused by Bacillus anthracis that is transmitted to human by contact with infected animals or their products.		
Cause → Bacteria – Bacillus anthracis MoT → Infected animals (cattle, sheep, goat etc.) I.P. → 1-7 days		Types → 3 routes of infection → Cutaneous / Inhalation / Gastrointestinal anthrax
Clinical Features → Cutaneous anthrax → Itchy sore similar to insect bite, may blister and forms ulcer, A scab often forms and dries & falls off within 2 weeks, Complete healing can take longer. Inhalation anthrax → Fever, malaise, headache, cough, Dyspnea, Chest pain Gastrointestinal anthrax → Abdominal pain, Diarrhea, Mouth sores, Nausea, Vomiting	Treatment → Penicillin G 2 million units I.M. 12 hourly or Streptomycin 500 mg I.M. 8 hourly Oral Penicillin for 7-10 days Symptomatic treatment	

SMALL POX		
A contagious disease, caused by the variola virus, which had high mortality, was eradicated worldwide by a global vaccination programme.		
Clinical Features → Prodromal symptoms → Fever, headache, prostration (lying stretched out on the ground) Signs & Symptoms → Skin rash which develops through macules and papules to vesicles and pustules, worst on face and distal extremities.		Investigations → Electron micrography and DNA detection test (PCR)

CHIKEN POX		
A highly contagious viral infection caused by varicella zoster virus, characterized by itchy, blister-like rash on the skin		
Cause → Virus – Varicella zoster I.P. → 14-16 days MoT → Droplet / Direct contact		Types →
Clinical Features → Prodromal stage → Fever, Backache, Malaise, Shivering Eruptive stage → Rash- (dew drop like), Rapid progression & spread of rashes	Diagnosis → Clinical diagnosis, Detection of antigen, DNA (PCR) or by Viral culture of aspirated vesicular fluid	Treatment → No specific treatment Isolation Disinfection Antiviral drug (Acyclovir) Symptomatic treatment

MEASLES		
A viral infection of respiratory system that is serious for small children but is easily preventable by vaccination.		
Cause → RNA Paramyxovirus I.P. → 10-14 day MoT → Droplet		Types → 3 stages → Prodromal / Eruptive / Post measles stage
Clinical Features → Prodromal stage → Fever, Sneezing, Red eyes, Photophobia, Koplik's spot Eruptive stage → Rash- dusky red/ maculo-papular Post measles stage → Loss of weight, Weakness, Diarrhea	Diagnosis → Clinical diagnosis Detection of antibody- Serum IgM, Sero Conversion or Salivary IgM	Treatment → Isolation Symptomatic treatment Vitamin A 100000 to 200000 IU orally Antibiotics for bacterial - complications MMR vaccination

SWINE FLU		
Swine flu (H1N1 flu) is a human respiratory viral infection caused by influenza strain that started in pigs.		
Clinical Features → Flu like symptoms (Fever, chills, sore throat, muscle pains, dyspnea, weakness)	Diagnosis → PCR	Treatment → Hospitalization & Isolation Maintain Fluid & Electrolyte Antiviral (Oseltamivir / Zanamivir)

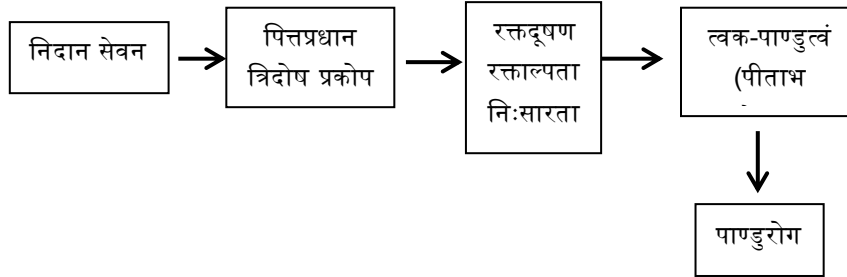
पाण्डु (Anaemia)

“पाण्डुत्वेनोपलक्षितो रोगः पाण्डुरोगः” the disease in which skin of the patient becomes pallor, is called pandu roga.

निदान →

आहारज निदान	अम्ल, लवण, क्षार, अत्युष्ण, विरुद्ध, असात्म्य भोजन निष्वाप, माष, पिण्याक, तिल, विदग्ध-अन्न अति सेवन मृदा सेवन, प्रमिताशन (एकरस सेवन)
विहारज	अतिव्यायाम, अतिमैथुन, दिवास्वप्न, वेग-धारण, पंचकर्म-विषमयोग
मानसिक	काम-चिन्ता-भय-क्रोध-शोक
निदानार्थकर रोग	रक्तार्श, रक्तपित्त, रक्तप्रदर, रक्तार्बुद, व्रण, विषमज्वर, जीर्णज्वर, ग्रहणी, उदररोग, गुल्म, अश्मरी, राजयक्ष्मा, प्रमेह, कामला etc.

सम्प्राप्ति →



पाण्डु-भेद	5	वातज – पित्तज – कफज – सन्निपातज – मृदभक्षणजन्य पाण्डु
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पूर्वरूप →

हृदयस्पन्दन – रौक्ष्य – स्वेदाभाव – श्रम

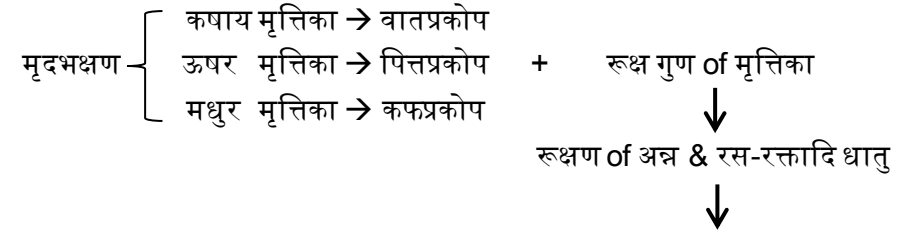
सामान्य लक्षण →

पाण्डुत्वम् (Pallority) – निःसारता (Loss of glow) – कर्णध्वेद (Tinnitus) – मन्दाग्नि – दुर्बलता – अन्नद्वेष – श्रम (Fatigue) – भ्रम (Vertigo) – गात्रशूल – ज्वर – श्वास – अक्षिकूट शोथ (Swelling around eyes) – पिण्डिकोद्वेहन (Cramps in calf region) etc.

विशेष लक्षण →

क्र.	पाण्डु-भेद	लक्षण
1	वातज पाण्डु	कृष्णपाण्डुत्व – रूक्ष अरुण अंग – अंगमर्द – रुजा – बलक्षय
2	पित्तज पाण्डु	पीत हरिताभ – ज्वर – दाह – तृष्णा – मूर्च्छा – पीतमूत्रशकृत
3	कफज पाण्डु	श्वेताभ – गौरव – अरुचि – श्वास – कास – शुक्लमूत्राक्षिवर्चः
4	सन्निपातज पाण्डु	त्रिदोषज मिश्रित लक्षण
5	मृदभक्षणजन्य पाण्डु	शून्यगण्डाक्षिकूटभूः शून्यपान्नाभिमेहनः । क्रिमिकोष्ठोऽतिसार्येत मलं सासृक् कफान्वितम् ॥

Samprapti of मृदभक्षणजन्य पाण्डुरोग →



पाण्डु ← ओज-बल-वर्ण क्षय ← चोषण-अभाव ← स्रोतोऽवरोध by अपक्व मृत्तिका

साध्यासाध्यता →

असाध्य लक्षण → Disease of long duration (चिरकालीन), generalized oedema (शोथाधिक्य), yellow vision (पीतानि पश्यति), greenish mucoid diarrhoea (सकफहरित अतिसार), छर्दि, मूर्च्छा, तृष्णा, अतिश्वेत, दीन etc.

उपद्रव →

अरुचि – पिपासाधिक्य (तृष्णा) – मूर्च्छा – क्लम – छर्दि – अतिसार – ज्वर – शिरःशूल – अग्निमान्द्य – शोष – श्वास – कास – स्वरभेद etc.

पाण्डु चिकित्सा सूत्र →

तत्र पाण्ड्वामयी स्निग्धस्तीक्ष्णैरूर्ध्वानुलोमिकैः ।

संशोध्यो मृदुभिस्तिकैः, कामली तु विरेचनैः॥ (च.चि.16/40)

पंचगव्यं महातित्तं कल्याणकमथापि वा ।

स्नेहनार्थं घृतं दद्यात् कामलापाण्डुरोगिणे ॥ (च.चि.16/43)

- निदान परिवर्जन
- संशोधन चिकित्सा
 1. आभ्यन्तर स्नेहन with पंचगव्य घृत / महातित्त घृत / कल्याणक घृत
 2. तीक्ष्ण वमन & विरेचन (गोमूत्रहरीतकी प्रयोग)
 3. रक्त बस्ति (अजा रक्त)
- संशमन चिकित्सा
 - पथ्याहार (पुराण शालि यव गोधूम – यूष of मुद्ग आढकी मसूर – जांगल मांसरस) & दोषानुसार उपचार
 - वातज → स्निग्ध-उष्ण औषधि
 - पित्तज → तिक्त रस & शीतवीर्य औषधि
 - कफज → कटु-तिक्त रस & उष्णवीर्य औषधि

मृदभक्षणजन्य पाण्डु चिकित्सा →

निपातयेत् शरीरात्तु मृत्तिकां भक्षितां भिषक् ।

युक्तिज्ञः शोधनैस्तीक्ष्णैः प्रसमीक्ष्य बलाबलम् ॥

शुद्धकायस्य सर्पीषि बलाधानानि योजयेत् ॥ (च.चि.16/117-118)

- निदान परिवर्जन
- तीक्ष्ण संशोधन to remove the eaten mud from the kotha
- औषधसिद्ध घृत प्रयोग for बलाधान

शमनौषधि →

1. रसौषधि → मात्रा = 125 - 250 mg. अनुपान = ऊष्णोदक / मधु / घृत
पुनर्नवा मण्डूर – नवायस लौह – लौह भस्म – स्वर्णमाक्षिक भस्म

2. वटी → मात्रा = 250 - 500 mg. अनुपान = ऊष्णोदक / मधु / घृत
मण्डूर वटक – आरोग्यवर्धिनी वटी – अभया वटी

3. चूर्ण → मात्रा = 3 - 6 g. अनुपान = ऊष्णोदक / मधु
त्रिफला चूर्ण – आमलकी चूर्ण

4. क्वाथ/आसव/अरिष्ट → मात्रा = 20 - 40 ml. अनुपान = जल
पुनर्नवाष्टक क्वाथ – फलत्रिकादि क्वाथ – पुनर्नवासव – लोहासव – विडंगारिष्ट

5. घृत/तैल → मात्रा = 10 - 20 ml. अनुपान = ऊष्णोदक
पंचगव्य घृत – महातित्त घृत – कल्याणक घृत – त्रिफला घृत

6. अवलेह → मात्रा = 20 - 30 g. अनुपान = दुग्ध / जल
द्राक्षावलेह – धात्र्यवलेह – विडंगावलेह

7. रसायन → मात्रा = रोगी & रोगबलानुसार
योगराज रसायन – त्रिफला रसायन – आमलकी रसायन – लौह रसायन

पथ्य आहार-विहार → यव – शालि – मुद्ग – मसूर – कूष्माण्ड – पटोल – जांगल मांस – गोमूत्र – हरिद्रा – मधु – घृत – तक्र – हरीतकी – आमलकी – शुण्ठी – मेथिका

अपथ्य आहार-विहार → तिल – माष – निष्पाव – सर्षप – सुरा – लवण – मत्स्य – अति
अम्बुपान – आतपसेवन – दिवास्वप्न – व्यायाम – चिन्ता – क्रोध

Anaemia

A reduction in the number of circulating red blood cells (RBCs) or reduction of Haemoglobin is termed as Anemia.

Normal levels	Male	Female
RBCs	4.5 -5.5 mill./cu.mm	4.1 -5.1 mill./cu.mm
Haemoglobin	14 -18 gm/dl	12 -16 gm/dl

Etiological Classification →

1. Anemia due to blood loss
 - a. Acute hemorrhage (e.g. trauma, road traffic accidents, surgery, menorrhagia, ectopic pregnancy etc.)
 - b. Chronic hemorrhage (e.g. GI ulcerations, parasitic infestations, hemorrhoids etc.)
2. Anemia due to premature destruction of RBCs (hemolysis)
 - a. Genetic cause, e.g. Sickel cell anemia, Thalassemia etc.
 - b. Acquired cause, e.g. Hypersplenism, Malaria, Malignancy, Blood transfusion reaction, SLE, Lead poisoning etc.
3. Anemia due to decreased production of red cells
 - a. Abnormal bone marrow, e.g. Aplastic anemia etc.
 - b. Nutritional deficiency
 - Iron deficiency
 - Folic acid deficiency
 - Vitamin B₁₂ deficiency

Morphological Classification →

1. Macrocytic Anemia
 - Megaloblastic anemia due to Vitamin B₁₂ and Folic acid deficiency.
2. Microcytic Hypochromic Anemia
 - Iron deficiency anemia
 - Abnormal globin synthesis – Thalassemia with or without Hemoglobinopathies
3. Normocytic Normochromic Anemia
 - Blood loss (hemorrhage)
 - Increased plasma volume (pregnancy, overhydration etc.)
 - Hypothyroidism, Adrenal insufficiency etc.
 - Renal disease / Liver disease / Cirrhosis etc.

Clinical Features →

- Pallor skin, nailbed and mucous membrane (conjunctiva etc.)
- Dyspnea on exertion – Fatigue – Weakness
- Palpitation – Headache – Edema on feet and ankle

Investigations →

CBC (RBC count, Hb, MCV, MCH, MCHC), Reticulocyte count, MP

Treatment →

- According to cause (e.g. Iron supplement in Iron deficiency anemia, Vit.B₁₂ & folic acid supplement in Megaloblastic anemia, Erythropoietic drugs, blood transfusion etc.)
- Symptomatic treatment.

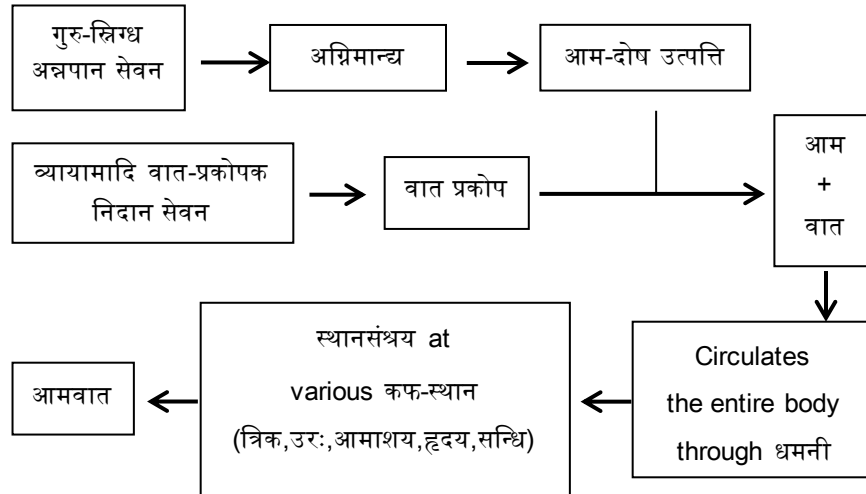
आमवात (Rheumatoid arthritis)

“आमेन सहितः वातः आमवातः” / “आमश्च वातश्च आमवातः” the disease in which both ama and vata are aggravated and affect various kapha-s-thanas like joints, heart etc.

निदान →

आहारज निदान	विरुद्ध आहार, स्निग्ध शीत आहार, अभिष्यन्दी विदाही गुरु अन्न सेवन विषमाशन, अजीर्णाशन, मंदाग्नि दधि, माष, दुग्ध, पायस, लकुच, तैल, मत्स्य, आनूप मांस सेवन दूषित, विषयुक्त & पर्युषित जल पान
विहारज	अतिव्यायाम, अव्यायाम (निश्चेष्ट), दिवास्वप्न, रात्रिजागरण, वेग-धारण
मानसिक	ईर्ष्या-द्वेष-काम-क्रोध-लज्जा-शोक-भय-चिन्ता-आलस्य-अवसाद
निदानार्थकर रोग	ग्रहणी दोष, रसावृत वात, अग्निमान्द्य, अजीर्ण, उरुस्तम्भ etc.

सम्प्राप्ति →



पूर्वरूप →

अग्निमान्द्य – आलस्य – लालास्राव – हृदयगौरव – निद्रानाश – अरति – लोमहर्ष

सामान्य लक्षण →

अंगमर्दोऽरुचिस्तृष्णाह्यालस्यं गौरवं ज्वरः ।

अपाकः शूनताऽङ्गानां आमवातस्य लक्षणम् ॥ (मा.नि.25/6)

Acute pain all over the body and joints (अंगमर्द), loss of taste (अरुचि), thirst (तृष्णा), lack of enthusiasm or laziness (आलस्य), heaviness in the body (गौरव), fever (ज्वर), indigestion (अपाक), oedema on the body (अंग-शूनता).

विशेष लक्षण →

क्र.	दोषानुसार भेद	लक्षण
1	वातोल्वण आमवात	तीव्र संधिशूल – निद्रानाश – विबंध – उत्साहहीन
2	पित्तोल्वण आमवात	अतिदाह – रक्तिमा – पिपासाधिक्य – मूर्च्छा – मृदुज्वर
3	कफोल्वण आमवात	स्तिमितता – गौरव – कण्डू – अग्निमान्द्य – बहुमूत्रता
4	सन्निपातज	त्रिदोषज मिश्रित लक्षण

क्र.	अवस्था भेद	लक्षण
1	आशुकारी / तीव्रावस्था (Acute phase)	हस्त पाद गुल्फ त्रिक जानु उरुप्रदेश large-joints pain with swelling, वृश्चिकदंशवत वेदना, दाह, अग्निमान्द्य, दौर्बल्य, भ्रम, मूर्च्छा, हृदग्रह, प्रातःकालीन जाड्यता
2	चिरकारी / जीर्णावस्था (Chronic phase)	अंग वैकल्य (Flexion deformity), खंज-अस्थि विकृति (e.g. Ulnar deviation, Swan neck deformity etc.), संधिजाड्यता (Ankylosis) etc.

साध्यासाध्यता →

साध्य → नवीन (newly manifested), एकदोषज, अल्प सन्धिगत

याप्य → द्विदोषज, अल्प सन्धिगत

असाध्य → सन्निपातिक & सर्वशरीर सन्धिगत, सर्वांग-शोथ, उपद्रवयुक्त

उपद्रव →

स्थायी वैकल्यता (Permanent deformity) – हृदय विकृति (Cardiac disorders - mitral stenosis) – स्तब्धता – गतिनाश – निद्रानाश etc.

आमवात चिकित्सा सूत्र →

लंघनं स्वेदनं तिक्तदीपनानि कटूनि च।

विरेचनं स्नेहपानं बस्तयश्चाममारुते ॥

रूक्षः स्वेदो विधातव्यो बालुकापोटलैस्तथा।

उपनाहाश्च कर्तव्यास्तेऽपि स्नेहविवर्जिता ॥ (योगरत्नाकर)

- निदान परिवर्जन
- संशोधन चिकित्सा

4. For आमपाचन → उपवास / लंघन-पाचन

(लंघन-पाचन with पंचकोल चूर्ण / अजमोदादि चूर्ण / त्रिकटु चूर्ण)

5. स्नेहन → Contraindicated for नव आमवात

For जीर्ण आमवात → आभ्यन्तर स्नेहन with एरण्ड तैल + गोमूत्र
बाह्य अभ्यंग with सैध्वादि तैल / बृहद्सैध्वादि तैल / प्रसारिणी तैल

6. रूक्ष स्वेदन with बालुका स्वेद / लवण पोट्टली स्वेद

For जीर्ण आमवात → स्निग्ध स्वेद with पत्र पिण्ड स्वेद / षष्टिक शालि पिण्ड स्वेद

7. विरेचन with एरण्ड तैल / हरीतकी चूर्ण / त्रिवृतादि चूर्ण

8. अनुवासन बस्ति with बृहद्सैध्वादि तैल / नारायण तैल

9. निरूह बस्ति with एरण्डमूलादि निरूह बस्ति / दशमूलादि निरूह बस्ति / क्षार बस्ति / वैतरण बस्ति

संशमन चिकित्सा →

1. रसौषधि → मात्रा = 125 - 250 mg. अनुपान = मधु

आमवातारि रस – मल्लसिन्दूर – समीरपन्न रस – योगेन्द्र रस – गोदन्ती भस्म

2. वटी → मात्रा = 250 - 500 mg. अनुपान = ऊष्णोदक / मधु / घृत
संजीवनी वटी – अग्रितुण्डी वटी – चित्रकादि वटी

3. गुग्गुलु → मात्रा = 500 mg. - 1 g. अनुपान = सुखोष्ण जल
सिंहनाद गुग्गुलु – योगराज गुग्गुलु – महायोगराज गुग्गुलु – गोक्षुरादि गुग्गुलु

4. चूर्ण → मात्रा = 3 - 6 g. अनुपान = ऊष्णोदक / मधु
पथ्यादि चूर्ण – पंचकोल चूर्ण – अजमोदादि चूर्ण – त्रिकटु चूर्ण

5. क्वाथ/आसव/अरिष्ट → मात्रा = 20 - 40 ml. अनुपान = जल
रास्ना पंचक क्वाथ – रास्ना सप्तक क्वाथ – दशमूल क्वाथ

6. लेप → मात्रा = यथावश्यक
दशांग लेप – हरिद्रा लेप – निर्गुन्डी पत्र लेप

7. अवलेह/पाक → मात्रा = 10 - 20 g. अनुपान = दुग्ध / उष्णोदक
एरण्डपाक – गोक्षुरादि अवलेह

8. रसायन → मात्रा = रोगी & रोगबलानुसार
अमृत भल्लतक – अश्वगन्धा रसायन – गुग्गुलु

पथ्य आहार-विहार → शुण्ठी – आर्द्रक – अजमोदा – मिश्रेया – मरिच – सैध्व – हिंगु –
लशुन – जीरक – शिगु – पटोल – कारवेल्लक – यव – कोद्रव – तक्र – उष्णोदक – एरण्ड
तैल – रूक्षण – स्वेदन – लंघन – चंक्रमण – मृदु व्यायाम

अपथ्य आहार-विहार → दधि – मत्स्य – गुड – दुग्ध – माष – आनूप मांस – पिष्टान्न –
विरुद्धाहार – पुर्वी वायु – आलस्य – रात्रिजागरण – वेगधारण – चिन्ता – शोक

एरण्ड तैल प्रयोग →

आमवातगजेन्द्रस्य शरीर वनचारिणः । निहन्त्यसावेक एव एरण्डस्नेदकेशरी ॥ (भै.र.)

Like a lion of the forest Eranda taila is capable of making the human body free from elephant like Amavata.

Rheumatoid Arthritis

RA is a chronic inflammatory disease of joints. It is an autoimmune disorder characterized by bilateral symmetrical joint disease, erosions and synovitis.

Most commonly, the wrist and hands are involved, with the same joints typically involved on both sides of the body. The disease may also affect other parts of the body.

Etiology →

RA is a systemic autoimmune disease. (Autoimmune disease – in which the body's immune system attacks healthy cells)

Some genetic and environmental factors affect the risk for RA →

Genetic → A family history of RA increases the risk.

Environmental → Smoking, Silica exposure etc.

Clinical Features →

- Pain and swelling in the joints (symmetrical – polyarthritis)
(Most commonly involved are the small joints of the hands, feet and cervical spine, but larger joints like the shoulder and knee can also be involved)
- Morning stiffness
- Anemia – Weight loss – Low grade fever
- Lymph node enlargement
- In chronic RA → deformities like Swan neck deformity, Ulnar deviation, Z-thumb etc.

Investigations →

Anemia –Hb% ↓

Raised ESR

Positive test for Rheumatoid factor (RA factor)

Positive CRP (C-reactive protein)

X-ray, CT scan, MRI, Ultrasound.

Treatment →

1) General Measures →

- a) Rest – Physical & mental
- b) Diet – Nutritious, fibrous diet
- c) Correction of anemia
- d) Moderate exercise
- e) Avoidance of cold, dampness, rainy day etc.

2) Medicines →

- a) Analgesics
Acetaminophen, Tramadol etc.
- b) Nonsteroidal anti-inflammatory drugs (NSAIDs)
Ibuprofen, Diclofenac sodium etc.
- c) Disease-modifying antirheumatic drugs (DMARDs)
Hydroxychloroquine, Methotrexate, Sulfasalazine etc.
- d) Corticosteroids

3) Surgical Treatment

- a) Prosthesis (Joint replacement)
- b) Synovectomy

मदात्यय (Alcoholism)

बुद्धिं लुम्पति यद् द्रव्यं मदकारि तदुच्यते ।

तमोगुणप्रधानं च यथामद्यसुरासदिकम् ॥ (शा.स.पू.4/22)

Dravyas with tamo-guna predominance, which destroys the intellect and causes mada and madatyaya are called madya or madakari.

ये विषस्य गुणाः प्रोक्तास्तेऽपि मद्ये प्रतिष्ठिताः ।

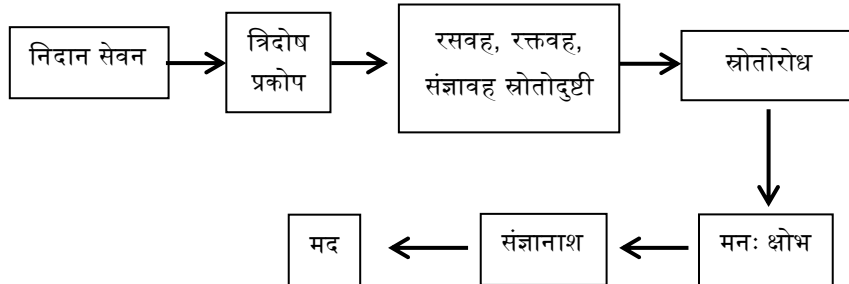
तेन मिथ्योपयुक्तेन भवत्युग्रो मदात्ययः ॥ (मा.नि.18/1)

Madya and visha are having opposite qualities of ojas. Improper or excessive consumption of madya, causes severe disease called madatyaya.

निदान →

अविधिपूर्वक मद्यपान – मात्राधिक्य मद्यपान – निर्जल मद्यपान – निराल कोष्ठ मद्यपान – अहितकर प्रकृति विपरीत मद्यपान – अल्प सत्व – पित्त प्रकृति etc.

सम्प्राप्ति →



सामान्य लक्षण →

स्मृति विभ्रंश – संमोह – हृदय व्यथा – अरुचि – तृष्णा – शिरःशूल – पार्श्वशूल – श्वास – कास – हिक्का – प्रजागर (निद्रानाश) – शरीर कम्प – छर्दि – भ्रम – प्रलाप

विशेष लक्षण →

क्र.	अवस्था भेद	लक्षण
1	प्रथम अवस्था (प्रथम मद)	Stage of excitement → Person feels happiness, interested in reading & singing, and more sleep. It does not affect intellect and memory.
2	द्वितीय अवस्था (द्वितीय मद)	Stage of mental confusion → Person is confused, irrelevant & irregular speech, irregular physical activities, excessive talk, and behaves like insane.
3	तृतीय अवस्था (मदान्तर)	Stage of delirium → Person becomes insane, does not respect elders, eats non-edible things, speaks secrets, talks funny etc.
4	चतुर्थ अवस्था (तृतीय मद)	Stage of coma (unconsciousness) → Person becomes motionless, unconscious, and does not perceive any stimulus.

क्र.	दोषानुसार भेद	लक्षण
1	वातज	हिक्का – श्वास – शिरःकम्प – पार्श्वशूल – अनिद्रा – प्रलाप
2	पित्तज	तृष्णा – दाह – ज्वर – अतिस्वेद – भ्रम – मूर्च्छा – अतिसार
3	कफज	छर्दि – अरुचि – हल्लास – तन्द्रा – स्तैमित्य – गौरव
4	सन्निपातज	त्रिदोषज मिश्रित लक्षण

अविधि मद्यपान जन्य विकार →

- पानात्यय (मदात्यय) – Acute Alcoholism
- परमद – Immediate after effects of Alcoholism (कफज लक्षण)
- पानाजीर्ण – Indigestion of Alcohol (पित्तज लक्षण)
- पानविभ्रम – Chronic Alcoholism

मद्यत्याग पश्चात् सहसा-मद्यपान जनित रोग →

- ध्वंसक – Person suffers from excessive expectoration but dryness in throat, excessive sleep and stupor.
- विक्षय – Severe bodyache, headache, and compression in chest, thirst, vomiting and fever.

साध्यासाध्यता →**असाध्य लक्षण →**

- If patient's lower lip is depressed.
- Feeling cold outside and hot inside the body.
- Face is shining like oil.
- Tongue, lips and teeth becomes कृष्ण or नील वर्ण (cyanosed).

उपद्रव →

हिक्का – छर्दि – शरीर कम्प – ज्वर – पार्श्वशूल – कास – भ्रम

मदात्यय चिकित्सा सूत्र →

सर्व मदात्ययं विद्यात् त्रिदोषामधिकं तु यम् ।

दोषं मदात्यये पश्येत् तस्यादौ प्रतिकारयेत् ॥ (च.चि.24/107)

All types of मदात्यय are त्रिदोषज, hence the predominant dosha (प्रधान दोष) is treated first.

कफस्थानानुपूर्व्या च क्रिया कार्या मदात्यये ।

पित्तमारुत पर्यन्तः प्रायेण हि मदात्ययः ॥ (च.चि.24/107)

कफस्थान (आमाशय) or कफ दोष is the first dosha that has to be treated, later पित्त and then वात.

Disorders caused by improper use of मद्य (wine) has to be treated by administration of मद्य (wine) in proper way.

→ वातज मदात्यय → लवण & अम्लद्रव्ययुक्त स्निग्धोष्ण वेशवार – स्निग्ध & आम्ल मांसरस – दाडिम रस – स्त्रीस्पर्श – वातनाशक बस्ति

→ पित्तज मदात्यय → वमन – द्राक्षा रस – सदाडिमरस शीत सक्तु – बहूदक मद्य

→ कफज मदात्यय → वमन – उपवास – उष्ण अन्न पान – स्त्रीस्पर्श

शमनौषधि →

बृहत्वातचिन्तामणि रस – उन्माद गजांकुश रस – त्रिकटु चूर्ण – अष्टांग लवण – सर्पगन्धा

चूर्ण – खर्जूरदि मंथ – अष्टांगावलेह – कूष्माण्डावलेह – ब्राह्मी तैल (बाह्य अभ्यंगार्थ)

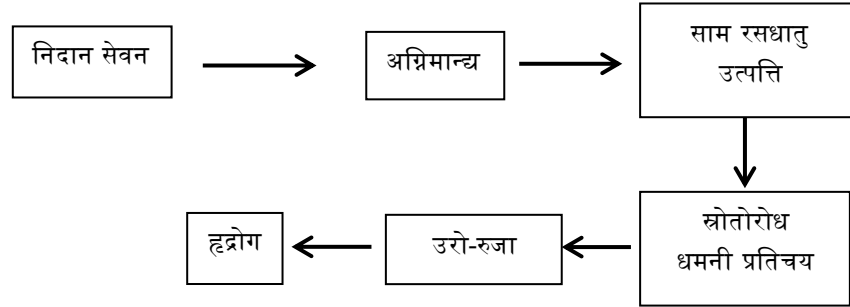
हृद्रोग (Heart diseases)

“हृदोरोगो हृद्रोगः” the diseases of hridaya (heart) are called hridroga.

निदान →

अतिव्यायाम – अति ऊष्ण तीक्ष्ण पदार्थ सेवन – अति विरेचन & अति बस्ति – चिन्ता – भय – त्रास – वेगधारण – कर्षण आहार विहार – कषाय तिक्त असात्म्य आहार

सामान्य सम्प्राप्ति →



भेद → (5) वातज – पित्तज – कफज – सन्निपातज – कृमिज हृद्रोग

सामान्य लक्षण →

वैवर्ण्य (Pallor/Cyanosis) – मूर्च्छा (Fainting) – ज्वर (Fever) – कास (Cough) – हिक्का (Hiccough) – श्वास (Dyspnoea) – आस्यवैरस्य (Altered taste of mouth) – तृष्णा (Polydipsia) – प्रमोह (Stupor) – छर्दि (Vomiting) – कफोत्क्लेश (Nausea) – रुजा (Chest pain) – अरुचि (Anorexia) etc.

विशेष लक्षण →

क्र.	हृद्रोग भेद	लक्षण
1	वातज	तोद – निर्मथन – दारण – स्फोटन – वेपथु – स्तम्भ – प्रमोह
2	पित्तज	तमःप्रवेश – नयनदाह – मोह – संत्रास – तृष्णा – ज्वर
3	कफज	हृदय स्तब्धता – गौरव – स्तिमितता – कफप्रसेक – अरुचि
4	सन्निपातज	त्रिदोषज मिश्रित लक्षण
5	कृमिज	तीव्र वेदना – तोद – कण्डू – उत्क्लेश – छीवन – अरुचि

साध्यासाध्यता →

साध्य → वातज, पित्तज & कफज हृद्रोग

याप्य → नव सन्निपातज & नव कृमिज

असाध्य → दुर्बल रोगी, उपद्रवयुक्त, अरिष्ट लक्षणयुक्त (e.g. स्तिमितता, आयताक्ष-dilated pupil, तीव्र उरोग्रह- tearing retrosternal pain etc.)

उपद्रव →

क्लम – अवसाद (depression) – भ्रम – मूर्च्छा – शोष etc.

हृद्रोग चिकित्सा सिद्धान्त →

- निदान परिवर्जन
- विश्राम (शारीरिक & मानसिक)
- दोषानुसार चिकित्सा

1. वातज हृद्रोग चिकित्सा →

- वमन with मदनफल चूर्ण + वचा + सैधव + तिल तैल
- Drink lukewarm तैल + सौवीरक + मस्तु + तक्र + सैधव
- अभ्यंग with पुनर्नवादि तैल
- पुष्करमूलादि चूर्ण – पुष्करमूलादि क्वाथ – पार्थाद्यरिष्ट – वासावलेह

2. पित्तज हृद्‌रोग चिकित्सा →

- प्रलेप with शीत वीर्य द्रव्य
- परिसेचन with शीतल औषधि
- विरेचन with द्राक्षा + सिता + मधु + परुषक
- पित्तशामक आहार – द्राक्षाद्य घृत – अर्जुन क्षीरपाक – शतपुष्प अर्क

3. कफज हृद्‌रोग चिकित्सा →

- स्वेदन
- वमन
- लंघन
- कफशामक आहार – कृष्णादि चूर्ण – पिप्पल्यादि चूर्ण – शिलाजतु रसायन

4. सन्निपातज हृद्‌रोग चिकित्सा →

- लंघन
- त्रिदोषशामक आहार
- हीन-मध्य-अतिवृद्ध दोषानुसार चिकित्सा

5. कृमिज हृद्‌रोग चिकित्सा →

- शोधन
- लंघन – पाचन
- कृमिहर चिकित्सा (निदान परिवर्जन – अपकर्षण – प्रकृति विघात)
- विडंगादि चूर्ण – पुष्करादि चूर्ण – पिप्पल्यादि चूर्ण – विडंगावलेह – अर्जुन घृत

Cardiovascular Diseases

Heart diseases that include diseased vessels, structural problems and blood clots. Most common types →

Coronary artery disease →

Coronary artery disease (CAD), also known as **ischemic heart disease** (IHD), refers to a group of diseases which includes stable angina, unstable angina, myocardial infarction, and sudden cardiac death.

Congenital heart disease →

CHD is an abnormality in the heart that develops before birth. Congenital Heart defects include Aortic Valve Stenosis (AVS), Atrial Septal Defect (ASD), Complete Atrioventricular Canal Defect (CAVC), Patent Ductus Arteriosus (PDA), Pulmonary Valve Stenosis, Tetralogy of Fallot, Ventricular Septal Defect (VSD) etc.

Arrhythmia →

Arrhythmia refers to irregular heartbeats → a group of conditions in which the heartbeat is irregular, too fast, or too slow.

More than 100 beats per minute in adults → Tachycardia.

Less than 60 beats per minute → Bradycardia.

Stroke →

Stroke is also called Cerebrovascular accident (CVA) in which poor blood flow to the brain results in cell death. There are two main types of stroke → ischemic and hemorrhagic.

Congestive heart failure →

CHF is when the heart is unable to pump sufficiently to maintain blood flow to meet the body's needs.

Peripheral artery disease →

PAD is also called peripheral vascular disease, which is characterized by narrowing of the arteries other than those that supply the heart or the brain. Peripheral artery disease most commonly affects the legs. When narrowing occurs in the heart it is called coronary artery disease, while in brain it is called cerebrovascular disease.

Narrowing of arteries is caused by Atherosclerosis, Artery-spasm etc. and the risk factors are Cigarette smoking, Diabetes, Hypertension, High blood cholesterol (Hyperlipidemia) etc.

Hypertension →

Abnormally high blood pressure is called hypertension. Normal level of blood pressure is 120/80 mmHg. More than 140/90 mmHg in those who are at risk and more than 160/100 mmHg in all are considered high BP. Hypertension is a major risk factor for stroke and MI.

Myocardial infarction →

Myocardial infarction (MI), commonly known as a heart attack, occurs when blood flow decreases or stops to a part of the heart, causing damage to the heart muscle.

The most common symptom is chest pain or discomfort which may travel into the shoulder, arm, back, neck, or jaw. The discomfort may occasionally feel like heartburn. Other symptoms may include shortness of breath, nausea, feeling faint, a cold sweat, or feeling tired.

An MI may cause heart failure, an irregular heartbeat, cardiogenic shock, or cardiac arrest.

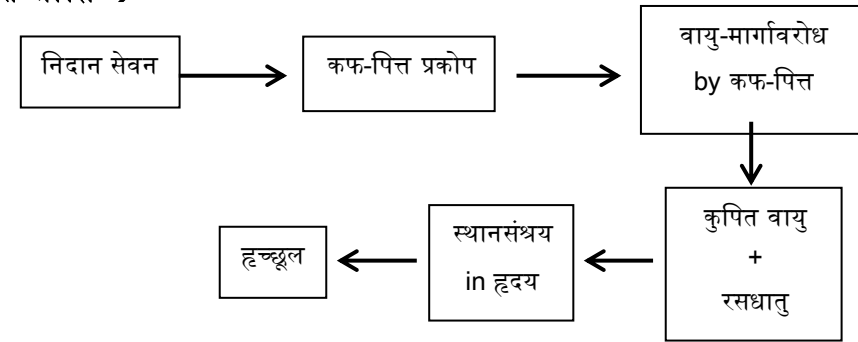
हृत्शूल (Angina)

Shoola in the hrid-pradesha due to any cause is called Hrid-shoola.

निदान →

अपानवात मल मूत्र वेगधारण – अतिभोजन – अजीर्णाध्यशन – विरुद्ध & असात्म्य भोजन – क्षुत्काले पानीयपानात् – पिष्टान्न शुष्कमांस सेवन etc.

सम्प्राप्ति →



सामान्य लक्षण →

- हृत्पीडा (Chest pain in left side)
- ऊच्छवास अवरोध (Difficulty in expiration)

हृत्शूल चिकित्सा सूत्र →

वातोपसृष्टे हृदये वामयेत् स्निग्धमातुरम् ।

द्विपञ्चमूलीक्वाथेन सस्नेह लवणेन च॥ (च.द.31/1)

Like Vataja Hridroga-Chikitsa, proper oleation (स्नेहन) and sudation (स्वेदन) should be given, and then emesis (वमन) should be administered with दशमूल क्वाथ+स्नेह+सैंधव

→ संशोधन चिकित्सा → स्नेहन – स्वेदन – वमन – शिरोधारा

→ संशमन चिकित्सा → हृदयार्णव रस – कस्तूरी भैरव रस – मुक्ता भस्म – प्रवाल पिष्टी – हिंवादि वटी – अर्जुन चूर्ण – अर्जुनारिष्ट – अर्जुन क्षीरपाक

अभ्यंगार्थ → कर्पूरादि तैल & लाक्षादि तैल

Angina Pectoris

Angina pectoris is defined as precordial pain brought on by effort, emotion or heavy meals, and is relieved by rest.

Etiology → This is due to myocardial ischemia (reduced blood flow to the heart) that can be caused by → Cardio-vascular diseases – Coronary thrombosis – Mental stress – Heavy meals – Old age etc.

Clinical Features →

Angina presents as squeezing, pressure, heaviness, tightness or pain in the chest. It may radiate to one or both arms, the neck, jaw or teeth.

Associated symptoms: Dyspnoea, nausea, sweatiness, and faintness.

Types of Angina →

1. Stable angina → induced by effort, relieved by rest.
2. Unstable angina → angina of increasing frequency or severity; occurs on minimal exertion or at rest; associated with risk of MI.
3. Decubitus angina → precipitated by lying flat.
4. Variant angina → caused by coronary artery spasm (rare).

Tests → ECG, Signs of past MI, Exclude precipitating factors → anaemia, diabetes, hyperlipidaemia, thyrotoxicosis, temporal arteritis.

Treatment → Avoid smoking, stress, over exertion & fatty diet.

- Nitroglycerin 0.3-0.6 mg sublingual & Aspirin 325-650 mg orally.
- Treat the underlying cause.

Hypertension

Abnormally high blood pressure is called hypertension. Normal level of blood pressure is 120/80 mmHg. More than 140/90 mmHg in those who are at risk (like existing vascular disease, diabetes, liver or renal or heart disease) and more than 160/100 mmHg in all are considered high BP.

HTN is caused by either (i) increased cardiac output, or (ii) increased peripheral vascular resistance.

Etiological Classification →

1. Essential hypertension (primary, cause unknown). ~95% of cases.
2. Secondary hypertension ~5% of cases. Causes include →
 - Renal disease → The most common secondary cause. 75% are from intrinsic renal disease: glomerulonephritis, polyarteritis nodosa (PAN), systemic sclerosis, chronic pyelonephritis, or polycystic kidneys. 25% are due to renovascular disease, most frequently atheromatous (cigarette smokers, eg with peripheral vascular disease) or rarely fibromuscular dysplasia (female).
 - Endocrine disease → Cushing's and Conn's syndromes, pheochromocytoma, acromegaly, hyperparathyroidism.
 - Others → Coarctation, pregnancy, steroids, MAOI, 'the Pill'.

Clinical Features →

Usually asymptomatic.

Signs and symptoms of underlying or associated diseases like renal disease, diabetes, Cushing's syndrome etc.

In chronic case → blurred vision, dizziness, headache & heart disease.

Investigations →

Measuring blood pressure with sphygmomanometer
 Blood test (CBC, Cholesterol – HDL, LDL, Triglycerides)
 Urine routine
 ECG

Treatment →

- Find and treat the underlying cause
- Antihypertensive drugs →
 - Diuretics
 - Loop diuretics e.g. Furosemide, Torasemide; Thiazide diuretics e.g. Hydrochlorothiazide; Potassium sparing diuretics e.g. Spironolactone etc.
 - Calcium channel blockers
 - Nifedipine, Amlodipine, Levamlodipine etc.
 - ACE inhibitors
 - Captopril, Enalapril, Ramipril etc.
 - Angiotensin II receptor antagonists
 - Valsartan, Telmisartan, Olmesartan etc.
 - Adrenergic receptor antagonists
 - β blockers e.g. Betaxolol, Atenolol, Labetalol, Propranolol etc. & α blockers e.g. Prazocin, Doxazosin etc.
 - Vasodilators
 - Sodium nitroprusside.

Hypotension

Hypotension is low blood pressure → A systolic blood pressure of less than 90 millimeters of mercury (mm Hg) or diastolic of less than 60 mm Hg (i.e. < 90/60 mmHg).

It is a physiological state, rather than a disease. Severely low blood pressure can deprive the brain and other vital organs of oxygen and nutrients, leading to a life-threatening condition called shock.

Causes → Low blood pressure can be caused by low blood volume (hypovolemia), electrolyte imbalance, hormonal changes, widening of blood vessels, medicine side effects, anemia, heart problems or endocrine problems.

Clinical Features →

- The primary symptoms of hypotension are lightheadedness or dizziness. If the blood pressure is sufficiently low, fainting may occur.
- Other symptoms may be headache, chest pain, shortness of breath, irregular heartbeat (arrhythmia), unconsciousness etc.

Treatment →

- The treatment for hypotension depends on its cause.
- General measures include → Intake of salt and sugar, and plenty of water or fruit juices, IV Fluids & Electrolytes, Vasopressors.
- The best way to determine if a person will benefit from fluids is by doing a passive leg raise followed by measuring the output from the heart.

Diseases of रक्तवह स्रोतस

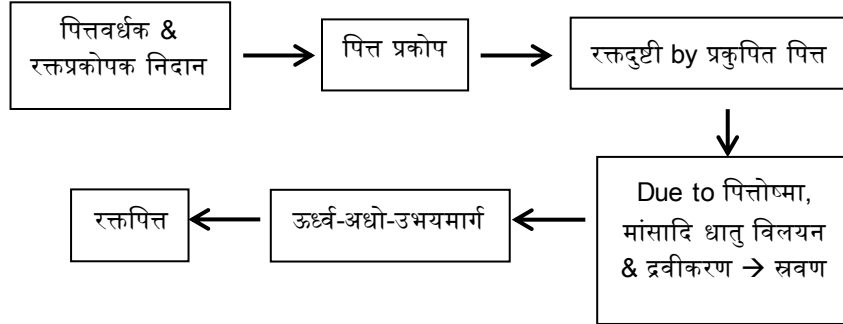
रक्तपित्त (Haemorrhagic Disorders)

“रक्तवत् पित्तमिति रक्तपित्तम्” the disease in which rakta gets vitiated due to aggravated pitta is called raktapitta.

निदान →

आहारज निदान	ऊष्ण, तीक्ष्ण, क्षार, अम्ल, लवण, & कटु पदार्थ अतिसेवन विदाही अन्नपान सेवन, दुग्ध+लवण & क्षारयुक्त शाक सेवन मूलक, सर्षप, लशुन, सुरा, सौवीरक अतिसेवन
विहारज	अति आतप सेवन, अति वायु सेवन, अति अध्वगमन, अतिव्यवाय
मानसिक	शोक-भयादि मानसिक हेतु

सम्प्राप्ति →



पूर्वरूप →

अंगसाद (Malaise) – शीतकामता – कण्ठधूमायन – वमन – लोहगन्धि निःश्वास

लक्षण →

क्र	दोषानुसार भेद	लक्षण
1	वातज रक्तपित्त	श्याव अरुण सफेन तनु रुक्ष रक्तस्राव
2	पित्तज रक्तपित्त	कृष्ण कषायाभ गोमूत्रवर्ण मेचक आगारधूमाभ अंजनाभ रक्तस्राव
3	कफज रक्तपित्त	पाण्डु सस्नेह सान्द्र पिच्छिल रक्तस्राव
4, 5, 6	द्वन्द्वज	द्विदोषज लक्षण
7	सन्निपातज	त्रिदोषज मिश्रित लक्षण

क्र.	गति-भेद	लक्षण
1	ऊर्ध्वग	कफसंसृष्ट – रक्तप्रवृत्ति through मुख, नासिका, नेत्र, कर्ण
2	अधोग	वातानुबन्धि – रक्तप्रवृत्ति through गुद, मेढ्र, योनिमार्ग
3	अभय (तिर्यक)	वात कफ – रक्तप्रवृत्ति through उर्ध्व / अधो-मार्ग / रोमकूप

साध्यासाध्यता →

साध्य → एकदोषज रक्तपित्त, ऊर्ध्वग रक्तपित्त, बलवान रोगी, नवीन & अल्पवेग रक्तपित्त, उपद्रव रहित.

याप्य → द्विदोषज रक्तपित्त, अधोग रक्तपित्त.

असाध्य → त्रिदोषज रक्तपित्त, उभयमार्गी रक्तपित्त, अतिवेग युक्त रक्तपित्त, अरिष्ट लक्षण.

उपद्रव →

दौर्बल्य – श्वास – कास – ज्वर – पाण्डु – दाह – मूर्च्छा – तृष्णा – शिरोभिताप etc.

रक्तपित्त चिकित्सा सूत्र →

स्तम्भन-निषेध : Initially, in a strong person (अक्षीण बल मांसस्य), the bleeding process should not be stopped by स्तम्भन चिकित्सा.

रक्तस्राव स्तम्भनजन्य उपद्रव → If bleeding is stopped it may lead to गलग्रह, पूतिनस्य, मूर्च्छा, अरुचि, ज्वर, गुल्म, प्लीहा, आनाह, किलास (श्वेतकुष्ठ), कुष्ठ, विसर्प, अर्श, भगन्दर, मूत्रकृच्छ्र, वर्णनाश, बुद्धि-इन्द्रियोपरोध etc.

अक्षीणबलमांसस्य यस्य सन्तर्पणोत्थितम् ।
 बहुदोषं बलवतो रक्तपित्तं शरीरिणः ॥
 काले संशोधनार्हस्यं तद्धेन्निरूपद्रवम् ।
 विरेचनेनोर्ध्वभागं अधोगं वमनेन च ॥ (च.चि.4/55-56)

In case of सन्तर्पणोत्थ रक्तपित्त, and if रोगी is बलवान्, and in बहुदोषावस्था but
 निरूपद्रव (no complication) → the patient should be treated with →
 संशोधन चिकित्सा → ऊर्ध्वग रक्तपित्त → विरेचन
 अधोग रक्तपित्त → वमन

लंघनं रक्तपित्तादौ तर्पणं वा प्रयोजयेत् ।
 उर्ध्वगे तर्पणं पूर्वं कर्तव्यं विरेचनम् ॥
 प्राग्धो गमने पेया वमनश्च यथा बलम् ।
 अवम्यविरेच्यश्च शमनीयैरुपाचरेत् ॥ (चि.दर्श.)

→ Depending on रोगी-रोग बल, physician should administer लंघन or तर्पण
 in the beginning.
 → उर्ध्वग रक्तपित्त → तर्पण & विरेचन
 → अधोग रक्तपित्त → पेया & वमन
 → बाल-वृद्ध-अवम्यविरेच्य रोगी → शमन चिकित्सा

In case of severe bleeding even if it is with ama, one should immediately
 use haemostatic herbs (रक्तस्तम्भक), as the blood is vital tissue in the
 body.

रक्तस्तम्भक द्रव्य →

दूर्वा-उशीर-रक्तचन्दन-पद्मक-कुटज-कांचनार-बिल्व-उदुम्बर-वासा-लोध्र-शाल्मली etc.

शमनौषधि →

1. रसौषधि → मात्रा = 125 - 250 mg. अनुपान = मधु / जल
 रक्तपित्तान्तक रस – पित्तान्तक रस – बोल पर्पटी – शंख भस्म – प्रवाल पिष्टी

2. वटी → मात्रा = 250 - 500 mg. अनुपान = ऊष्णोदक / मधु / घृत
 एलादि वटी – तीक्ष्णादि वटी

3. चूर्ण → मात्रा = 3 - 6 g. अनुपान = ऊष्णोदक / मधु
 लाक्षा चूर्ण – उशीरादि चूर्ण – चन्दनादि चूर्ण – आटरुषकादि चूर्ण

4. क्वाथ/आसव/अरिष्ट → मात्रा = 20 - 40 ml. अनुपान = जल
 आटरुषकादि क्वाथ – वासकादि क्वाथ – उशीरासव – लोधासव – द्राक्षादि अरिष्ट

5. घृत → मात्रा = 10 - 30 ml. अनुपान = उष्ण जल / दुग्ध
 वासा घृत – शतावरी घृत

6. अवलेह/खण्ड → मात्रा = 20 - 30 g. अनुपान = दुग्ध / जल
 खण्ड कूष्माण्डावलेह – वासाखण्ड – वासावलेह – उदुम्बरावलेह

पथ्य आहार-विहार → पुराण शालि – लाजा – गोधूम – मुद्ग – मसूर – पटोल – मूलक –
 पलाण्डु – कूष्माण्ड – अजादुग्ध – गोदुग्ध – घृत – जांगल मांसरस – आमलकी – कदली –
 द्राक्षा – मृद्विका – मिश्री – शीतजल स्नान – शीत विहार – चन्दन लेप – मुक्ता मणिधारण –
 चन्द्रकिरण सेवन – वमन विरेचन नस्यादि

अपथ्य आहार-विहार → तिल – माष – कुलत्थ – सर्षप – गुड – दधि – क्षार – लवण – कतु
 तीक्ष्ण विदाही अन्नपान – मत्स्य – आतप सेवन – स्वेदन – वेगधारण – क्रोध

Hemorrhagic Disorders

Bleeding disorders (Coagulopathy) →

Bleeding disorders can be inherited or acquired. Some bleeding disorders can occur due to conditions such as thrombocytopenia, anemia, vitamin K deficiency, leukemia, cirrhosis of liver and HIV.

They can also result from certain medications that are used to thin the blood, including antiplatelet and anticoagulant medications. e.g. Aspirin, Clopidogrel (antiplatelet), Warfarin (anticoagulant) etc.

Examples of inherited bleeding disorders include Hemophilia A, Hemophilia B, and von Willebrand Disease.

Clinical Features →

Bruising – Petechial bleeding – Nosebleeds (Epistaxis) – Earbleeds (Otorrhagia) – Bleeding gums – Hematemesis – Hematuria – Melina – Menorrhagia – Excessive bleeding following surgery etc.

Investigations →

CBC (Low platelet level – thrombocytopenia)

BT CT or Prothrombin time (increased)

Treatment →

- Treat the basic cause of hemorrhage.
- Maintain Fluid & Electrolytes balance with IV Fluids.
- Plasma expanders or Blood transfusion (as indicated).
- Haematinic & Hematopoietic drugs.
- Injection vitamin K

कामला (Jaundice)

“कामान् लाति इति कामला”

The disease in which all likes and dislikes disappear.

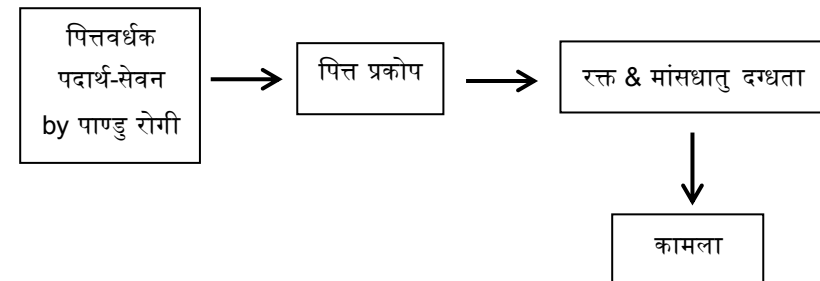
कामला is characterized by हारिद्र-पीत वर्णता of त्वचा-नेत्र-नख-मुख.

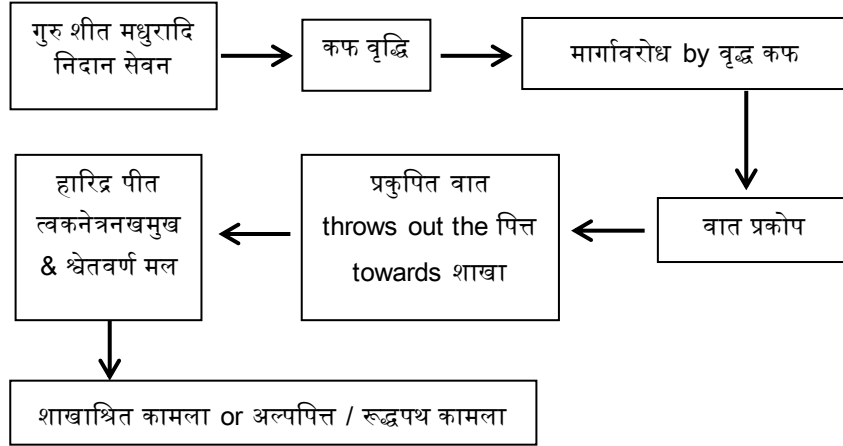
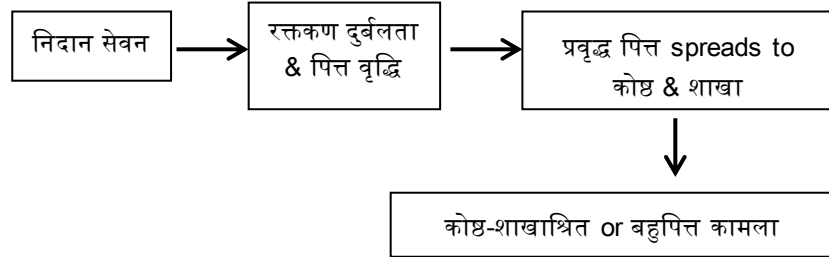
निदान →

आहारज निदान	अम्ल, लवण, क्षार, अत्युष्ण, विरुद्ध, असात्म्य भोजन निष्पाव, माष, पिण्याक, तिल, विदग्ध-अन्न अति सेवन रूक्ष, शीत, गुरु, मधुर पदार्थ अतिसेवन
विहारज	अतिव्यायाम, अतिमैथुन, दिवास्वप्न, वेग-धारण, पंचकर्म-विषमयोग
मानसिक	काम-चिन्ता-भय-क्रोध-शोक

If the patient of anemia (पाण्डुरोगी) consumes पित्तवर्धक आहार, he may develop the disease- कामला.

समान्य सम्प्राप्ति →



Samprapti of शाखाश्रित कामला →**Samprapti of कोष्ठ शाखाश्रित कामला →****सामान्य लक्षण →**

हारिद्र पीत त्वकनेत्रनखमुख – रक्त पीत शकृन्मूत्र – इन्द्रिय दौर्बल्यता – दाह – अविपाक – अरुचि – सदन – काश्य etc.

विशेष लक्षण →

1	शाखाश्रित कामला	श्वेतवर्च (तिलपिष्टनिभ पुरीष) – हारिद्र पीत त्वकनेत्रनखमुख – विबन्ध – आटोप – अग्निमान्द्य – अरुचि – दौर्बल्य
2	कोष्ठाश्रित कामला	हारिद्र त्वकनेत्रनखमुख – रक्तपीत शकृन्मूत्र – दाह – अविपाक – दौर्बल्य – सदन – अरुचि – कर्षण

Acharya Vagbhata classified Kamala as →

1. परतन्त्र कामला → secondary to पाण्डु (Haemolytic jaundice)
2. स्वतन्त्र कामला → manifests primarily (Hepatocellular jaundice)

साध्यासाध्यता →

असाध्य लक्षण → Kamala is incurable if urine and stool becomes black or dark yellow (कृष्ण पीत वर्ण), and /or eyes, mouth and face becomes red (रक्त वर्ण), or Kamala associated with शोथ, मूर्च्छा, दाह, तृष्णा, नष्टाग्नि etc.

उपद्रव →

कुम्भकामला – पानकी – हलीमक – शोथ – मूर्च्छा – तृष्णा – नष्टाग्नि – नष्टसंज्ञा

कामला चिकित्सा सूत्र →

संशोध्यो मृदुभिस्तिकैः कामली तु विरेचनैः । (च.चि.16/40)

पंचगव्यं महातित्तं कल्याणकमथापि वा ।

स्नेहनार्थं घृतं दद्यात् कामलापाण्डुरोगिणे ॥ (च.चि.16/43)

→ निदान परिवर्जन

→ मृदु विरेचन with तित्तरस द्रव्य

→ For स्नेहपान → पंचगव्य घृत – महातित्त घृत – कल्याणक घृत

→ After स्नेहपान → गोमूत्र प्रयोग (for 15 days)

1. गोमूत्र + त्रिफला क्वाथ
2. गोमूत्र + गोगुग्ध
3. गोमूत्र + माहिष दुग्ध

शाखाश्रित कामला चिकित्सा →

कटु तीक्ष्णोष्ण लवणैर्भूशाम्लैश्चाप्युपक्रमः ।

आ पित्तरागाच्छकृतो वायोश्चाप्रशमाद् भवेत् ॥

स्वस्थानमागते पित्ते पुरीषे पित्त रंजिते ।

निवृतोपद्रवस्य स्यात् पूर्वः कामलिको विधिः ॥ (च.चि.16/130-131)

Remove obstruction by using कफनाशक औषध → कटु तीक्ष्ण उष्ण लवण द्रव्य, by which शाखागत पित्त comes to कोष्ठ, and पुरीष gets its normal colour. When obstruction is removed, general line of treatment of kamala should be adopted.

शमनौषधि →

नवायस लौह – धात्री लौह – विडंगादि लौह – पुनर्नवा मण्डूर – मण्डूर वटक – शिलाजतु वटक – आरोग्यवर्धिनी वटी – भूम्यामलकी – कालमेघ स्वरस (भूनिम्ब) – त्रिफला चूर्ण – पंचतित्त घृत – आमलक्यादि लेह – धात्र्यवलेह

कुम्भकामला

“कालान्तरात् खरीभूता कृच्छ्रा स्यात् कुम्भकामला” this is chronic stage of kamala which develops when koshta-shakhashrita kamala lasts for more than one year. Pitta vitiates the mamsa dhatu and body becomes dry and rough.

लक्षण →

कृष्ण पीत शकृन्मूत्र – अत्यधिक शोथ – रूक्षता – पर्वभेद

साध्यासाध्यता →

असाध्य लक्षण → छर्दि – हृल्लास – अरुचि – क्लम – ज्वर – श्वास – अतिसार

पानकी

This is सातिसार कामला → Chronic Jaundice complicated with Diarrhoea.

लक्षण →

अतिसार – संताप – बहिरन्तश्च पीतता – पाण्डुता

हलीमक

This stage of the disease develops as serious complication of Pandu or Kamala. There is predominance of Vata and Pitta in pathogenesis. Sushruta explained it under the title ‘लाघरक’.

लक्षण →

हरित श्याव पीत वर्णता – बलोत्साह क्षय – मन्दाग्नि – मन्दज्वर – अंगमर्द – श्वास – अप्रहर्ष – तृष्णा – अरुचि – भ्रम

हलीमक चिकित्सा →

- स्नेहन by using माहिष घृत cooked with माहिष दुग्ध & गुडूची स्वरस
- विरेचन by using त्रिवृत् चूर्ण in आमलकी स्वरस
- यापना वस्ति
- अगस्त्यहरीतकी प्रयोग
- वात-पित्तनाशक मधुररस प्रधान आहार

Jaundice

Jaundice, also known as icterus, is a yellowish or greenish discoloration of the skin and sclera (whites of the eyes) due to high bilirubin levels.

Types →

1. Pre-hepatic jaundice → caused by anything that causes an increased rate of hemolysis (breakdown of red blood cells).
2. Hepatic jaundice → Hepatocellular (hepatic) jaundice can be caused by acute or chronic hepatitis, hepatotoxicity, cirrhosis, drug-induced hepatitis and alcoholic liver disease.
3. Post-hepatic jaundice → also called obstructive jaundice, is caused by an interruption to the drainage of bile containing conjugated bilirubin in the biliary system. The most common causes are gallstones in the common bile duct, and pancreatic cancer in the head of the pancreas.
4. Neonatal jaundice → Neonatal jaundice is usually harmless: this condition is often seen in infants around the second day after birth, lasting until day 8 in normal births, or to around day 14 in premature births. It is treated with phototherapy.

Two types – physiological and pathological → Jaundice is considered pathologic if it presents within the first 24 hours after birth, the total serum bilirubin level rises by more than 5 mg per dL per day or is higher than 15 mg per dL.

Treatment of Jaundice → Hepatoprotective drugs should be used & hepatotoxic drugs and alcohol should be completely avoided.

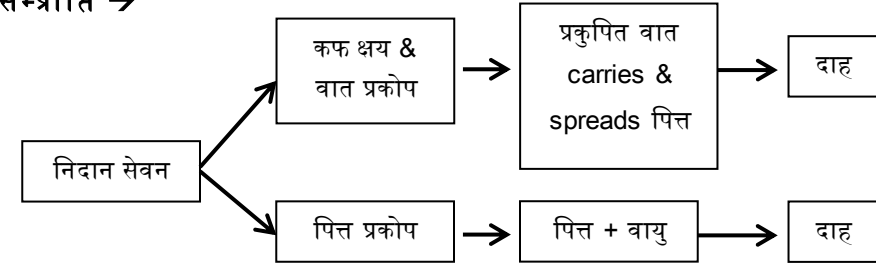
दाह (Burning sensation)

Daha means burning sensation. In this disease there is burning of the whole body.

निदान →

पित्तवर्धक आहार विहार – उष्ण तीक्ष्ण मद्यपान – तृष्णा – रक्तपूर्ण कोष्ठ – क्षत – मर्माभिघात – धातुक्षय – आषयापकर्ष (वातज दाह) etc.

सम्प्राप्ति →



भेद → पित्तज – मद्यज – तृष्णा निरोधज – रक्तज – रक्तपूर्ण कोष्ठज – क्षतज – मर्माभिघातज – धातुक्षयज

दाह चिकित्सा सूत्र →

दाहेऽतिशिशिरं तोयं क्रिया कार्या सुशीलता ।

सर्वांगे चन्दनालेपश्चन्द्रकस्तूरीका युतः ॥ (यो.र.1/10)

→ निदान परिवर्जन

→ शीतोपचार (बाह्य & आभ्यन्तर) – पित्तज ज्वर & रक्तपित्त चिकित्सा

→ शीत जल सेवन (पान & स्नान) – चन्दन लेप – चन्द्रकिरण सेवण – मुक्ता मणि धारण – नीलकमल युक्त शीतल शय्या

शमनौषधि → दाहान्तक रस – चन्द्रकला रस – शंख भस्म – प्रवाल पिष्टी – मुक्ता पिष्टी – मृत संजीवनी वटी – चन्दनादि चूर्ण – चान्दनासव – धान्यक हिम – शतधौत घृत – चन्दनादि तैल (अभ्यंगार्थ)

मद-मूर्च्छा-सन्यास

मद →

Mada means neurosis or intoxication which leads to confused state of mind. The person gets excited and feels happy and joyful after consuming madakari dravyas like madya, bhanga etc. मद, मूर्च्छा & सन्यास are three stages of the disease.

लक्षण →

क्र	भेद	लक्षण
1	वातज मद	Improper but stuttering talk, constant talk, fast movements and the body becoming extreme dry.
2	पित्तज मद	Anger, harmful talk, aggressiveness.
3	कफज मद	Lethargy, excess sleep and improper but slow talk.
4	सन्निपातज मद	Mixed symptoms of all above are seen.
5	रक्तज मद	This results due to excessive loss of blood from the body. All the symptoms of Pittaja mada are seen.
6	मद्यज मद	Improper movement and talk.
7	विषज मद	Due to poisoning this type is developed in which there are tremors and excess sleep are seen.

मूर्च्छा →

सुखदुःखव्यपोहाच्च नरः पतति काष्ठवत् ।

मोहो मूर्च्छेति तां प्राहुः (सु.उ.46/7)

All the senses, including happiness and sadness are lost and the person lies down like wood, then it is called as murccha.

निदान →

अति धातुक्षय – अति दोष प्रकोप – आघात – वेगधारण – अल्प सत्व etc.

पूर्वरूप →

हृत्पीडा – जृम्भा – ग्लानि – संज्ञा दौर्बल्य

लक्षण →

क्र	भेद	लक्षण
1	वातज मूर्च्छा	The patient while becoming unconscious sees various colours in the sky like blue, black, reddish. During attack there is – वेपथु, अंगमर्द, हृत्पीडा, श्याववर्णता
2	पित्तज मूर्च्छा	The patient sees green, red, yellow colour in the sky. During attack there is – स्वेद, तृष्णा, दाह, पीतवर्णता
3	कफज मूर्च्छा	The patient feels that sky has become cloudy and darkness is produced everywhere. During attack there is – गौरव, स्तैमित्य, प्रसेक, हल्लास
4	सन्निपातज मूर्च्छा	Mixed symptoms of all above are seen
5	रक्तज मूर्च्छा	(due to smell or sight of blood) स्तब्धांगता – स्तब्ध दृष्टि – गूढ उच्छ्वास
6	मद्यज मूर्च्छा	The patient lies down with irrelevant speech.
7	विषज मूर्च्छा	Specific symptoms of poison with वेपथु-तृष्णा-तम

सन्यास →

When the extremely aggravated doshas get localized at the vital centers, destroys the function of speech, body and mind. The patient becomes unconscious and lies down like a wood, as if dead. This is fatal disease, causes death if not treated immediately.

मूर्च्छा चिकित्सा सिद्धान्त →

During attack (वेगावस्था), physician should try to bring back the consciousness of the patient by following measures →

- तीक्ष्ण अंजन (शिरिषादि अंजन / मधुकादि अंजन)
- अवपीडन नस्य – धूमपान
- सूची वेधन – केशलुध्नन – नखान्तरे पीडन – कपिकच्छू घर्षण
- दाह कर्म

After regaining consciousness, patient should be treated with following measures →

- स्नेहन – स्वेदन – बलानुसार संशोधन
- आश्र्वासन
- व्यायाम
- मूर्च्छा-भेदानुसार चिकित्सा →
 - वातज, पित्तज & कफज मूर्च्छा → ज्वरघ्न कषाय प्रयोग
 - रक्तज मूर्च्छा → शीतल उपचार
 - मद्यज मूर्च्छा → वमन with वामक औषधि+मद्य
 - विषज मूर्च्छा → विषघ्न औषध प्रयोग

शमनौषधि →

मूर्च्छान्तक रस – स्मृति सागर रस – योगेन्द्र रस – मेध्य वटी – अश्वगन्धारिष्ट

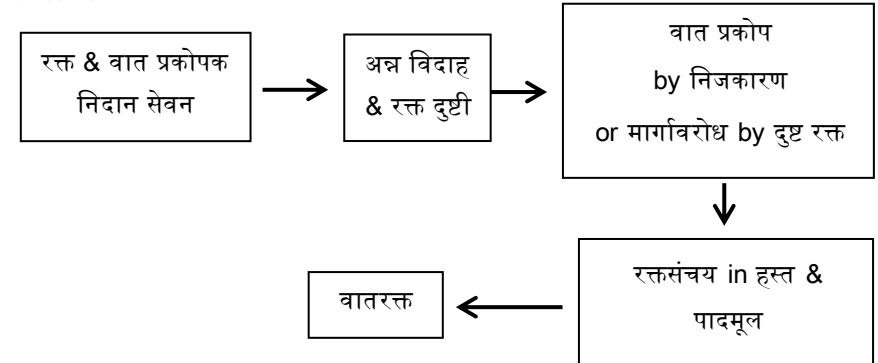
वातरक्त (Gout)

Synonyms → वातशोणित – वातरक्त – आढ्यवात – खुडरोग – वातबलास

“वातस्यावरणेन बलमत्स्यस्मिन् शोणित इति वातबलासः” this disease is manifested due to vata-avarana and rakta-dushti. Vitiated vata and vitiated rakta cause this disease hence it is called as Vatarakta.

निदान →

आहारज निदान	अम्ल लवण कटु क्षार स्निग्ध उष्ण आहार अतिसेवन क्लिन्न आहार, शुष्कान्न, औद्धिद आनूप मांस अतिसेवन विरुद्ध आहार, अध्यशन, अजीर्णाशन तिलपिष्ट, मूलक, कुलत्थ, माष, दधि, इक्षु, अरनाल, सौवीरक
विहारज	दिवास्वप्न, रात्रिजागरण, अम्बुक्कीडा, अभिघात, अतिव्यवाय
प्रकृतिज	सुकुमार, स्थूल, अव्ययामी or सुखी पुरुष taking मिथ्या आहार-विहार

सम्प्राप्ति →

स्थान →

तस्य स्थानं करौ पादावंगुल्यः सर्वसन्धयः → hands, feet, fingers & all joints.

मूल स्थान → हस्त & पाद

पूर्वरूप →

अतिस्वेद or अस्वेद – काष्ण्य – स्पर्श अज्ञत्वं – क्षतेऽतिरुक् – सन्धिशैथिल्यता – सदन – पिडकोद्गम – सन्धि तोद स्फुरण भेद – गुरुता – सुप्ति

सामान्य लक्षण →

मूषिकविषवत वेदना

Severe pain & gouty swelling in the affected joints

Hot, red and tendor joints

Patient will not tolerate touch to the affected parts

विशेष लक्षण →

क्र.	वातरक्त भेद	लक्षण
1	उत्तान वातरक्त (superficial)	Characterized by कण्डू (itching), दाह (burning sensation), रुक् (pain), तोद (pricking sensation), आयाम (stretching), and कुञ्चन (contraction) of affected joint. Skin becomes श्याव or रक्त or ताम्र वर्ण.
2	गम्भीर वातरक्त (deep)	Characterized by शोथ (swelling), स्तब्धता (stiffness), कठिनता (hardness), and अति रुक् (severe pain), in the deeper tissues of the body. Skin becomes ताम्र वर्णी with दाह, तोद, स्फुरण, पाक.

क्र.	दोषानुसार भेद	लक्षण
1	वातज वातरक्त	सिरा आयाम – शूल – स्फुरण – तोद – कृष्ण श्याववर्ण शोथ
2	पित्तज वातरक्त	दाह – वेदना – मूर्च्छा – स्वेदाधिक्य – तृष्णाधिक्य – उष्मा
3	कफज वातरक्त	स्तैमित्य – गौरवता – स्निग्धता – मन्दवेदना – कण्डू
4	रक्तज वातरक्त	कण्डू क्लेदयुक्त शोथ – वेदनायुक्त संधिशोथ – चिमचिमायमन
5	द्वन्द्वज वातरक्त	द्विदोषज लक्षण
6	सन्निपातज वातरक्त	त्रिदोषज मिश्रित लक्षण

साध्यासाध्यता →

साध्य लक्षण → एकदोषज, नवीन, उपद्रव रहित वातरक्त

याप्य लक्षण → द्विदोषज, अल्प उपद्रव युक्त वातरक्त

असाध्य लक्षण → त्रिदोषज, चिरकालीन, उपद्रव युक्त वातरक्त, विवर्ण, स्फुटित, बहुस्रावी, पूयस्रावी, मांसक्षय, इंद्रिय ताप, अर्बुद.

उपद्रव →

अनिद्रा – अरोचक – श्वास – मांसकोथ – शिरोग्रह – मूर्च्छा – मद – विसर्प – अंगुलिचक्रता – दाह – मर्मग्रह – अर्बुद etc.

वातरक्त चिकित्सा सिद्धान्त →

1. संशोधनचिकित्सा: विरेच्य स्नेहयित्वाऽऽदौ स्नेहयुक्तैर्विरेचनैः।

रूक्षैर्वा मृदुभिः शस्तमसकृद्वस्तिकर्म च ॥

सेकाभ्यंगप्रदेहान्नस्नेहाः प्रायोऽविदाहिनः ॥ (च.चि.29/41)

In रूक्ष शरीर → स्नेहन – स्नेहयुक्त मृदुविरेचन – रक्तमोक्षण

In स्निग्ध शरीर → रूक्ष मृदुविरेचन (without स्नेहन) – रक्तमोक्षण

अनुवासन & निरुह बस्ति should be used according to conditions.

घृत मिश्रित क्षीर बस्ति is best.

रक्तमोक्षण → In वातरक्त, Blood-letting can be done with शृंग (horn), जलौका (leech), अलाबु (bottle-gourd), प्रच्छेदन (small incisions) or सिराव्यध (venesection).

Avoid blood-letting in a patient who is weak (कृश), ununctuous (रूक्ष) and has predominance of vayu (वातप्रधान).

2. संशमन चिकित्सा :

वातरक्तान्तक रस – विश्वेश्वर रस – शिलाजतु – कैशोर गुग्गुलु – अमृता गुग्गुलु – हरीतकी चूर्ण – निम्बादि चूर्ण – मंजिष्ठादि चूर्ण – मंजिष्ठाद्यरिष्ट – मंजिष्ठादि क्वाथ – बला घृत – जीवनीय घृत – एरण्डतैल+दुग्ध

अभ्यंग with पिण्ड तैल – सुकुमार तैल – अमृताद्य तैल

लेप with हरिद्रा fried in घृत – उपनाह with वेशवार

Gout

Build up of crystals of uric acid in the joints leads to Gouty arthritis. It is caused by excess uric acid in the bloodstream (hyperuricaemia).

Clinical Features →

- Acute arthritis affects the big toe (MTR) in 75% of attacks. Occasionally ankle, other toes, knees or fingers are affected. Joint is extensively painful, red, hot, swollen and very tender.
- Tophi may occur on the outer ears, hands and feet.
- Hypertension, obesity, renal uric acid stone may occur.

Investigations → Serum uric acid (>6mg /dl), Raised ESR, X-ray.

Treatment →

Analgesics, NSAIDs, Urate lowering drugs (Allopurinol, Probenacid) etc.

प्लीहादोष

Over eating (अत्यशन), excessive travelling or jerking (यानात संक्षोभ), excessive activities (अति चेष्टा), excessive sexual intercourse (अति व्यवाय), lifting heavy weights (भार), excessive walking (अध्व), excessive vomiting (वमन), or emaciated by other diseases (व्याधि कर्षण) cause प्लीहा दोष.

Aggravated vata vitiates rakta and produces this disease. Spleen gets enlarged and pain is produced in the upper left quadrant of the abdomen.

Types → 5 → वातज – पित्तज – कफज – रक्तज – सन्निपातज

क्र	भेद	लक्षण
1	वातज	Slight enlargement of spleen with pain, distension of abdomen and constipation.
2	पित्तज	The enlarged spleen is soft to touch and there are symptoms like fever, burning, thirst, jaundice and anemia.
3	कफज	The size of enlargement of spleen is very big. Other symptoms are nausea, vomiting and edema on the body.
4	रक्तज	All the pittaja-symptoms and there is tenderness over the enlarged spleen.
5	सन्निपातज	Mixed symptoms of all V-P-K doshas.

चिकित्सा →

→ निदान परिवर्जन (Treat the cause of Enlargement of Spleen)

→ वामबाहु कुर्पराभ्यन्तर सिराव्यध

→ शमनौषधि →

प्लीहारि रस – रोहितकादि वटी – चित्रकादि वटी – शिग्रु क्वाथ – रोहित अभया क्वाथ –

रोहितकारिष्ट – कुमारी आसव

यकृतदोष

Over eating (अत्यशन), excessive travelling or jerking (यानात संक्षोभ), excessive activities (अति चेष्टा), excessive sexual intercourse (अति व्यवाय), lifting heavy weights (भार), excessive walking (अध्व), excessive vomiting (वमन), or emaciated by other diseases (व्याधि कर्षण) cause प्लीहा दोष and /or यकृत दोष.

Yakrita dosha leads to abnormal enlargement of liver which results in palpable lump or swelling in right hypochondrium.

लक्षण →

- Tenderness and swelling in right hypochondrium.
- Swelling may not be visible but enlarged liver can be palpated.
- Jaundice – Anemia – Anorexia – Nausea – Vomiting – Fever etc.

चिकित्सा →

→ निदान परिवर्जन (Treat the cause of Enlargement of Liver)

→ दक्षिणबाहु कुर्पराभ्यन्तर सिराव्यध

→ शमनौषधि →

भूम्यामलकी – कालमेघ स्वरस (भूनिम्ब) – भृंगराज चूर्ण – अविपत्तिकर चूर्ण –

आरोग्यवर्धिनी वटी - रोहितकारिष्ट – कुमारी आसव – वर्धमान पिप्पली रसायन

Hepatoprotective drugs →

भूम्यामलकी – भूनिम्ब – कटुकी – कुमारी – रोहीतक – शरपुंखा etc.

Hepatitis

Hepatitis refers to an inflammatory condition of the liver. It's commonly caused by a viral infection, but there are other possible causes of hepatitis. These include autoimmune hepatitis and hepatitis that occurs as a secondary result of medications, drugs, toxins, and alcohol.

Some people have no symptoms whereas others develop yellow discoloration of the skin (icterus) and whites of the eyes (pallor), poor appetite, anorexia, vomiting, tiredness, abdominal pain, or diarrhea. Over time the chronic form may progress to scarring of the liver, liver failure, or liver cancer.

➤ Viral Hepatitis →

Viral infections of the liver that are classified as hepatitis include hepatitis A, B, C, D, and E. Hepatitis A is always an acute, short-term disease, while hepatitis B, C, and D are most likely to become ongoing and chronic. Hepatitis E is usually acute but can be particularly dangerous in pregnant women.

Hepatitis A is caused by an infection with the hepatitis A virus (HAV). This type of hepatitis is most commonly transmitted by consuming food or water contaminated by feces from a person infected with hepatitis A.

Hepatitis B is transmitted through contact with infectious body fluids, such as blood, vaginal secretions, or semen, containing the hepatitis B virus (HBV). Injection drug use, having sex with an infected partner, or sharing razors with an infected person increase the risk of getting hepatitis B.

Hepatitis C comes from the hepatitis C virus (HCV). Hepatitis C is transmitted through direct contact with infected body fluids, typically through injection drug use and sexual contact.

Hepatitis D is a serious liver disease caused by the hepatitis D virus (HDV). HDV is transmitted through direct contact with infected blood. Hepatitis D is a rare form of hepatitis that only occurs in conjunction with hepatitis B infection. The hepatitis D virus can't multiply without the presence of hepatitis B.

Hepatitis E is a waterborne disease caused by the hepatitis E virus (HEV). Hepatitis E is mainly found in areas with poor sanitation and typically results from ingesting fecal matter that contaminates the water supply.

➤ **Bacterial Hepatitis** →

Bacterial infection of the liver commonly results in pyogenic liver abscesses, acute hepatitis, or granulomatous (or chronic) liver disease.

➤ **Non-infectious hepatitis** →

Alcoholic Hepatitis → Excessive alcohol consumption can cause liver damage and inflammation. Over time, it can cause permanent damage and lead to liver failure and cirrhosis.

Toxic and drug-induced hepatitis → Many chemical agents, including medications, industrial toxins, and herbal and dietary supplements, can cause hepatitis. Many types of drugs can cause liver injury, including the analgesic paracetamol; antibiotics such as isoniazid, nitrofurantoin, Rabin Singh

amoxicillin-clavulanate, erythromycin, and trimethoprim-sulfamethoxazole; anticonvulsants such as valproate and phenytoin; cholesterol-lowering statins; steroids such as oral contraceptives and anabolic steroids; and highly active anti-retroviral therapy used in the treatment of HIV/AIDS.

Autoimmune hepatitis is a chronic disease caused by an abnormal immune response against liver cells. The disease is thought to have a genetic predisposition.

Investigations →

CBC, LFT (Serum bilirubin, SGOT, SGPT), USG, Liver biopsy etc.

Normal levels →

Serum bilirubin → Direct bilirubin → less than 0.3 mg/dL

Total bilirubin → 0.1 to 1.2 mg/dL

AST (SGOT) → 5 to 40 u/lit.

ALT (SGPT) → 7 to 56 u/lit.

AST – Aspartate Aminotransferase, SGOT – Serum glutamic oxaloacetic transaminase, ALT- Alanine Aminotransferase, SGPT – Serum glutamic pyruvic transaminase → these are enzymes produced by liver.

Treatment →

Hepatitis A, B, and D are preventable with immunization. Antiviral medications may be used to treat chronic cases of viral hepatitis. Autoimmune hepatitis may be treated with medications to suppress the immune system. A liver transplant may also be an option in certain cases.

Liver Cirrhosis

Cirrhosis of liver refers to irreversible scarring or fibrosis of liver, which is late stage of chronic or severe liver diseases.

Etiology →

Alcohol – Chronic infection – Viral hepatitis – Obstructive jaundice – Congenital hepatic fibrosis – Nutritional deficiencies etc.

Clinical Features →

- Hepatomegaly
- Portal hypertension
- Ascites
- Nausea, vomiting, anorexia, diarrhea
- Spider naevi around the face, neck and shoulder

Investigations →

LFT, USG, CT scan, Liver biopsy etc.

Treatment →

There is no specific treatment for cirrhosis. Liver cirrhosis is irreversible, therefore treatment aims on preventing its progression and maintaining the healthy liver tissue.

- Treat the underlying cause
- Hepatoprotective drugs
- Healthy diet
- Avoid alcohol and smoking

Leukemia

Leukemia, also known as Blood cancer, is cancer of the body's blood-forming tissues, including the bone marrow and the lymphatic system.

Leukemia is defined as a malignant progressive disease in which the bone marrow and other blood-forming organs produce increased numbers of immature or abnormal leucocytes (WBCs).

Classification →

Clinically and pathologically, leukemia is subdivided into a variety of large groups. The first division is between its acute and chronic forms:

Four major kinds of leukemia		
Cell type	Acute	Chronic
Lymphocytic leukemia (or "lymphoblastic")	Acute lymphoblastic leukemia (ALL)	Chronic lymphocytic leukemia (CLL)
Myelogenous leukemia ("myeloid" or "nonlymphocytic")	Acute myelogenous leukemia (AML or myeloblastic)	Chronic myelogenous leukemia (CML)

Clinical Features → Symptoms may include bleeding and bruising problems, feeling tired, fever, and an increased risk of infections. These symptoms occur due to a lack of normal blood cells. Diagnosis is typically made by blood tests or bone marrow biopsy.

Treatment → Treatment may involve some combination of chemotherapy, radiation therapy, targeted therapy, and bone marrow transplant, in addition to supportive care and palliative care as needed.

कुष्ठ (Skin diseases)

“कुष्णाति सर्वं वपुः इति कुष्ठम्” Kustha is characterized by deformity or discolouration of the body (skin diseases).

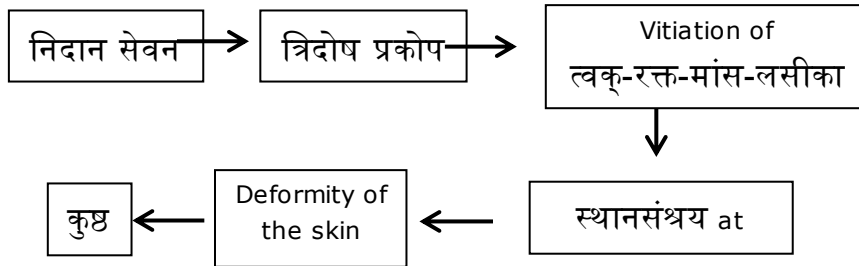
निदान →

विरुद्ध आहार – अति द्रव स्निग्ध गुरु आहार सेवन – छर्दि वेगधारण – अतिव्यायाम
आतपसेवन अतिभोजन – अजीर्ण भोजन – नवान्न दधि मत्स्य अतिलवण अम्ल सेवन –
दिवास्वप्न – पाप कर्म etc.

पूर्वरूप →

स्पर्श अज्ञत्व – अतिस्वेद – अस्वेद – वैवर्ण्य – कण्डू – कोठ – तोद – सुप्तता – लोमहर्ष

सम्प्राप्ति →



सप्तदृश्य (7 entities affected in kustha) →

वात – पित्त – कफ – त्वक् (रस) – रक्त – मांस – लसीका

Types of कुष्ठ : 18

महाकुष्ठ	7	कपाल – औदुम्बर – मण्डल – ऋष्यजिह्वा – पुण्डरीक – सिद्धम – काकण
क्षुद्रकुष्ठ	11	एककुष्ठ – चर्मकुष्ठ – किटिभ – विपादिका – अलसक – दद्रू – चर्मदल – पामा – विस्फोट – शतारु – विचर्चिका

क्र.	कुष्ठ भेद	दोष	लक्षण
1	कपाल	वात	Patches resemble broken pieces of earthen pot, blackish or reddish, dry, rough, thick
2	औदुम्बर	पित्त	Patches resemble the fruits of Udumbara, the hair on the patches turn brown
3	मण्डल	कफ	Multiple circular patches matted with each other, whitish or reddish, stable, compact
4	ऋष्यजिह्वा	वातपित्त	Patches resemble the tongue of rishya (a type of antelope with blue testicles)
5	पुण्डरीक	पित्तकफ	Patches resemble the leaf of lotus, elevated patches, white in colour with red edges
6	सिद्धम	वातकफ	Patches resemble the flower of Alabu, and when rubbed it emits small particles of skin as dust, generally located in chest region
7	काकण	त्रिदोष	Extremely painful, does not get suppured, all the VPK lakshanas may be present

8	एककुष्ठ	वातकफ	Patches resemble the scales of fish
9	चर्मकुष्ठ	वातकफ	As thick as the skin of the elephant
10	किटिभ	वातकफ	Blackish brown, rough, hard to touch
11	विपादिका	वातकफ	Cracks in palms and soles
12	अलसक	वातकफ	Nodular growth with itching and redness
13	दद्रू	पित्तकफ	Itchy circular patches with elevated edges
14	चर्मदल	पित्तकफ	Reddish, itchy, boils, painful cracks
15	पामा	पित्तकफ	Whitish, reddish or blackish multiple eruptions with severe itching
16	विस्फोट	पित्तकफ	Whitish or reddish boils
17	शतारु	पित्तकफ	Reddish brown painful burning eruptions covering large area of the skin
18	विचर्चिका	कफ	Blackish brown eruptions with itching and excessive exudation

दोषजन्य लक्षण in कुष्ठ :

क्र.	दोष	लक्षण
1	वातप्रधान कुष्ठ	रूक्षता शोष तोद शूल संकोच पारुषता खरभाव श्याव-अरुणवर्णता
2	पित्तप्रधान कुष्ठ	दाह पाक राग परिस्राव विस्त्रगन्ध क्लेद अंगपतन
3	कफप्रधान कुष्ठ	श्वेत्य (श्वेतवर्ण) शैत्य (शीतता) कण्डू स्थिरता गुरु-स्निग्ध-उत्सेध क्लेद जन्तुभिः अभिभक्ष्यं (body parts eaten by microorganisms)

साध्यासाध्यता →

साध्य → कपाल, औदुम्बर, ऋष्यजिह्वा, पुण्डरीक, सिध्म कुष्ठ, एकदोषज, वातकफज कुष्ठ

कृच्छ्रसाध्य → मण्डल कुष्ठ, द्विदोषज, पित्तज, रक्त & मांसगत कुष्ठ

असाध्य → काकणक कुष्ठ, त्रिदोषज, उपद्रवयुक्त कुष्ठ

उपद्रव →

पूयस्राव – अंगभेद – अंग अवयव पतन – ज्वर – तृष्णा – दाह – अतिसार – दौर्बल्य

चिकित्सा →

Though kustha is a tridoshaja vyadhi, it should be treated according to dosha predominant. प्रधान दोष should be treated first, then अप्रधान (अनुबन्ध) दोष should be treated.

कुष्ठ चिकित्सा सूत्र →

वातोत्तरेषु सर्पिः वमनं श्लेष्मोत्तरेषु कुष्ठेषु ।

पित्तोत्तरेषु मोक्षो रक्तस्य विरेचनं चाग्रे ॥ (च.चि.7/39)

→ In predominance of वात → घृतपान (स्नेहन)

→ In predominance of पित्त → विरेचन & रक्तमोक्षण

→ In predominance of कफ → वमन

पश्चात् पश्चाच्छर्दनान्यभ्युपेयान् मासान्मासात् संसनं स्राव्यं रक्तं वत्सरे हि द्विरल्पं नस्यं दद्याच्च त्रिरात्रान्-त्रिरात्रान् ॥ (सु.चि.9/43)

Acharya Sushruta says that वमन कर्म should be administered once in 15 days, विरेचन कर्म once in 1 month, रक्तमोक्षण once in 6 months, and अवपीडन नस्य once in 3 days.

रक्तमोक्षण in कुष्ठ →

- In महाकुष्ठ → रक्तमोक्षण with सिराव्यध
- In क्षुद्रकुष्ठ → रक्तमोक्षण with प्रच्छदान
- The patches of kushtha which are स्थिर (stable) कठिन मण्डल (hard & rounded) is subjected to प्रस्तर & नाडी स्वेद and rubbed with कूर्चशस्त्र so that the vitiated blood oozes out & get eliminated.
- The elevated patches of kushtha are fomented (स्वेदन) with hot pottalis containing meat of marshy & aquatic animals (आनुप-औदक मांस). Thereafter, the blood is eliminated by scrapping (लेखन) with sharp edge scalp (तीक्ष्ण शस्त्र).

In अल्पदोषयुक्त कुष्ठ, blood is eliminated by scratching patches and by applying horn (शृंग), gourd (अलाबु) and leech (जलौका).

शमनौषधि →

1. रसौषधि → मात्रा = 125 - 250 mg. अनुपान = मधु / घृत / गोमूत्र
तालकेश्वर रस – शुद्ध गंधक – गंधक रसायन – स्वर्णमाक्षिक भस्म – कुष्ठ कुठार रस
2. वटी / गुग्गुलु → मात्रा = 250 - 500 mg. अनुपान = ऊष्णोदक / मधु
आरोग्यवर्धिनी वटी – योगराज गुग्गुलु – पंचतित्त घृत गुग्गुलु
3. चूर्ण → मात्रा = 3 - 6 g. अनुपान = ऊष्ण जल
पंचनिम्ब चूर्ण – मंजिष्ठादि चूर्ण – त्रिफलादि चूर्ण
4. क्वाथ/आसव/अरिष्ट → मात्रा = 20 - 40 ml. अनुपान = जल
महामंजिष्ठादि क्वाथ – खदिराष्टक क्वाथ – खदिरारिष्ट – कनकबिन्दुरिष्ट – मध्वासव
5. घृत/तैल → मात्रा = 10 - 20 ml. अनुपान = ऊष्णोदक
पंचतित्त घृत – महातित्तक घृत – कनकक्षीरी तैल – करंज तैल – निम्ब तैल
6. अवलेह/पाक → मात्रा = 10 - 20 g. अनुपान = दुग्ध / उष्णोदक
अमृत भल्लातक लेह – भल्लातकावलेह – धात्र्यवलेह

बाह्य प्रयोगार्थ → यथा आवश्यकतानुसार

कुष्ठाद्य तैल – कनकक्षीरी तैल – सर्षप तैल – करंज तैल – निम्ब तैल – खदिरसार तैल – कुष्ठादि लेप – मनःशिलादि लेप – करञ्जादि लेप – करवीरादि लेप – चतुर्विध कुष्ठनाशक लेप (शिरीषत्वक लेप – कर्पासपुष्प लेप – आरग्वधपत्र लेप – काकमाची लेप) – भृष्ट सर्षप कल्क

पथ्य आहार-विहार → तिक्त लघु आहार – मुद्ग – गोधूम – यव – पुराण घृत – निम्ब हरिद्रा आमलकी खदिर – पटोल – कारवेल्लक – मधु – लघु व्यायाम – स्नान – अभ्यंग

अपथ्य आहार-विहार → नवान्न – पिष्टान्न – विरुद्धाहार – दधि – मत्स्य – तिल – माष – गुड – विदाही अभिष्यन्दी गुरु स्निग्ध आहार – दिवास्वप्न – वेगधारण

श्वित्र / किलास (Vitiligo-Leucoderma)

“किलेन श्वैत्येन असति इति किलासः” / “श्वेतेते इति श्वित्रम्” Shvitra kusta or Aruna kusta or Kilasa is characterized by abnormal white pigmentation of the skin.

निदान →

विरुद्ध आहार – अति द्रव स्निग्ध गुरु आहार सेवन – छर्दि वेगधारण – अतिव्यायाम आतपसेवन अतिभोजन – अजीर्ण भोजन – नवान्न दधि मत्स्य अतिलवण अम्ल सेवन – दिवास्वप्न – पाप कर्म etc.

पूर्वरूप →

स्पर्श अज्ञत्व – अतिस्वेद – अस्वेद – वैवर्ण्य – कण्डू – कोठ – तोद – सुप्तता – लोमहर्ष

सामान्य लक्षण →

Shvitra starts with a white patch anywhere on the body, and then it spreads gradually. White pigmentation of the skin is characteristic feature of Shvitra. It is non-infectious and non-contagious skin disease.

भेद → 1. व्रणज & 2. दोषज श्वित्र

व्रणज श्वित्र → 3 → रक्ताश्रित – मांसाश्रित – मेदाश्रित

दोषज श्वित्र → 3 → वातज – पित्तज – कफज किलास

विशेष लक्षण →

क्र	भेद	नाम	लक्षण
1	वातज (रक्ताश्रित)	दारुण	अरुण वर्ण – परुष – रोमनाशक – मण्डलाकार
2	पित्तज (मांसाश्रित)	वारुण	कमलपत्रवत् – ताम्रवर्ण – परिदाह – रोमशातन
3	कफज (मेदाश्रित)	श्वित्र	श्वेत – स्निग्ध – कण्डू – बहलगौरव – घन

साध्यासाध्यता →

साध्य लक्षण → Newly developed and colour of hair on the patches is not changed, then it is curable.

असाध्य लक्षण → Chronic (more than 1 year) and lips & finger tips are affected, then it is incurable.

श्वित्र चिकित्सा सिद्धान्त →

→ वमन – विरेचन

→ After स्नेहन → मलप्य स्वरस (काकोदुम्बर) with गुड is given for संसन → Afterwards patient should be exposed to sun (आतपसेवन)

→ बाकुची तैल – खदिर क्वाथ – मनःशिलादि लेप etc.

Shvitra is considered as the disease caused by sinful acts (पापकर्म),

therefore it should also be treated with दैवव्यपाश्रय चिकित्सा →

व्रत – सेवा – दान – धार्मिक अनुष्ठान – शिव गणेश & भास्कर पूजन etc.

शमनौषधि →

तालकेश्वर रस – श्वित्रारि रस – गंधक रसायन – स्वर्णमाक्षिक भस्म – आरोग्यवर्धिनी वटी – कैशोर गुग्गुलु – पंचनिम्ब चूर्ण – मंजिष्ठादि चूर्ण – मंजिष्ठादि क्वाथ – खदिर क्वाथ – खदिरारिष्ट – पंचतित्त घृत – कुष्ठहर तैल – कुष्ठहर लेप

शीतपित्त-उदरद-कोठ (Urticaria)

शीतपित्त → शीतमारुतसंस्पर्शात्प्रदुष्टौ कफमारुतौ ।

पित्तेन सह सम्भूय बहिरन्तर्विसर्पतः ॥ (मा.नि.50/1))

Exposure to cold, suppression of natural urges and other reason cause vitiation of Kapha and Vata. This then vitiates Pitta and then produces rash on the skin, which is called as Sheetapitta. It is characterized by वरटीदृष्ट संस्थान शोथ & तोद (pain and rashes as if bitten by black-bee).

उदरद → Vata is predominant in Sheetapitta, whereas Kapha is predominant in Udarda. There will be more pain in Sheetapitta and more itching in Udarda (characterized by कण्डूयुक्त शोथ).

कोठ → Kotha is caused by improper administration of Vamana (वमनकर्म अयोग & मिथ्यायोग). It is characterized by कण्डूयुक्त रक्त शोथ.

उत्कोठ → If there is continuous recurrence of kotha, then it is termed as 'उत्कोठ'.

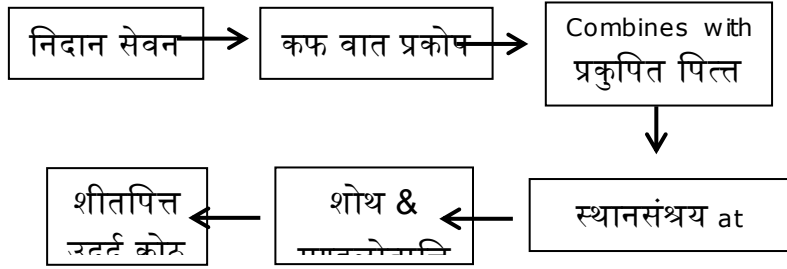
निदान →

शीतवायु स्पर्श – छर्दि वेग धारण – वमनकर्म अयोग – शीतोष्ण प्रभाव – विरुद्ध आहार – असात्म्य आहार सेवन

पूर्वरूप →

अतितृष्णा – अरुचि – हृल्लास – अंगसाद – गौरवता – रक्तलोचनता

सम्प्राप्ति →



लक्षण →

वरटीदष्ट संस्थान शोथ (शीतपित्त/उदरद) – तोदबाहुल्य (शीतपित्त) – कण्डूबाहुल्य (उदरद) – कण्डूयुक्त रक्तशोथ (कोठ) – छर्दि – ज्वर – दाह

चिकित्सा →

निदान परिवर्जन → अभ्यंग (with सर्प तैल) → स्वेदन (उष्णोदक परिषेक) → वमन (with पटोल+निम्ब+वासा) → विरेचन (with त्रिफला+गुडूची+पिप्पली / नवकार्षिक कषाय) → रक्तमोक्षण

शमनौषधि → सूतशेखर रस – प्रवाल भस्म – अमृता सत्व – आरोग्यवर्धिनी वटी – कैशोर गुग्गुलु – नवकार्षिक गुग्गुलु – त्रिकटु चूर्ण – पंचशिरीष चूर्ण – मंजिष्ठा रिष्ट
For लेप → दूर्वादिलेप – सैधवादि योग – सिद्धार्थादि योग

विसर्प (Erysipelas)

“विविधं सर्पणात् विसर्पः” Disease which spreads fast like snake-poison.

विसर्प भेद (Types)	1. वातज	2. पित्तज	3. कफज	4. सन्निपातज
	5. वात-पित्तज (आग्नेय विसर्प)	6. कफ-पित्तज (कर्दम विसर्प)	7. कफ-वातज (ग्रन्थि विसर्प)	

3 types depending on location (स्थान) →

1. आभ्यन्तर-difficult to cure 2. बाह्य-curable 3. उभय-incurable

सप्तदुष्य (विसर्पजनक सप्त धातु) →

वात – पित्त – कफ – त्वक् (रस) – रक्त – मांस – लसीका

लक्षण →

अनुन्नत शोफ – राग – दाह – ज्वर – मूर्च्छा

क्र.	विसर्प-भेद	लक्षण
1	वातज विसर्प	रूक्ष श्याम अरुण वर्ण – शोथ – अस्थिसन्धि शूल – कम्प – ज्वर
2	पित्तज विसर्प	ताम्रहरितपीतकृष्ण वर्ण – दाह – स्वेद – तृष्णा – मूर्च्छा – ज्वर
3	कफज विसर्प	श्वेतपाण्डु कृच्छ्रपाकी स्फोट – स्राव – गौरव – अरुचि – शीतज्वर
4	सन्निपातज विसर्प	सर्वलिंगव्यापी – सर्वधातुव्यापी – अचिकित्स्य
5	आग्नेय विसर्प	अग्निदग्ध समान स्फोट – कृष्ण रक्तवर्ण – वात पित्त लक्षण
6	कर्दम विसर्प	कुणपगन्धी – स्विन्नक्लिन्नपूतिमांसत्वक – कफ पित्त लक्षण
7	ग्रन्थि विसर्प	कृच्छ्रपाकी ग्रन्थिमाला – तीव्ररुजा – कफ वात लक्षण

साध्यासाध्यता →

एकदोषज → साध्य; सन्निपातज, आग्नेय & कर्दम → असाध्य

ग्रन्थि विसर्प without complications → साध्य

ग्रन्थि विसर्प with complications → असाध्य

उपद्रव →

ज्वर – अतिसार – छर्दि – तक दारण – मांस दारण – क्लम – अरुचि – अविपाक

विसर्प चिकित्सा सूत्र →

पूर्वमेव विसर्पेषु कुर्यात् लंघन रूक्षणे।

विरेक वमन आलेप सेचन असृग्विमोक्षणैः ॥

उपाचरेद् यथादोषं विसर्पानविदाहिभिः ॥ (यो.र.)

→ लंघन – रूक्षण

→ वमन – विरेचन – रक्तमोक्षण

→ आलेपन – सेचन

If सामदोष in कफस्थान → लंघन – वमन – तिक्तरसयुक्त औषध – रूक्ष शीत प्रलेप

If सामदोष in पित्तस्थान → लंघन – वमन – तिक्तरससेवन – विरेचन – रक्तमोक्षण

If सामदोष in वातस्थान → विरूक्षण चिकित्सा

ग्रन्थिविसर्पचिकित्सा → रूक्षण – लंघन – प्रदेह with पञ्चवल्कल क्वाथ – वमन विरेचन – रक्तमोक्षण with सिराव्यध or जलौका – कफवातनाशक उपचार

रक्तमोक्षण in विसर्प →

As रक्त is the site (आश्रय) of विसर्प, among all the treatments रक्तमोक्षण is the best.

शमनौषधि →

रसमाणिक्य – गंधक रसायन – संशमनी वटी – आरोग्यवर्धिनी वटी – कैशोर गुग्गुलु – चन्दनादि क्वाथ – मुस्तादि क्वाथ – किरातादि क्वाथ – मंजिष्ठाद्यरिष्ट – खदिरारिष्ट – पंचतित्त घृत – महातित्त घृत – शतधौत घृत – करञ्जादितैल – दशांग लेप

क्षुद्र रोग (Minor diseases)

Acharya Shushruta has explained 44 minor diseases →

अजगल्लिका – यवप्रख्या – अंधालजी – विवृता – कच्छपिका – वल्मीक – इंद्रवृद्धा – पनसिका – पाषाणगर्दभ – जालगर्दभ – कक्षा – विस्फोटक – अग्निरोहिणी – चिप्प – कुनख – अनुशयी – विदारिका – शर्करार्बुद – पामा – विचर्चिका – रकसा – पाददारिका – कदर – अलस – इन्द्रलुप्त – दारुणक – अरुषिका – पालित – मसूरिका – यौवनपिडका – पद्मिनीकण्टक – जतुमणि – मशक – चर्मकील – तिलकालक – न्यच्छ – व्यंग – परिवर्तिका – अवपाटिका – निरुद्धप्रकश – सन्निरुद्ध गुद – अहिपूतन – वृषणकच्छ – गुदभ्रंश

क्र	क्षुद्ररोग	लक्षण	चिकित्सा
1	अजगल्लिका	Vata-Kapha disorder occurring in children, characterized by a rash on skin which is waxy, painless papules of size of green gram.	in अपक्व अजगल्लिका → - रक्तमोक्षण (जलौका) - लेप → शुक्ति+यवक्षार+जल वासापत्र+इंद्रयवकल्क - in पक्वावस्था, treat it like व्रण
2	यवप्रख्या	Kapha-Vata disorder characterized by papules of the size of barley, hard and localized in muscles.	- स्थानीय स्वेदन - विम्लापन - लेप → मनःशिला + हरताल + कुष्ठ + देवदारु - कांचनार गुग्गुलु

क्र	क्षुद्ररोग	लक्षण	चिकित्सा
3	अन्धालजी (अन्त्रालजी)	Kapha-Vata disorder characterized by hard papule without opening and with little pus inside.	- स्थानीय स्वेदन - विम्लापन - लेप → मनःशिला + हरताल + कुष्ठ + देवदारु - गुडूची & त्रिफला
4	विवृता (Boils)	pidaka like pakwa-udumbara, round in shape having wide opening with severe burning.	- निदान परिवर्जन - पित्तज विसर्प चिकित्सा - व्रण रोपण with मधुरद्रव्य सिद्ध घृत
5	कच्छपिका	5 or 6 papules in group, painful, very hard and resembles shell of tortoise.	- स्वेदन - लेप → हरिद्रा + कुष्ठ + मनःशिला+हरताल+देवदारु - in पक्कावस्था → भेदन & व्रण रोपण
6	वल्मीक (Actino-mycosis)	Like a granthi (tumor), knotty indurated eruptions resembling ant's hill increases gradually in size. Common sites are nape of the neck, shoulder, axillae, hands, feet and joints.	- शस्त्रकर्म चिकित्सा (valmika is removed surgically and then क्षार or अग्निकर्म should be carried out) - अर्बुदवत शोधन & रोपण
7	इन्द्रवृद्धा	Vata-Pitta disorder characterized by multiple small papules resembling central part of lotus.	- निदान परिवर्जन - पित्तज विसर्प चिकित्सा

क्र	क्षुद्ररोग	लक्षण	चिकित्सा
8	पतसिका	Hard, painful papules (pidaka) appear inside ears, outside or around the ears.	- स्वेदन - लेप → हरिद्रा + कुष्ठ + मनःशिला+हरताल+देवदारु - गंधक रसायन & त्रिफला
9	पाषाणगर्दभ (Parotitis)	Swelling appears at the mandibular angle (hanu-sandhi pradesha), which is hard and slightly painful.	- निदान परिवर्जन - जलौकावचरण - लेप (मनःशिला + कुष्ठ + देवदारु) - कांचनार गुग्गुलु
10	जालगर्दभ (Cellulitis)	Thin swelling which spreads like visarpa, associated with fever and burning.	- निदान परिवर्जन - पित्तज विसर्प चिकित्सा
11	कक्षा (Herpes Zoster)	Black coloured and painfull swelling or blisters appear at the side of arms, shoulders and axillae due to Pitta-dosha.	- निदान परिवर्जन - पित्तज विसर्प चिकित्सा - दशांग लेप - व्रण रोपण with मधुर द्रव्य सिद्ध घृत
12	विस्फोटक (Bullous Eruption)	Blisters associated with burning sensation and fever. Can affect one part or all of the body.	- निदान परिवर्जन - पित्तज विसर्प चिकित्सा - रक्तमोक्षण (सिराव्यध) - तिक्त षट्फल घृत
13	अग्निरोहिणी	Blisters in axillae, associated with fever, burning sensation and feeling as if body is burning. It kills the person within 7, 10 or 15 days.	- निदान परिवर्जन - पित्तज विसर्प चिकित्सा - लंघन-रूक्षण-रक्तमोक्षण

क्र	क्षुद्ररोग	लक्षण	चिकित्सा
14	चिप्प (Whitlow)	Due to Pitta-Vata, pus develops at the base of the nails with severe pain and swelling.	- निदान परिवर्जन - रक्तमोक्षण (जलौका) - पित्त शामक चिकित्सा - सुदर्शन लेप - गुडूची
15	कुनख (कुलीन)	Due to injury, nails and the muscles around the nails becomes blackish, dry and rough. Pt. feels pain and burning sensation.	- चिप्प रोग चिकित्सा - पित्त-वात शामक चिकित्सा - छेदन & व्रण रोपण (affected nail should be removed)
16	शर्करावृद्ध (Sebacious Horn)	Aggravated Kapha-Vata and meda gets localized in mamsa, sira and snayu and produces pidaka. When this pidakas get burst and open, the exudates resemble madhu, ghrita or vasa. Later on Vata dries up the mamsa and produces cyst which is called as Sharkara. When foul smelling and blood stained secretions comes out it is called as Sharkararbuda.	- निदान परिवर्जन - मेदज अवृद्ध चिकित्सा - स्वेदन-भेदन-व्रणरोपण - कैशोर गुग्गुलु

क्र	क्षुद्ररोग	लक्षण	चिकित्सा
17	अनुशयी (Cold Abscess)	Due to Kapha prakopa, mild swelling develops slowly, without discolouration.	- कफज विद्रधि चिकित्सा - in पक्कावस्था → भेदन & व्रण रोपण
18	विदारिका	Due to tridosha-prakopa, round papules (pidaka) develops in axillae or groins resembling Vidari-kanda.	- रक्तमोक्षण (जलौका) - लेप → शिशु + देवदारु or निम्ब - in पक्कावस्था → भेदन & व्रण रोपण
19	पामा (Scabies)	Whitish, reddish or blackish multiple eruptions with severe itching.	- निदान परिवर्जन - कुष्ठहर चिकित्सा - करंज तैल अभ्यंग
20	विचर्चिका (Eczema)	Blackish brown eruptions with itching and excessive exudation (oozing).	- निदान परिवर्जन - कुष्ठहर चिकित्सा - अर्क तैल अभ्यंग
21	रकसा	Pidaka all over body with itching and without secretion is called as Rakasa.	- निदान परिवर्जन - कुष्ठ चिकित्सा
22	पाददारिका (विपादिका)	Those who walk bare footed, develop fissure \s on foot due to aggravation of Vata. There is severe pain and sometimes bleeding from these fissures.	- निदान परिवर्जन - अभ्यंग-स्वेदन-लेप - लेप → मधुच्छिष्ट + वसा + मज्जा + घृत - उपोदिकादि क्षार तैल

क्र	क्षुद्ररोग	लक्षण	चिकित्सा
23	कदर (Corn)	Due to friction or constant contact with any hard object sole of the feet, a hard corn is developed.	- निदान परिवर्जन - गोधूमज्जा लेप - शस्त्रकर्म & अग्निकर्म
24	अलस (Dhobies Itch)	When the legs remain constantly in water, exuding ulcers develop between the toes, with itching, burning sensation and pain.	- निदान परिवर्जन - प्रलेप → हरताल लेप करंजादि लेप
25	इन्द्रलुप्त (Alopecia)	Aggravated Pitta-Vata causes hair fall. Hair fall of beards is called as Indralupta, hair fall of scalp is called as Khalitya, and hair fall throughout body is called as Ruhya.	- निदान परिवर्जन - स्नेहन-स्वेदन - रक्तमोक्षण - धतूरेपत्र स्वरस लेप - भल्लातकादि लेप - भृंगराज घृत & तैल - रसायन प्रयोग
26	दारुणक (Seborrheic Dermatitis)	The skin of the scalp gets cracked, becomes dry and rough with itching. This resembles dandruff.	- निदान परिवर्जन - स्नेहन-स्वेदन - रक्तमोक्षण - निम्बू रस प्रयोग - गुंजा तैल + भृंगराज तैल
27	अरुणिका	Due to Kapha-Rakta-Krimi, small ulcers on scalp with many openings & secretions	- निदान परिवर्जन - शोधन - लेप

क्र	क्षुद्ररोग	लक्षण	चिकित्सा
28	पालित	The heat produced by anger, grief, strain and other causes of Pitta vitiation cause greying of hairs.	- निदान परिवर्जन - शोधन - नस्य with निम्ब तैल or नीलिन्यादि तैल - शिरो अभ्यंग with भृंगराज or महाभृंगराज तैल or महानील तैल - रसायन प्रयोग
29	मसूरिका (Small Pox)	Eruptions like मसूर (lentil) appear on the skin & fever is one of the major symptoms. It resembles Small pox.	- निदान परिवर्जन - कुष्ठनाशक लेप - कफज & पित्तज विसर्पवत चिकित्सा - ज्वर चिकित्सा
30	यौवनपिडका (Acne Vulgaris)	Also called as मुखदूषिका. This is disease of young people. Papules appear on face resembling शाल्मली कण्टक (the sprouts on the bark of Shalmali tree).	- शोधन चिकित्सा - रक्तमोक्षण - प्रलेप → शाल्मली कण्टक+दुग्ध धान्यक+वचा+लोध्र+कुष्ठ - अभ्यंग → कुंकुमाद्य तैल मंजिष्ठाद्य तैल
31	पद्मिनीकण्टक	Circular white patches studded with papules and itching on skin are caused by vitiation of Kapha-Vata..	- वमन - निम्ब कल्क + घृत + मधु - हरिद्रा खण्ड

क्र	क्षुद्ररोग	लक्षण	चिकित्सा
32	जटुमणि (Congenital Mole)	Also known as Lakshma. It is congenital deformity in the form of slightly elevated, painless patch on the skin.	- शस्त्रकर्म - क्षार & अग्निकर्म
33	मषक (Elevated Mole)	Painless, black, round nodules on skin resembling black-gram.	- शस्त्रकर्म - क्षार & अग्निकर्म
34	तिलकालक (Non-Elevated Moles)	Black spots resembling sesame seeds (tila).	- शस्त्रकर्म - क्षार & अग्निकर्म
35	चर्मकील (Warts)	Small nail like growths (कील) on the skin.	- शस्त्रकर्म - क्षार & अग्निकर्म
36	न्यच्छ (लाञ्छन)	These are congenital, painless, broad or small, blue or black patches present on skin as birth-sign.	- शोधन - प्रलेप → क्षीरीवृक्ष त्वक + दुग्ध
37	व्यंग	Due to anger or physical exertion aggravated Vata-Pitta gets localized in the face. This produces painless, thin, black patches on the skin of face.	- निदान परिवर्जन - शोधन (रक्तमोक्षण) - लेप → रक्तचन्दनादि लेप अर्जुनत्वक+मंजिष्ठा+मधु - अभ्यंग → कुंकुमाद्य घृत & तैल

क्र	क्षुद्ररोग	लक्षण	चिकित्सा
38	अवपाटिका (Tear of Prepuce)	Due to masturbation or forcible sexual intercourse with women having narrow vagina, the skin on the glans penis gets ulcerated.	- निदान परिवर्जन - स्वेदन with उष्ण घृत - लेप
39	परिवर्तिका (Para-Phimosis)	Due to friction or squeezing injury, the skin on the glans penis gets upturned. This causes pain if Vata is predominant and swelling if Kapha is predominant.	- निदान परिवर्जन - स्वेदन with उष्ण घृत - reposition the prepuce (foreskin). - again स्वेदन - वात नाशक बस्ति
40	निरुद्धप्रकश (Phimosis)	The skin over the glans penis gets tightened and causes obstruction to the flow of the urine. It is very painful condition.	- निदान परिवर्जन - शस्त्रकर्म - dilatation of urethra - wound-healing
41	सन्निरुद्ध गुद (Stricture of the Rectum)	By constantly indulging in the suppression of the urge of defecation, Vata gets aggravated, it blocks the rectum narrowing its lumen. This results in obstruction to the passage of faeces.	- असाध्य - निदान परिवर्जन - स्थानिक अभ्यंग with वातनाशक तैल - dilatation of rectum - wound-healing

क्र	क्षुद्ररोग	लक्षण	चिकित्सा
42	गुद भ्रंश (Rectal Prolapse)	A person who is weak and unctuous, when he strains excessively during defaecation or has a severe attack of diarrhea, in him the rectum comes out (prolapse), to be termed as guda-bramsha.	- स्नेहन & स्वेदन - After lubrication & sudation, the prolapsed part should be carefully pushed back & repositioned. - Gophana-bandha (T-bandage with hole)
43	अहिपूतन (Diaper Rash)	Due to bad hygiene in the region of anus and urinary passage in children, there develops itching. Constant itching may result in formation of pustules, which leads to ulcer formation.	- निदान परिवर्जन - धात्री शोधन - स्थानिक प्रक्षालन - maintain hygiene - keep the diaper-area clean & dry
44	वृषणकच्छ	In men, due to improper cleaning of the scrotal region, itching develops which leads to ulcer formation.	- निदान परिवर्जन - पामा & अहिपूतनवत चिकित्सा - कर्पूरादि लेप

National Health Programmes

National Malaria Control Programme (NMCP)

Malaria is a mosquito-borne infectious disease affecting humans and other animals caused by parasitic protozoa belonging to the Plasmodium type. Causative agents are P.falciparum, P.vivax, P.ovale, and P.malariae. The disease is most commonly transmitted by an infected female Anopheles mosquito.

Malaria is one of the serious public health problems in India. At the time of independence malaria was contributing 75 million case with 0.8 million deaths every year prior to the launching of National malaria control programme in 1953.

Sequence of MCP in India →

1953 : National malaria control programme (NMCP) was launched.

1958 : National malaria eradication programme (NMEP) was launched.

1971 : Urban malaria scheme (UMS) was established.

1995 : Malaria action plan (MAP) was launched.

1999 : The Govt. of India decided to drop the term (NMEP) and renamed it as national antimalarial programme (NMAP).

Strategies for prevention & control →

- Early case detection and prompt treatment.
 - Antimalarial drugs (e.g. chloroquine) and funds for training volunteers are provided by the Govt. of India.

- Integrated vector management
 - Indoor residual spraying
 - Insecticide treated bed nets
 - Use of larvivorous fish

Antimalarial Drug Policy →

- Chloroquine (25 mg/kg for 3 days) + Primaquine (0.75 mg/kg for one day), or

Artesunate (4 mg/kg for 3 days) + Pyrimethamine (1.25 mg/kg as single dose) + Primaquine (0.75 mg/kg as single dose), or

Chloroquine (25mg/kg for 3 days) + Primaquine (0.25mg/kg for 14 days)

Along with antimalarial drugs, Antibiotics, Antipyretics, Antiemetics, PPIs, multivitamins and Hepatoprotective drugs are also used.

Ayurvedic management →

- Ayush-64 (Polyherbal nontoxic antimalarial drugs developed by CCRS) have been proven antimalarial, which is given in the dose of 2 tab (500 mg) thrice a day with hot water.
- Godanti mishrana in the dose of 125 mg TID or QID is found effective.
- Sudarshana churna (3-6 gm) or Sudarshana ghanavati (500 mg) is given TID with warm water.
- Tulasi patra svarasa, Ghritakumari svarasa, Bhumyamalaki, Pippali, Musta, Parpataka, and Patola are beneficial in malaria.

National Filaria Control Programme (NFCP)

The filariasis is caused by parasitic nematodes (worms) – *Wuchereria bancrofti*, which is transmitted by the bites of infected mosquitoes (*Culex*, *Anopheles* & *Aedes*) to humans. Filariasis is characterized by lymphangitis, lymphadenitis, and elephantiasis of genitals, legs and arms.

1995 : National filaria control programme (NFCP) was launched.

Objectives of NFCP →

- Reduction of the problem in un-surveyed areas.
- Filariasis control in urban areas through anti-larval & anti parasitic measures.

Control strategy →

- Vector control through anti-larval spray / application of appropriate larvicides at weekly intervals.
- Biological control through larvivorous fishes.
- Antiparasitic measures through diagnosis & treatment of microfilaria carriers and cases.
- IEC → Information, Education & Communication to generate community awareness.

Treatment of filarial →

- DEC (diethyl carbamazine) + albendazole
or Albendazole + Ivermectin
- Doxycycline

Ayurvedic management →

The lymphatic filariasis of conventional system of medicine is very near to Shlipada of Ayurveda. Ayurvedic drugs are found effective in the prevention & cure of early cases of filariasis →

- Nityananda rasa – 2 tab (500mg each) TID for 2 weeks
- Saptaparna ghanavati – 2 tab (500mg each) TID for 2 weeks
- Ayush-55 – 2 tab (500mg each) TID for 2 weeks
- Sudarshana churna – 3 gm TID with warm water after meal
- In order to reduce swelling, drugs like Gomutra-Haritaki, Punarnavadi guggulu, Gokshuradi guggulu are used.
- Blood letting with the application of Jalauka is also found effective.

Kala Azar Control Programme

Kala-azar is a chronic disorder, which is caused by an intracellular protozoan; known as *Leishmania* species. It is transmitted to human by bite of female phlebotomus sandfly.

The classical presentation of Kala-azar → fever, splenomegaly and hepatomegaly associated with anemia, weight loss & weakness.

Post Kala-azar dermal lesions may occur after the apparent cure of systemic disease. If untreated it may be fatal.

1990-91 : Kala-azar control programme (NFCP) was launched.

2010 : It was presumed that India become free from Kala-azar by the year 2010. But in clinical set up many cases are now reported in the successive years.

Strategy for control of Kala-azar →

Interruption of transmission by reducing vector population through indoor residual insecticides.

Early diagnosis and complete treatment of Kala-azar including PK 39 rapid diagnostic kits and oral drug Miltefosine for their treatment.

Ayurvedic management →

The following ayurvedic drugs are helpful in prevention & cure of early cases of Kala-azar →

- Tamra bhasma – 60 mg BD with honey, after meal.
- Yakritaplihari Lauha – 150 mg with honey, after meal.
- Rohitakarishtha or Amritarishta 40 ml BD with equal quantity of water, after meal.
- Drugs like Godanti bhasma, Svarna gairika, Tribhuvana kirti rasa can be used alone or in combination with above drugs.

National Vector Borne Disease Control Programme (NVBDP)

NVBDP is launched in 2003-04 by merging National anti-malaria control programme, National Filaria Control Programme and Kala Azar Control programmes. Japanese B Encephalitis and Dengue / DHF have also been included in this. Program Directorate of NAMP is the nodal agency for prevention and control of major Vector Borne Diseases.

List of Vector Borne Diseases Control Programme Legislations →

- 1) National Anti - Malaria programme
- 2) Kala - Azar Control Programme

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- 3) National Filaria Control Programme
- 4) Japanese Encephalitis Control Programme
- 5) Dengue and Dengue Hemorrhagic fever

National Leprosy Eradication Programme (NLEP)

Leprosy is caused by Mycobacterium leprae. In Vedic references leprosy is mentioned as kustha roga. It has the maximum social stigma. There are many misconceptions about disease like past sins, hereditary and incurability etc. that causes social aversion & ostracism against leprosy patients leading to the high deformity. Now by the scientific inventions leprosy has been identified as infectious disease; which can be cured & eradicated.

Programme related to leprosy →

The establishment of the Indian council of British Empire leprosy relief association was in 1925, which laid the foundation of organized leprosy work in India. After Independence, this association was renamed as Hind Kushtha Nivarana Sangha in 1974. Followings are the Landmark of Leprosy eradication programme →

1955 : National Leprosy Control Programme was established →

Dapsone as monotherapy was the back bone of NLCP.

1982 : Multidrug therapy (MTP) came into use.

1983 : Govt. of India launched the National Leprosy Eradication Programme (NLEP) with the objective of eliminate leprosy as public health problem by the year 2000 AD.

Objectives of NLEP →

- Early detection and treatment
- Interrupting transmission
- Prevent deformities
- Eradicate leprosy

Treatment plan →

MDT (Rifampicin, Clofazimine, & Dapsone) is available free of cost on all working days at all state centers, PHC, Govt. dispensaries and hospitals.

Ayurvedic management → Kustha chikitsa should be adopted with assessment of dosha and dushya.

National Tuberculosis Control Programme (NTCP)

Tuberculosis is a widespread, infectious disease caused by various strains of mycobacteria, usually *Mycobacterium tuberculosis*. Generally affects the lungs, but can also affect other parts of body.

1962 : National Tuberculosis Control Programme (NTCP) was launched.

1992 : The Govt. of India, WHO and World Bank together reviewed the NTP.

1993 : Revised National Tuberculosis Control Programme (RNTCP) was introduced.

The salient features of the strategy of RNTCP are →

- Achievement of at least 85 per cent cure rate of infectious case through supervised Short Course Chemotherapy involving peripheral health functionaries;
- Detecting 70% of the estimated cases – through quality sputum microscopy.
- Involvement of NGOs; Information, Education and communication and improved operational research.

Long term objectives → To reduce tuberculosis in the community to that level when it ceases to be a public health problems

Short term objectives → To detect maximum number of TB cases and treat them effectively. To vaccinate newborn & infants with BCG.

Treatment of TB →

Directly observed treatment short-course (DOTS) is the name given to the TB control strategy recommended by the World Health Organization.

- New patients with pulmonary TB should receive a regimen containing 6 months of rifampicin → 2HRZE / 4HR.
- New patients with pulmonary TB may receive a daily intensive phase followed by three times weekly continuation phase [2HRZE / 4(HR)₃] provided that each dose is directly observed.
- TB patients returning after defaulting or relapsing from their first treatment course may receive the retreatment regimen containing first-line drugs → 2HRZES / 1HRZE / 5HRE

[H → Isoniazid, R → Rifampicin, Z → Pyrazinamide, E → Ethambutol, S → Streptomycin]

Ayurvedic management →

Tuberculosis is correlated with Rajayakshma in which 'dhatukshaya' is the main reason to initiate pathogenesis and this may lead to 'ojokshaya' i.e. reduced immune power. The drugs which improve immune power, provide nutritional pool & improve Agni are found effective for its management along with ongoing conventional anti-TB drugs.

- Kanchanara guggulu & Triphala guggulu are used to treat TB affecting the glands.
- Chyavanaprasha avaleha to improve immunity.
- Sitopaladi churna, Talisadi churna, Svarna vasant malati, Abhraka bhasma etc. are helpful in the treatment of TB.

National AIDS control programme (NACP)

1992 : First National AIDS control programme (NACP-I) was launched.

1999 : NACP-II was launched.

2004 : Anti-retroviral treatment (ART) initiated.

The Ministry of Health & Family Welfare has setup National AIDS control organization (NACO) as a separate wing to implement and closely monitor the various components of the programme.

The aim of this programme is to prevent further transmission of HIV infection and to minimize the socio-economic impact resulting from HIV infection.

National AIDS Prevention & Control Policy (NAPCP) →

1. Blood safety programme.
2. Counselling & HIV testing : with support of NACO.
3. Condom promotion : with support of NACO.
4. STD control programme.
5. HIV surveillance..
6. School AIDS education programme.
7. IEC – information , education & communication with support of NACO.

ART (Anti-retroviral treatment) →

The World Health Organization (WHO) has recommended a combination of antiretroviral drugs for people starting HIV treatment →

- TDF (tenofovir)
- either 3TC (lamivudine) or FTC (emtricitabine)
- and EFV (efavirenz).

There are many other antiretroviral drugs used in ART.

Although a cure for HIV does not yet exist, ART can keep an HIV +ve patient healthy for many years, and greatly reduces the chance of transmitting HIV to their partner(s) if taken consistently and correctly.

Ayurvedic management of AIDS →

Drugs like Chyavanaprasha, Amalaki rasayana, Ashwagandha rasayana, Shilajatu etc. are effective to boost immunity. Certain drugs like Triphala, Haridra, Tulasi etc. are potent antiviral. So these drugs can be used along with ART.

Different Systems of Medicine

Yoga

The word योग has its origin from a Sanskrit root 'युज्'. 'यु' means both to unite and to separate; to unite with real and to separate from unreal. To this when consonant 'ज' is combined it means energy or direction. Hence the meaning of Yoga is discipline or integration.

- युज्यते अनेन इति योगः ।

That which unites atma with paramatma.

- योगः चित्तवृत्ति निरोधः ।

Cessation of mental modifications due to mind, intellect & ego.

Types of Yoga →

राजयोग-हठयोग-मंत्रयोग-लययोग-ज्ञानयोग-कर्मयोग-भक्तियोग

Rajayoga → One endowed with knowledge about eternal soul (nitya) & non eternal body (anitya), conquers unhappiness produced due to Trigunas ie. satva, rajas & tamas and Tribhavas i.e. raga, dvesha & trishna by his spiritual power, philosophical power and gains victory over Chittavritti. This is called Rajayoga.

Hathayoga → With the help of gross body control of Chittavritti by the practice of Pranayama, Asana etc. by a strong person is called as Hathayoga.

Gyanayoga → The person devoid of Irshya, Dvesha, Kama, Krodha etc. gets unite with Paramatma by gaining knowledge about self ie. Atmagyana.

Karmayoga → One should always perform his duties without any expectation of fruit, it should not be happen that one does not do any work.

अष्टांग योग →

यम-नियम-आसन-प्राणायाम-प्रत्याहार-धारण-ध्यान-समाधि

1. यम → (Abstinence)

अहिंसा-सत्य-अस्तेय-ब्रह्मचर्य-अपरिग्रह

2. नियम → (Observance)

शौच-संतोष-तपस्-स्वाध्याय-ईश्वर प्रणिधान

3. आसन → (Postures): स्थिरं सुखं आसनम् ।

सूर्यनमस्कार → प्रणामासन – हस्तोत्थानासन – पादहस्तासन – अश्वसंचालनासन – पर्वतासन – अष्टांगनमस्कार – भुजंगासन – पर्वतासन – अश्वसंचालनासन – पादहस्तासन – हस्तोत्थानासन – प्रणामासन

Standing postures → अर्धकटिचक्रासन – पादहस्तासन – अर्धचक्रासन – त्रिकोणासन

Sitting postures → स्वस्तिकासन – गोमुखासन – पद्मासन – वज्रासन – भद्रासन – शशांकासन – सिंहासन – उष्ट्रासन – पश्चिमोत्तानासन – सुप्तवज्रासन – अर्धमत्स्येन्द्रासन – सिद्धासन

Supine postures → पवनमुक्तासन – सर्वांगासन – मत्स्यासन – हलासन – चक्रासन – शवासन – सेतुबंधासन – शीर्षासन

Prone postures → भुजंगासन – शलभासन – धनुरासन – मकरासन – मयूरासन

4. प्राणायाम → (Breathing exercise): प्राणस्य आयामः प्राणायामः ।

Stages: रेचक – पूरक – कुम्भक (आभ्यन्तर & बाह्य कुम्भक)

Types of Pranayama: सूर्यभेदन – उज्जयी – सीत्कारी – शीतली – भस्त्रिक – भ्रामरी – मूर्च्छा – प्लावनी

5. प्रत्याहार → (Control over sensory faculties)

6. धारण → (Concentration)

7. ध्यान → (unbroken flow of awareness/ meditation)

a. सगुण ध्यान

b. निर्गुण ध्यान

8. समाधि → (oneness with the mediated object)

a. सम्प्रज्ञात समाधि

b. असम्प्रज्ञात समाधि

षट्कर्म →

Regular practice of Shatkarma leads to purification of the body.

1. धौति: help in clearing the impurities of the body

- अन्तर्धौति – वात सार, वारि सार, वह्नि सार, बहिष्कृत
- दंतधौति – दंत मूल, जिह्वा मूल, कर्णरंध्र, कपाल रंध्र
- हृदधौति – वस्त्र धौति, दण्ड धौति, वमन धौति
- मूलशोधन

2. बस्ति: जलबस्ति – स्थलबस्ति

3. नेति: सुत्रनेति – जलनेति

4. नौलि: Moving stomach & intestines

5. त्राटक: continuously looking at some small object till eye tears.

6. कपालभाती: वामकर्म – व्युत्कर्म – शीत्कर्म

Yoga therapy →

Yoga therapy is the application of Yogic principles to a particular person with the objective of achieving a particular spiritual, psychological, or physiological goal.

The application of Yoga therapy is from one or more of three perspectives:

1. The use of Yoga to gain a sense of power, i.e., to develop muscular power, the power to concentrate, the power to do difficult postures, the ability to work over and extended period of time, etc. This is called the application of shakti-krama.
2. The use of Yoga to heal specific problems, such as eliminating impurities in the organs (doshas) or energy centers (cakras) and channels (nâdīs) of the body. This is chikitsâ-krama.
 - If sickness is present, it needs to be cured (chikitsâ).
 - If sickness is not present, protection is necessary (rakshana).
 - If sickness is not present and one has learned how to protect oneself, training is necessary (shikshana).
3. The use of Yoga to go beyond the physical to understand what is beyond the limited sense of self; to know one's true self as unchanging Witnessing Presence (Purusha) of all that is changing (prakriti). This is called the application of âdhyâtmika-krama.

Naturopathy

Naturopathy refers to treating the diseases using natural substances like sunlight, water, soil, air, natural food and herbs, and exercise.

- The healing power of nature → Vismedicatrix naturae.
- Body has vital capacity, which can cure diseases by nature.

Use of water for treatment of diseases → Hydrotherapy

Use of sunlight for treatment of diseases → Heliotherapy

Use of soil / mud for treatment of diseases → Mud therapy

Use of food for treatment of diseases → Diet therapy

Use of fasting for treatment of diseases → Fasting therapy

Hydrotherapy

Importance of water → It is universal solvent – inexpensively available in large quantity – non toxic – soothing – it promptly absorbs and holds heat etc.

Principles of Hydrotherapy →

- Local use of warm water causes relaxation of muscular tissue and dilatation of blood vessels thus enhancing the local blood circulation.
- Local use of cold water causes contraction of blood vessels etc.

Usage of water in hydrotherapy →

Internal → drinking – enema – steam inhalation

External → bathing – compression – pack

Bathing →

On the basis of temperature → hot – cold – alternate - neutral

On the basis of body part → whole body – localized

Localized → Sitz bath – spinal bath – head bath – foot bath etc.

Steam bath → full body steam bath – steam inhalation – immersion bath – sauna bath

Whirlpool bath → bath in moving (current) of hot water

Heliotherapy

Importance of Sunlight → it is prime source of energy – Most vital element required for all living beings – Vitamin D is synthesized by skin in presence of sun light etc.

Types of Sun-bath → Morning sunbath – Mild sunbath – Waking sunbath – Trunk pack sunbath – Whole pack sunbath – Kuhne's sunbath

Mud therapy

Mud pack / mud mask – mud bath – internal use

Yellow mud → contains sulphure, beneficial for skin conditions

Black mud → beneficial for poisoning

White mud → beneficial for neuro, hepatic diseases

Red mud → fatigue – pain – rheumatic conditions

Sandy mud → snake bite – gastroenteritis

Unani

The word Unani means Greek. In early Islamic period the Muslims took over the medical system of Greece.

Principles of Unani medicine →

1. Arkan (Elements) → The human body contains four elements → Air, Earth, Fire, Water.
2. Mizaj (Temperament) → In the Unani system, the temperament of the individual is very important as it is considered to be unique. The individual's temperament is believed to be the result of the interaction of the elements. The temperament can be real equitable where the four elements used are in equal quantities.
3. Akhlat (Humours) → They are 4 → blood (dam or khun), phlegm (balgam), black bile (sawada), and yellow bile (saffra).
4. Aza (Organs) → These are the various organs of the human body. The health or disease of each individual organ affects the state of health of the whole body.
5. Arwah (Spirits) → Ruh (Spirit) is a gaseous substance, obtained from the inspired air, it helps in all the metabolic activities of the body. It burns the akhlat latifah to produce all kinds of quwa (powers) and hararat ghariziyah, it is the source of vitality for all the organs of the body. These are considered to be the life force and are, therefore, important in the diagnosis and treatment of disease.

6. Quwa (Faculties) → These are three kinds of power → 1. Quwa Tabiyah or Natural power, 2. Quwa Nafsaniyah or Psychic power. 3. Quwa Haywaniyah or Vital power.
7. Afaal (Functions) → Quwa Haywaniyah or Vital power.

Prevention of diseases →

For preventing the diseases and promotion of health, 'sat jaruriye' (six important things) have been explained →

1. Air
2. Food and drinks
3. Bodily movement and repose
4. Psychic movement and repose
5. Sleep and wakefulness
6. Evacuation and retention

Treatment →

- This should be given taking into consideration the strength of the digestive fire. Herbs and foods used should be of the opposite qualities and when dosha (akhlat) are aggravated they should be taken out of the body by purificatory procedures.
- After diagnosing the disease, Usoole Ilaj (principle of management) of disease is determined on the basis of etiology in the following pattern:
 - Izalae Sabab (elimination of cause)
 - Tadeele Akhlat (normalization of humors)
 - Tadeele Aza (normalization of tissues/organs)

Siddha

Siddha medicine is a system of traditional medicine originating in ancient Tamilakam (Tamil Nadu) in South India.

Traditionally, it is taught that the **siddhas** laid the foundation for this system of medication. Siddhars were spiritual adepts who possessed the **ashta siddhis**, or the eight supernatural powers. **Agastya** is considered the first siddha and the guru of all siddhars; the siddha system is believed to have been handed over to him by **Murugan**, son of Shiva and Parvati.

प्रयोजन of सिद्ध →

1. देह सिद्धि → With the help of पारद, strengthening the body by avoiding जरा, रोग & मृत्यु and making it immortal.
2. लौह सिद्धि → With the help of metals and chemicals producing superior metals, also called धातुवाद.

The drugs used by the Siddhars could be classified into three groups: thavaram (herbal product), thadhu (inorganic substances) and jangamam (animal products).

The treatment in siddha medicine is aimed at keeping the three humors in equilibrium and maintenance of seven elements. So proper diet, medicine and disciplined regimens of life are advised for a healthy living, and to restore equilibrium of humors in diseased condition. Treatment is classified into 3 categories: devammaruthuvum (Divine method); manuda maruthuvum (rational method); and asura maruthuvum (surgical method).

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Physiotherapy

Physiotherapy is a system of medicine for complete health care of all age groups in modern age. Physiotherapy is a treatment measure of physical and electrical means to accelerate the patient's recovery from injuries and diseases that hazards the normal lifestyle. Physical therapy is used to improve a patient's quality of life through examination, diagnosis, prognosis, physical intervention, and patient education. It is performed by physiotherapists.

Acc. to WHO "Physiotherapists assess, plan and implement rehabilitative programs that improve or restore human motor functions, maximize movement ability, relieve pain syndromes, and treat or prevent physical challenges associated with injuries, diseases and other impairments".

Modes of physiotherapy →

- Exercise therapy
- Massage
- Electrotherapy
- Electromagnetic therapy
- Actinotheraoy
- Heliotherapy
- Infrared therapy
- Ultrasonic therapy
- Wax bath therapy etc.

Rehabilitation

Rehabilitation is defined as "a set of measures that assist individuals, who experience or are likely to experience disability, to achieve and maintain optimum functioning in interaction with their environments".

Rehabilitation refers to →

- **Rehabilitation (neuropsychology)**, therapy to regain or improve neurocognitive function that has been lost or diminished.
- **Rehabilitation (wildlife)**, treatment of injured wildlife so they can be returned to the wild.
- **Drug rehabilitation**, medical or psychotherapeutic treatment for dependency on substances such as alcohol and drugs.
- **Occupational rehabilitation**, therapy to return injured workers to an appropriate level of work activity.
- **Physical medicine and rehabilitation**, a branch of medicine that aims to enhance and restore functional ability to those with physical impairments or disabilities.
- **Physical rehabilitation**, physical therapy using mechanical force and movement.
- **Psychiatric rehabilitation**, a branch of psychiatry dealing with restoration of mental health and life skills after mental illness.
- **Vision rehabilitation**, rehabilitation to improve vision or low vision
- **Vocational rehabilitation**, process which enables persons with impairments or disabilities to maintain or return to employment or occupation.

Diseases of Endocrine gland

Endocrine system is group of ductless glands that regulate body processes by secreting chemical substances called hormones.

Endocrine disorders may be subdivided into three groups →

- Endocrine gland hyposecretion (leading to hormone deficiency)
- Endocrine gland hypersecretion (leading to hormone excess)
- Tumours (benign or malignant) of endocrine glands

Endocrine disorders are often quite complex, involving a mixed picture of hyposecretion and hypersecretion because of the feedback mechanisms involved in the endocrine system. For example, most forms of hyperthyroidism are associated with an excess of thyroid hormone and a low level of thyroid stimulating hormone.

Thyroid Gland

The thyroid gland is an endocrine gland in the neck, consisting of two lobes connected by an isthmus. It is found at the front of the neck, below the Adam's apple. The thyroid gland secretes thyroid hormones, which primarily influence the metabolic rate and protein synthesis. The hormones also have many other effects including those on development. The thyroid hormones **triiodothyronine (T3)** and **thyroxine (T4)** are created from iodine and tyrosine. The thyroid also produces the hormone **calcitonin**, which plays a role in calcium homeostasis.

Hyperthyroidism (Thyrotoxicosis) →

Hyperthyroidism is the condition that occurs due to excessive production of thyroid hormone by the thyroid gland.

Thyrotoxicosis is the condition that occurs due to excessive thyroid hormone of any cause and therefore includes hyperthyroidism.

Signs and symptoms vary between people and may include irritability, muscle weakness, sleeping problems, a fast or irregular heartbeat, heat-intolerance, diarrhea, enlargement of the thyroid, and weight loss.

Causes of thyrotoxicosis →

- Graves' disease
- Multinodular goiter
- Toxic adenoma
- Inflammation of the thyroid
- Eating too much iodine and too much synthetic thyroid hormone
- A less common cause is a pituitary adenoma.

Investigations →

Typically blood tests show a low thyroid stimulating hormone (TSH) and raised T3 or T4. Radioiodine uptake by the thyroid, thyroid scan, and TSI antibodies may help determine the cause.

Treatment of thyrotoxicosis →

Antithyroid drugs → Carbimazole / Methimazole / Propylthiouracil

β-Blockers → Propranolol, Labetalol etc.

Surgery → Partial or complete thyroidectomy

Thyroid storm →

Thyroid storm presents with extreme symptoms of hyperthyroidism. It is treated aggressively with resuscitation measures along with a combination of the above modalities including: an intravenous beta blockers such as propranolol, followed by a thioamide such as methimazole, an iodinated radiocontrast agent or an iodine solution if the radiocontrast agent is not available, and an intravenous steroid such as hydrocortisone.

Graves' disease →

Graves' disease, also known as toxic diffuse goiter, is an autoimmune disease that affects the thyroid. It frequently results in and is the most common cause of hyperthyroidism.

Symptoms of the resultant hyperthyroidism are mainly insomnia, hand tremor, hyperactivity, hair loss, excessive sweating, itching, heat-intolerance, weight loss despite increased appetite, diarrhea, frequent defecation, palpitations, periodic partial muscle weakness or paralysis in those especially of oriental descent, and skin warmth and moistness. Further signs that may be seen on physical examination are most commonly a diffusely enlarged (usually symmetric), nontender thyroid, lid lag, excessive lacrimation due to Graves' ophthalmopathy, arrhythmias of the heart, such as sinus tachycardia, atrial fibrillation, and premature ventricular contractions, and hypertension. People with hyperthyroidism may experience behavioral and personality changes, including: psychosis, mania, anxiety, agitation, and depression.

Goiter →

A goiter is a swelling in the neck resulting from an enlarged thyroid gland. The term is also used to describe an enlarged thyroid.

Cause →

- Iodine deficiency (most common cause)
- Congenital hypothyroidism
- Pituitary disease (hypersecretion of TSH)
- Graves' disease (hyperthyroidism)
- Acute or chronic inflammation (thyroiditis)
- Thyroid cancer (malignant tumor)

Types of Goiter →

1. Simple goiter (iodine-deficiency / puberty goiter / multinodular goiter)
2. Toxic goiter (Graves disease / secondary thyrotoxicosis)
3. Neoplastic goiter (benign adenoma / malignant tumor)
4. Thyroiditis (acute bacterial / autoimmune thyroiditis)

Clinical Features →

A goiter, associated with hypothyroidism or hyperthyroidism, may be present with symptoms of the underlying disorder.

Treatment →

- Treat the cause
(iodine supplement / levothyroxine etc. in hypothyroidism, and antithyroid drugs carbimazole / Methimazole etc. in hyperthyroidism, antibiotics & antiinflammatory drugs in thyroiditis)
- In severe cases or neoplasm, a partial or complete thyroidectomy

Hypothyroidism →

Hypothyroidism, also called underactive thyroid or low thyroid, is a disorder of the endocrine system in which the thyroid gland does not produce enough thyroid hormone.

Causes → Worldwide, too little iodine in the diet is the most common cause of hypothyroidism. In countries with enough iodine in the diet, the most common cause of hypothyroidism is the autoimmune condition Hashimoto's thyroiditis. Less common causes include: previous treatment with radioactive iodine, injury to the hypothalamus or the anterior pituitary gland, certain medications, a lack of a functioning thyroid at birth, or previous thyroid surgery.

Symptoms → Poor ability to tolerate cold, a feeling of tiredness, constipation, depression, and weight gain. Occasionally there may be swelling of the front part of the neck due to goitre. Untreated hypothyroidism during pregnancy can lead to delays in growth and intellectual development in the baby or cretinism.

The diagnosis of hypothyroidism, when suspected, can be confirmed with blood tests measuring thyroid-stimulating hormone (TSH) and thyroxine levels.

Prevention & Treatment → Salt iodization has prevented hypothyroidism in many populations. Hypothyroidism can be treated with levothyroxine. The dose is adjusted according to symptoms and normalization of the thyroxine and TSH levels.

Myxedema →

Myxedema or myxoedema is a term used synonymously with severe hypothyroidism. However, the term is also used to describe a dermatological change that can occur in hypothyroidism and some forms of hyperthyroidism.

Myxedema is a condition marked by thickening and swelling of the skin caused by insufficient production of thyroid hormones by the thyroid gland. Myxedema is also associated with other symptoms of underactive thyroid like poor ability to tolerate cold, a feeling of tiredness, constipation, depression, and weight gain.

Myxedema coma is a loss of brain function as a result of severe, longstanding low level of thyroid hormone in the blood (hypothyroidism). Myxedema coma is rare but often fatal.

Treatment of myxoedema →

Primary treatment is prompted by the administration of adequate doses of either the thyroid hormone levo-thyroxine given intravenously or by giving L-triiodothyronine via a nasogastric tube. It is essential to identify and treat the condition precipitating the coma.

Cretinism →

Congenital iodine deficiency syndrome, previously known as Cretinism, is a condition of severely stunted physical and mental growth owing to untreated congenital deficiency of thyroid hormone (congenital hypothyroidism) usually owing to maternal hypothyroidism.

Parathyroid Gland

Parathyroid glands are four small glands of the endocrine system which regulate the calcium level in the blood. Parathyroid glands are located in the neck behind the thyroid.

Parathyroid glands regulate the calcium by producing a hormone called Parathyroid Hormone (PTH).

Hyperparathyroidism →

Hyperparathyroidism is an increased parathyroid hormone (PTH) levels in the blood.[4] This occurs either from the parathyroid glands inappropriately making too much PTH (primary hyperparathyroidism) or other events triggering increased production by the parathyroid glands (secondary hyperparathyroidism).

Most people with primary disease have no symptoms at the time of diagnosis. In those with symptoms the most common is kidney stones with other potential symptoms including weakness, depression, bone pains, confusion, and increased urination. Both types increase the risk of weak bones.

Primary hyperparathyroidism in 80% of cases is due to a single benign tumor known as a parathyroid adenoma with most of the rest of the cases due to a multiple benign tumors.

Secondary hyperparathyroidism typically occurs due to vitamin D deficiency, chronic kidney disease, or other causes of low blood calcium.

Treatment of hyperparathyroidism →

Primary hyperparathyroidism can be cure by surgical removal of the parathyroid tumor (parathyroid adenoma).

The treatment of secondary hyperparathyroidism consists of a low phosphorus diet, phosphate binders (Calcium salts, Sevelamer Hydrochloride etc.), vitamin D derivatives, calcimimetics (cinacalcet), and even parathyroidectomy.

Hypoparathyroidism →

Hypoparathyroidism is the state of decreased secretion or activity of parathyroid hormone (PTH). This leads to decreased blood levels of calcium (hypocalcemia) and increased levels of blood phosphorus (hyperphosphatemia).

Low levels of calcium in the blood, often causes cramping and twitching of muscles or tetany (involuntary muscle contraction), and other symptoms like fatigue, headaches, bone pain and insomnia. Crampy abdominal pain may occur.

Treatment of hypoparathyroidism →

- Precautions are taken to prevent seizures or larynx spasms.
- In severe hypocalcemia, IV Calcium (e.g. as Calcium gluconate).
- Injections of recombinant human parathyroid hormone.
- Long-term treatment of hypoparathyroidism is with vitamin D analogs and calcium supplementation.

Pituitary Gland

The pituitary gland, or hypophysis, is an endocrine gland about the size of a pea and weighing 0.5 grams in humans. It is a protrusion off the bottom of the hypothalamus. The hypophysis rests upon the hypophysial fossa of the sphenoid bone in the center of the middle cranial fossa and is surrounded by a small bony cavity (sella turcica).

The anterior pituitary synthesizes and secretes→

- Growth hormone (GH)
- Thyroid stimulating hormone (TSH)
- Adrenocorticotropin (ACTH)
- gonadotropic hormone i.e., both luteinizing hormone (LH) and follicle stimulating hormone (FSH)
- Prolactin

The posterior pituitary stores and secretes (but does not synthesize) the following important endocrine hormones: →

- Antidiuretic hormone (ADH) / also known as Vasopressin
- Oxytocin

Hyperpituitarism →

Hyperpituitarism is a condition due to the primary hypersecretion of pituitary hormones, it typically results from a pituitary adenoma.

Symptoms include hirsutism, visual field loss or double vision, Excessive sweating, lethargy, headache, muscle weakness etc.

The four most common types of hyperpituitarism are prolactinoma, corticotropinoma (Cushing's disease), somatotropinoma (gigantism), and thyrotropinoma.

Gigantism → also known as giantism, is a condition characterized by excessive growth and height significantly above average. In humans, this condition is caused by over-production of growth hormone.

"Gigantism refers to abnormally high linear growth (see the image below) due to excessive action of insulinlike growth factor I (IGF-I) while the epiphyseal growth plates are open during childhood. Acromegaly is the same disorder of IGF-I excess but occurs after the growth plate cartilage fuses in adulthood.

Treatment with Pegvisomant is beneficial for the pediatric gigantism patients.

Thyrotropinoma → A pituitary tumor that secretes thyroid-stimulating hormone (TSH) is called a thyrotropinoma (or sometimes a TSHoma). An excess of amount of thyroid hormone causes hyperthyroidism.

The primary treatment for a thyrotropinoma is surgery to remove the tumor.

Cushing's disease → Cushing's disease is a cause of Cushing's syndrome characterised by increased secretion of adrenocorticotrophic hormone (ACTH) from the anterior pituitary (secondary hypercortisolism).

Treatment → Surgical resection of ACTH-secreting pituitary adenoma & Radiation therapy.

Prolactinoma → A prolactinoma is a benign tumor (adenoma) of the pituitary gland that produces a hormone called prolactin. It is the most common type of functioning pituitary tumor.[1] Symptoms of prolactinoma are too much prolactin in the blood (hyperprolactinemia), or those caused by pressure of the tumor on surrounding tissues.

Treatment of Hyperpituitarism →

- Treatment (for hyperpituitarism) in the case of prolactinoma consists of long-term medical management. Dopamine agonists are strong suppressors of PRL secretion and establish normal gonadal function.
- Treatment for gigantism begins with establishing target goals for IGF-1, trans-sphenoidal surgery (somatostatin receptor ligands-preoperatively) and postoperative imaging assessment.
- For Cushing's disease there is surgery to extract the tumor; after surgery, the gland may slowly start to work again, though not always.

Hypopituitarism →

Hypopituitarism is the decreased (hypo) secretion of one or more of the eight hormones normally produced by the pituitary gland.

If there is decreased secretion of one specific pituitary hormone, the condition is known as selective hypopituitarism. If there is decreased secretion of most or all pituitary hormones, the term panhypopituitarism is used.

Symptoms → Growth hormone (GH) deficiency leads to a decrease in muscle mass, central obesity and impaired attention and memory. Children experience growth retardation and short stature.

Thyroid-stimulating hormone (TSH) deficiency leads to hypothyroidism. Typical symptoms are tiredness, intolerance to cold, constipation, weight gain, hair loss and slowed thinking, as well as a slowed heart rate and low blood pressure. In children, hypothyroidism leads to delayed growth and in extreme inborn forms to a syndrome called cretinism.

Adrenocorticotrophic hormone (ACTH) deficiency leads to adrenal insufficiency, a lack of production of glucocorticoids such as cortisol by the adrenal gland. If the problem is chronic, symptoms consist of fatigue, weight loss, failure to thrive (in children), delayed puberty (in adolescents), hypoglycemia, anemia and hyponatremia. If the onset is abrupt, collapse, shock and vomiting may occur. ACTH deficiency is highly similar to primary Addison's disease.

Deficiency of luteinizing hormone (LH) and follicle-stimulating hormone (FSH), leads to different symptoms in men and women. Women experience oligo- or amenorrhea and infertility. Men lose facial, scrotal and trunk hair, as well as suffering decreased muscle mass and anemia. Both sexes may experience a decrease in libido and loss of sexual function, and have an increased risk of osteoporosis.

Prolactin (PRL) plays a role in breastfeeding, and inability to breastfeed may point at abnormally low prolactin levels.

Antidiuretic hormone (ADH) deficiency leads to diabetes insipidus (DI).

Oxytocin (OXT) deficiency generally causes few symptoms, as it is only required at the time of childbirth and breastfeeding.

Treatment of Hypopituitarism →

- Hormone replacement therapy →
 - Growth Hormone (GH)
 - Levothyroxine for TSH-deficiency
 - Corticosteroids (usually hydrocortisone or prednisone) for ACTH-deficiency
 - Sex hormones (testosterone in men and estrogen or a combination of estrogen and progesterone in women) for Gonadotropin deficiency
 - Desmopressin for ADH-deficiency

Diabetes insipidus →

Excessive urination and extreme thirst as a result of inadequate secretion of the pituitary hormone ADH (Vasopressin) or lack of the normal response by the kidney to ADH, is called diabetes insipidus (DI). Signs of dehydration may also appear in some individuals, since the body cannot conserve much (if any) of the water it takes in.

Types → Central DI, Gestational DI, Nephrogenic DI etc.

Treatment involves drinking sufficient fluids to prevent dehydration. Other treatments depend on the type.

In central and gestational DI treatment is with desmopressin.

Nephrogenic DI may be treated by addressing the underlying cause or the use of a thiazide, aspirin, or ibuprofen.

Pancreas

The pancreas is a glandular organ of the digestive system and endocrine system. It secretes pancreatic enzymes (amylase, lipase, protease) as exocrine gland, and pancreatic hormones (insulin, glucagon, somatostatin, pancreatic polypeptide) as an endocrine gland.

Diabetes mellitus →

Diabetes mellitus is metabolic clinical syndrome characterized by hyperglycemia due to absolute or relative deficiency of insulin and by long term complications involving the eyes, kidneys, nerves and blood vessels.

Types of DM →

IDDM (Type-1 DM) → Insulin Dependent Diabetes Mellitus

NDDM (Type-2 DM) → Non Insulin Dependent Diabetes Mellitus

MRDM → Malnutrition Related Diabetes Mellitus

MODY → Maturity Onset Diabetes of Young (Juvenile DM)

Clinical Features →

- Classic hyperglycemic triad → Polyuria – Polydipsia – Polyphagia
- Weight loss
- Tiredness / debility
- Blurred vision
- Genital itching
- Delayed wound-healing

- Symptoms may develop rapidly (weeks or months) in type 1 DM, while they usually develop much more slowly and may be subtle or absent in type 2 DM.

Complications →

- Diabetic ketoacidosis (DKA)
- Cardiovascular disease (angina, atherosclerosis etc.)
- Diabetic neuropathy (Nerve damage)
- Diabetic nephropathy (Kidney damage)
- Diabetic retinopathy (Eye damage)
- Diabetic foot ulcers
- Skin lesions etc.

Investigations →

- Fasting blood glucose level (FBS) > 126 mg/dL
(FBS less than 100 mg/dL is normal. FBS from 100 to 125 mg/dL is considered prediabetes. If it's 126 mg/dL or higher on two separate tests, diabetes is confirmed.)
- Post-prandial blood glucose level (PPBS) > 200 mg/dL
- Random blood glucose level (RBS) > 200 mg/dL & symptoms
- Urinary glycosuria – present

Treatment →

- Diet control
- Daily exercise & Life style modifications
- Medications – Insulin / metformin / sulphonylureas
- Mass Education / Counselling to the patient

Adrenal Glands

The adrenal glands (also known as suprarenal glands) are endocrine glands that produce a variety of hormones including adrenaline and the steroids aldosterone and cortisol. They are found above the kidneys. Each gland has an outer cortex which produces steroid hormones, and an inner medulla which produces epinephrine and norepinephrine (adrenaline and noradrenaline).

Hyperadrenalism → is a condition in which the adrenal glands are secreting excessive amounts of the hormones that they produce. This may involve an overproduction of androgen hormones, corticosteroids or aldosterone. Excessive production of each of these hormones results in different symptoms that may include facial hair growth, high blood pressure, muscle weakness, hair loss, weight gain and more.

Cushing's syndrome →

Cushing's syndrome involves an excess amount of cortisol that may develop as a result of a tumor, genetic condition or as a side effect of certain medications.

Signs and symptoms may include high blood pressure, abdominal obesity but with thin arms and legs, reddish stretch marks, a round red face, a fat lump between the shoulders, weak muscles, weak bones, acne, and fragile skin that heals poorly. Women may have more hair and irregular menstruation. Occasionally there may be changes in mood, headaches, and a chronic feeling of tiredness.

Treatment of Cushing's syndrome →

- If due to medications, these can often be slowly stopped.
- If caused by a tumor, it may be treated by a combination of surgery, chemotherapy, and/or radiation.
- Medications to control excessive production of cortisol at the adrenal gland include ketoconazole, mitotane and metyrapone.
- Mifepristone is approved for people with Cushing syndrome who have type 2 diabetes or glucose intolerance.

Hypoadrenalism →

Hypoadrenalism is a condition caused by underactive adrenal glands. Some common symptoms of hypoadrenalism are fatigue, dizziness, muscle weakness, nausea and loss of appetite.

Addison's disease →

Addison's disease, also known as primary adrenal insufficiency and hypocortisolism, is a long-term endocrine disorder in which the adrenal glands do not produce enough steroid hormones.

Symptoms generally include abdominal pain, weakness, and weight loss. Darkening of the skin in certain areas may also occur. Under certain circumstances, an adrenal crisis may occur with low blood pressure, vomiting, lower back pain, and loss of consciousness.

Treatment for Addison's disease involves replacing the missing cortisol, sometimes in the form of hydrocortisone tablets, or prednisone tablets in a dosing regimen that mimics the physiological concentrations of cortisol.

Immunodeficiency Disorders

Immunodeficiency disorders result in a full or partial impairment of the immune system. Primary immunodeficiencies are the result of genetic defects, and secondary immunodeficiencies are caused by environmental factors, such as HIV/AIDS or malnutrition.

There are two types of immunodeficiency disorders →

1. Primary immunodeficiency (PID) → inherited immune disorders resulting from genetic mutations, usually present at birth and diagnosed in childhood.

Primary immunodeficiencies are classified by the main component of the immune system that is deficient, absent, or defective →

- Humoral immunity deficiencies
- Cellular immunity deficiencies
- Combined humoral and cellular immunity deficiencies
- Phagocytic cells deficiencies
- Complement proteins deficiencies

2. Secondary immunodeficiency (SID) → acquired immunodeficiency as a result of disease or environmental factors, such as HIV, malnutrition, or medical treatment (e.g. chemotherapy).

Causes include →

- Systemic disorders (eg, diabetes, undernutrition, HIV infection)
- Immunosuppressive treatments (eg, cytotoxic chemotherapy, bone marrow ablation before transplantation, radiation therapy)
- Prolonged serious illness

→ Secondary immunodeficiency also occurs among critically ill, older, or hospitalized patients. Prolonged serious illness may impair immune responses; impairment is often reversible if the underlying illness resolves.

Signs and symptoms of immunodeficiency disorders can include →

- Frequent and recurrent viral and bacterial infections such as pneumonia, bronchitis, sinus infections, ear infections, meningitis or skin infections etc.
- Inflammation and infection of internal organs.
- Blood disorders, such as low platelet counts or anemia.
- Some can develop autoimmune disorders.

Treatment →

- Prevention of infection
- Healthy diet
- Hygiene
- Treatment for immunodeficiency disorders commonly includes antibiotics and immunoglobulin therapy. Other antiviral drugs, amantadine and acyclovir, or a drug called interferon are used for treatment of the viral infections caused by immunodeficiency disorders.

AIDS (acquired immunodeficiency syndrome) is a syndrome caused by a virus called HIV (human immunodeficiency virus). The disease alters the immune system, making people much more vulnerable to infections and diseases. This susceptibility worsens if the syndrome progresses.

Etiology of AIDS →

HIV is a retrovirus that infects and replicates primarily in human CD4+ T cells and macrophages. HIV can be transmitted via blood, blood products, sexual fluids, other fluids containing blood, and breast milk. Most individuals are infected with HIV through sexual contact, before birth or during delivery, during breast-feeding, or when sharing contaminated needles and syringes (intravenous drug users).

Clinical features of AIDS →

The early signs of HIV may appear as symptoms similar to those caused by the flu. Symptoms include frequent and recurrent infections, sore throat, mouth ulcers, fever, fatigue, pneumonia, swollen lymph nodes, severe weight loss etc.

Investigations →

- ELISA test
- Tri-dot test
- Less than 200 CD4+ T cells per cubic millimeter of blood

Prevention of HIV / AIDS →

- Safe sex (e.g. using condom during intercourse)
- Using disposable gloves when at risk of direct contact with blood
- Using disposable syringes and blades
- Avoiding sharing syringes and blades
- Taking optimum care during pregnancy like an HIV woman should take ART properly, caesarean section is preferred and breastfeeding is avoided.

Autoimmune disorders

Autoimmune disorders are caused by an overactive immune system. It is a disease in which body's immune system attacks its own tissue mistakenly.

The cause is generally unknown. Some autoimmune diseases such as SLE run in families, and certain cases may be triggered by infections or other environmental factors.

There are more than 80 different autoimmune diseases. Most of them are long-term illnesses, with the severity of symptoms changing over time. Some of the more common autoimmune diseases include →

- Rheumatoid arthritis (RA)
- Systemic lupus erythematosus (SLE)
- Multiple sclerosis (MS)
- Psoriasis
- Grave's diseases
- Inflammatory bowel disease (IBD)
- Diabetes mellitus type 1
- Celiac disease

Treatment depends on the type and severity of the condition. Nonsteroidal anti-inflammatory drugs (NSAIDs) and immunosuppressants are often used. Intravenous immunoglobulin may also occasionally be used. While treatment usually improves symptoms they do not typically cure the disease.

Management of Emergency conditions

Acute Haemorrhage

Hemorrhage or bleeding is an escape of blood from ruptured blood vessel. Hemorrhage can refer to blood loss inside the body, called internal bleeding, or it can refer to blood loss outside of the body, called external bleeding.

Causes →

- Trauma / injury
- Bleeding peptic ulcer
- Bleeding esophageal varices
- Low platelet count (thrombocytopenia)

Symptoms →

- Tachycardia and hypotension
- Cold and clammy skin
- Pallor and restlessness
- Hypovolemic shock

Treatment →

1. Stop bleeding by →
 - Pressure & bandaging / suturing of the wound
 - Elevation of the affected part
2. Maintain body fluid & blood volume by →
 - IV fluids & electrolytes
 - Blood transfusion, if needed
3. Treat the cause

Rabin Singh

Hypertensive Emergencies

Hypertensive crisis is an umbrella term for hypertensive urgency and hypertensive emergency. These two conditions occur when blood pressure becomes very high, possibly causing organ damage.

Hypertensive urgency occurs when blood pressure spikes – blood pressure readings are 180/110 mmHg or higher – but there is no damage to the body's organs.

Hypertensive emergency means blood pressure is so high that organ damage can occur. This is managed in an ICU of a hospital.

Symptoms of hypertensive emergency include:

- Chest pain
- Headache
- Arrhythmias
- Dyspnea
- Nosebleeds
- Severe anxiety
- Altered mental status

Organ damage associated with hypertensive emergency may include:

- Bleeding into the brain (hemorrhagic stroke)
- Heart failure
- Heart attack
- Fluid in the lungs (pulmonary edema)
- Aneurysm (aortic dissection)
- Eclampsia (occurs during pregnancy)

Diagnosis →

- Regular monitoring of blood pressure (hypertensive emergency is when blood pressure is more than 180 / 120 mm Hg)
- History taking
- Eye exam to look for swelling and bleeding
- Blood and urine testing
- ECG

Treatment →

- IV sodium nitroprusside injection.
- Other intravenous agents like nitroglycerine, labetalol etc. can also be used.
- In addition, non-pharmacological treatment could be considered in cases of resistant malignant hypertension due to end stage kidney failure, such as surgical nephrectomy, laparoscopic nephrectomy, and renal artery embolization in cases of anesthesia risk.
- It is also important that the blood pressure is lowered smoothly, not too abruptly. The initial goal in hypertensive emergencies is to reduce the pressure by no more than 25% (within minutes to 1 or 2 hours), and then toward a level of 160/100 mm Hg within a total of 2–6 hours. Excessive reduction in blood pressure can precipitate coronary, cerebral, or renal ischemia and, possibly, infarction.

Acute abdominal pain

An acute abdomen refers to a sudden, severe abdominal pain. It is in many cases a medical emergency, requiring urgent and specific diagnosis. Several causes need surgical treatment.

Causes →

- Acute appendicitis
- Acute peptic ulcer and its complications
- Acute Gastritis
- Acute Pancreatitis
- Acute Peritonitis (including hollow viscus perforation)
- Biliary colic
- Ureteric colic
- Renal colic
- Bowel obstruction
- Acute intestinal ischemia
- Ruptured spleen

Diagnosis →

The initial assessment should attempt to determine if the patient has an acute surgical problem that requires immediate and prompt surgical intervention, or urgent medical therapy. History, Physical examination, Abdominal scan (USG / CT scan / MRI) etc. can help to diagnose cause.

Differential Diagnosis →

The location of abdominal pain is the most useful initial feature to help narrow differential. These can be classified based upon quadrant or region affected →

Epigastric region → Peptic ulcers, Cholecystitis, Pancreatitis, MI

Left hypochondrium region → Gastric ulcer, Pneumonia, Splenic rupture

Right hypochondrium region → Hepatitis, Cholecystitis, Pneumonia

Peri-umbilical region → Small bowel obstruction, Large bowel obstruction, Abdominal aortic aneurysm, Appendicitis

Left lumbar region → Pyelonephritis, Renal colic, Ureteric colic

Right lumbar region → Pyelonephritis, Renal colic, Ureteric colic

Hypogastric region → IBD, Gynecological, UTI

Left iliac region → Testicular torsion, Inguinal hernia, Bowel obstruction

Right iliac region → Appendicitis, Testicular torsion, Inguinal hernia

Treatment →

- Hospitalization
- IV or IM Analgesics or Sedatives to relieve pain
Antispasmodics like dicyclomine with paracetamol or mefenamic acid
- Antibiotics to treat infections (if any)
- IV Fluids & Electrolytes
- Find and treat the underlying cause (The definitive management of acute abdomen depends largely on the cause)
- Surgery (if needed)

Lithotomy or lithotripsy in case of calculus (stone),

Appendectomy in case of appendicitis,

Cholecystectomy in case of cholecystitis or gallstone,

Herniectomy or herniorrhaphy in case of hernia,

Colectomy in case of diverticulitis etc.

Anuria / Oliguria

Oliguria is the low output of urine.

In humans, it is clinically classified as an output less than 400 ml/day (but more than 80 ml/day). The decreased output of urine may be a sign of dehydration, kidney failure, hypovolemic shock, multiple organ dysfunction syndrome, urinary obstruction / urinary retention, DKA, pre-eclampsia, and urinary tract infections etc.

Beyond oliguria is anuria, which represents an absence of urine, clinically classified as below 80 ml/day.

Diagnosis →

Ultrasound examination of the kidney to rule out obstructive processes.

The mechanisms causing oliguria can be categorized globally in three different categories:

- **Prerenal:** in response to hypoperfusion of the kidney (e.g. as a result of dehydration by poor oral intake, cardiogenic shock, diarrhea, G6PD deficiency, massive bleeding or sepsis)
- **Renal:** due to kidney damage (severe hypoperfusion, rhabdomyolysis, medication)
- **Postrenal:** as a consequence of obstruction of the urine flow (e.g. enlarged prostate, tumour compression urinary outflow, expanding hematoma or fluid collection)

Treatment →

IV Furosemide – Catheterization – Surgical intervention (if needed) –

Treat the cause like urinary obstruction, renal insufficiency etc.

Congestive Heart Failure

CHF is when the heart is unable to pump sufficiently to maintain blood flow to meet the body's needs.

Types of CHF →

1. **Left-sided CHF** is the most common type of CHF. It occurs when the left ventricle doesn't properly pump blood. As the condition progresses, fluid can build up in the lungs, which makes breathing difficult. There are two kinds of left-sided heart failure:
 - a. **Systolic heart failure** occurs when the left ventricle fails to contract normally. This reduces the level of force available to push blood into circulation. Without this force, the heart can't pump properly.
 - b. **Diastolic failure**, or diastolic dysfunction, happens when the muscle in the left ventricle becomes stiff. Because it can no longer relax, the heart can't quite fill with blood between beats.
2. **Right-sided CHF** occurs when the right ventricle has difficulty pumping blood to the lungs. Blood backs up in the blood vessels, which causes fluid retention in lower extremities, abdomen, and other vital organs.

It's possible to have left-sided and right-sided CHF at the same time. Usually, the disease starts in the left side and then travels to the right when left untreated.

Causes →

- Coronary artery disease
- Heart attack (occurs when a coronary artery becomes suddenly blocked, stopping the flow of blood to the heart muscle)
- Cardiomyopathy – Valve disease – Hypertension – Kidney disease – Congenital heart disease – Diabetes etc.

Symptoms →

Acute attack includes rapid or irregular heartbeat (arrhythmias) associated with shortness of breath (dyspnea), chest pain or tightness, weakness or fainting.

Chronic symptoms include dyspnea at rest, chest tightness, fatigue, weakness, edema in legs, ankles & feet etc.

Treatment →

- Resuscitation.
- Treat high blood pressure.
- Treat lipid disorders.
- Discontinue alcohol or illegal drug use.
- An ACE inhibitor or an angiotensin II receptor blocker is prescribed if patient has coronary artery disease, diabetes, high blood pressure, or other vascular or cardiac conditions. β blockers may be prescribed if patient has high blood pressure or a previous heart attack.
- Surgical intervention (if needed) →
 - Angioplasty
 - Open heart surgery (Valve repair)
 - Heart bypass surgery (Coronary Artery Bypass Grafting)

Myocardial Infarction

Myocardial infarction (MI), commonly known as a heart attack, occurs when blood flow decreases or stops to a part of the heart, causing damage to the heart muscle.

The most common symptom is chest pain or discomfort which may travel into the shoulder, arm, back, neck, or jaw. The discomfort may occasionally feel like heartburn. Other symptoms may include shortness of breath, nausea, feeling faint, a cold sweat, or feeling tired.

An MI may cause heart failure, an irregular heartbeat, cardiogenic shock, or cardiac arrest.

Investigations →

- ECG
- Angiogram

Treatment →

- Resuscitation
- The pain associated with myocardial infarction should be treated with Nitroglycerin 0.3-0.6 mg sublingually.
- Aspirin with Clopidogrel, an antiplatelet/anticoagulant, is given.
- Percutaneous coronary intervention (PCI) is the treatment of choice for STEMI if it can be performed in a timely manner, ideally within 90–120 minutes.
- Coronary Artery Bypass Grafting (CABG).

Shock

Shock is the life-threatening condition in which the circulatory system fails to maintain adequate blood flow in the body.

Shock is the state of low blood perfusion to tissues resulting in widespread reduction in delivery of oxygen and nutrients.

Types of shock →

Cardiogenic shock – Hypovolemic shock – Anaphylactic shock – Septic shock – Neurogenic shock

Clinical Features →

Anxiety, Sweating, Fainting, Pallor, Deep breathing,
Cold and clammy extremities,
Cyanosis (central as well as peripheral),
Fall of BP & increase the pulse rate (hypotension & tachycardia),
Thread pulse with low volume,
Oliguria.

Treatment →

- Head low position to improve blood volume towards heart and brain.
- Wrapping of limbs with blankets to raise the temperature.
- Inj. Atropine 0.6 mg IV
- Inj. Morphine IV
- IV RL / D5 / DNS as per condition, IV Mannitol / Inj. Hydrocortisone
- Antibiotics – Removal of septic focus (if septic shock)
- Intracardiac injection of Adrenaline

Syncope

Syncope is a temporary loss of consciousness usually related to insufficient blood flow to the brain. It's also called fainting. It most often occurs when blood pressure is too low (hypotension) and the heart doesn't pump enough oxygen to the brain.

Underlying causes →

Neurocardiogenic syncope results when something triggers a short-term malfunction of the autonomous nervous system (ANS). It is also known as neurally mediated syncope (NMS).

Possible triggers include:

- Sudden exposure to an unpleasant sight or experience, or suddenly seeing something that is unpleasant or shocking, such as blood
- Sudden emotional upset, for example when receiving bad news
- Extreme embarrassment
- Standing still for a long time
- Being in a hot and stuffy place for a long time

Orthostatic hypotension can happen when a person faints after stands up rapidly from a seated or lying down position.

Triggers include:

Dehydration – Untreated Diabetes – Some medications like diuretics, β blockers like antihypertensive drugs – Alcohol etc.

Cardiac syncope → An underlying heart problem can cause a drop in blood and oxygen supply to the brain.

Possible conditions include:

- Arrhythmias, or abnormal heart beat
- Stenosis, a blockage of the heart valves
- Hypertension, or high blood pressure
- A heart attack, when a heart muscle dies because of a lack of blood and oxygen

Management →

- Make the patient lie flat with raised legs
- Fruit juice or glucose drink or IV Fluids if needed
- Reassurance
- Diagnose and treat the underlying cause
- If there is no sign of breathing and circulation, start cardiopulmonary resuscitation (CPR).
- If the patient was injured and fainting is due to blood loss, stop bleeding by pressure, bandage etc. and treat the hemorrhagic shock.

Convulsions

Convulsion refers to a sudden, violent, irregular movement of the body, caused by involuntary contraction of muscles and associated especially with brain disorders such as epilepsy, the presence of certain toxins or other agents in the blood, or fever in children.

Management →

- Maintenance of adequate airway, breathing and circulation (ABCs).
- IV Benzodiazepines → Lorazepam, Diazepam, or Midazolam.
- Diagnose and treat the underlying cause.

Hyperpyrexia

Hyperpyrexia is an extreme elevation of body temperature which, depending upon the source, is classified as a core body temperature greater than or equal to 40.0 or 41.5 °C (104.0 or 106.7 °F).

Causes →

Intracranial hemorrhage – Viral or bacterial infections – Sepsis – Kawasaki syndrome – Thyroid storm etc.

Management →

- A cool bath or cold, wet sponges put on the skin.
- Liquid hydration through IV Fluids or from drinking.
- Fever-reducing medications, such as dantrolene.

Hyperglycaemia

Hyperglycemia doesn't cause symptoms until glucose values are significantly elevated – above 200 milligrams per deciliter (mg/dL).

Acute hyperglycemia involving glucose levels that are extremely high is a medical emergency and can rapidly produce serious complications (such as fluid loss through osmotic diuresis). It is most often seen in persons who have uncontrolled insulin-dependent diabetes.

Classic hyperglycemic triad → Polyuria – Polydipsia – Polyphagia.

Management →

- Acute hyperglycemia can be treated by direct administration of insulin in most cases. Severe hyperglycemia can be treated with oral hypoglycemic therapy and lifestyle modification.
- A combination of proper diet, regular exercise, and insulin or other medication such as metformin, sulphonylureas etc.

Status Asthmaticus

Acute severe asthma is an acute exacerbation of asthma that does not respond to standard treatments of bronchodilators (inhalers) and corticosteroids.

Symptoms include chest tightness, rapidly progressive dyspnea (shortness of breath), dry cough, labored breathing, and extreme wheezing. On examination, the respiratory rate may be elevated (more than 25 breaths per minute), and the heart rate may be rapid (110 beats per minute or faster). Reduced oxygen saturation levels (but above 92%) are often encountered.

Management →

- Positive-pressure therapy, including mechanical ventilation.
- Intravenous (IV) medications (e.g. magnesium sulfate), aerosolized medications to dilate the airways (bronchodilation) (e.g., albuterol or ipratropium bromide / salbutamol)
- Intravenous corticosteroids and methylxanthines are often given.

Acute Respiratory Distress Syndrome (ARDS)

ARDS is a rapidly progressive disease occurring in critically ill patients. The main complication in ARDS is a buildup of fluid in the alveoli of the lungs. It results in less oxygen reached to organs (hypoxia).

ARDS is not a particular disease; rather, it is a clinical condition triggered by various pathologies such as trauma, pneumonia, and sepsis.

Management → Supplemental oxygen – Mechanical ventilation – IV Fluids – Medications like sedatives, antibiotics, anti-inflammatory etc.

Drowning

Drowning is defined as respiratory impairment from being in or under a liquid (submersion). Near-drowning is a term typically used to describe almost dying from suffocating under water; it is the last stage before fatal drowning.

Management →

- Cardiopulmonary resuscitation (CPR)
- The airway-breathing-circulation (ABC) sequence should be followed, rather than starting with compressions as is typical in cardiac arrest, as the basic problem is lack of oxygen.
- If the person has near-drowned in cold water, remove their wet clothes and cover them in warm blankets or clothing to prevent hypothermia.
- Hypotension remaining after oxygenation may be treated by rapid crystalloid infusion.

Electric shock

Electrical injury is a physiological reaction caused by electric current passing through the (human) body. Electric shock occurs upon contact of a body part with any source of electricity that causes a sufficient magnitude of current to pass through the victim's flesh, viscera or hair.

Management →

- Begin CPR if the person shows no signs of circulation, such as breathing, coughing or movement.

- Treatment of electric shock depends on the severity of the burns or the nature of other injuries found.
- Topical antibiotic ointments and dressings are applied for minor burns. For severe burns, surgery with skin grafting may be required. Sometimes amputation may also be needed.
- As with conventional thermal injury, electrical injuries cause massive fluid shifts with extensive tissue damage and acidosis; therefore, monitoring a patient's hemodynamics is important. A Foley catheter is helpful in monitoring urine output and, therefore, tissue perfusion.
- IV Fluids & electrolytes to maintain fluid & electrolyte balance
- Antibiotics to avoid sepsis.
- Analgesics and anti-inflammatory drugs for symptomatic relief.